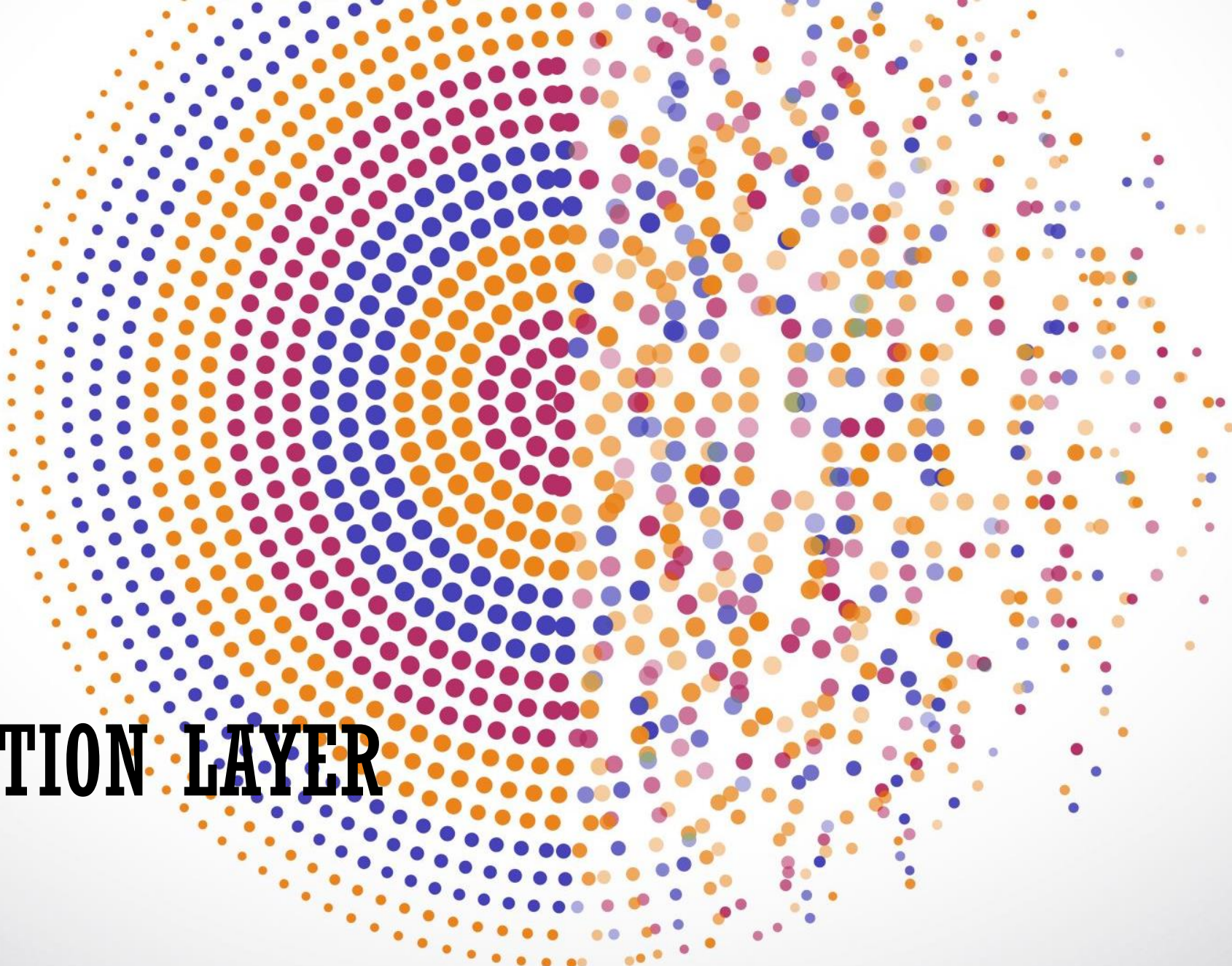


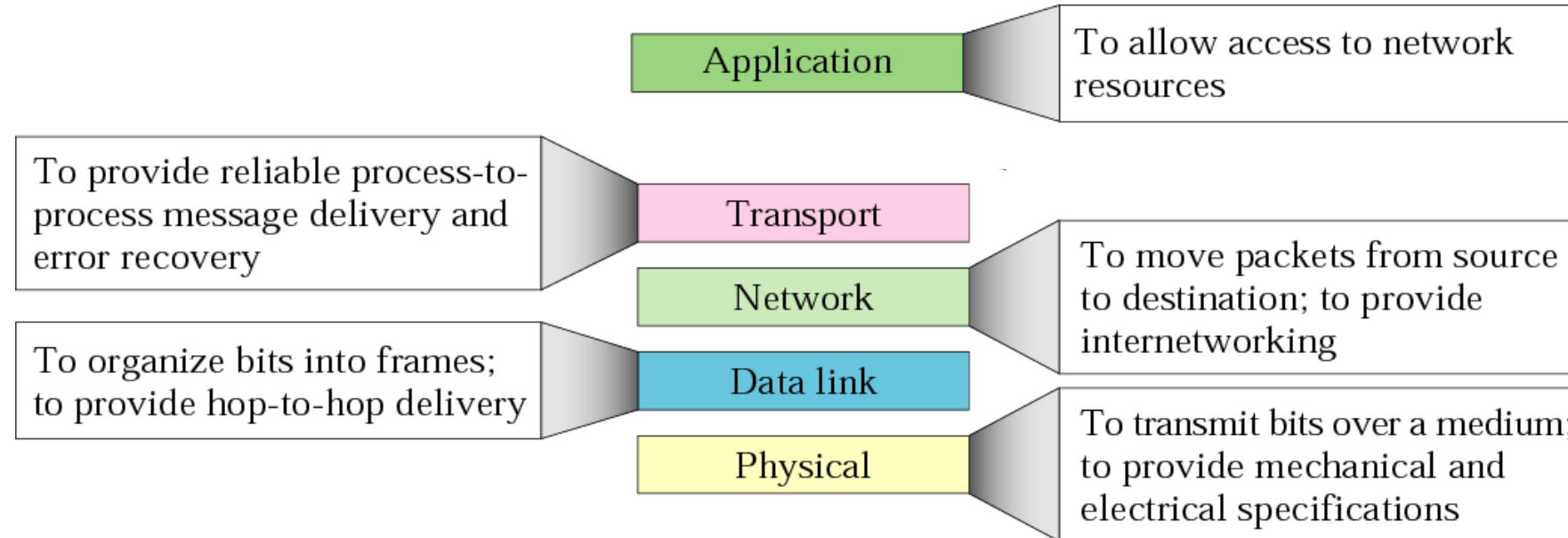
CN55 | UNIT 5



APPLICATION LAYER

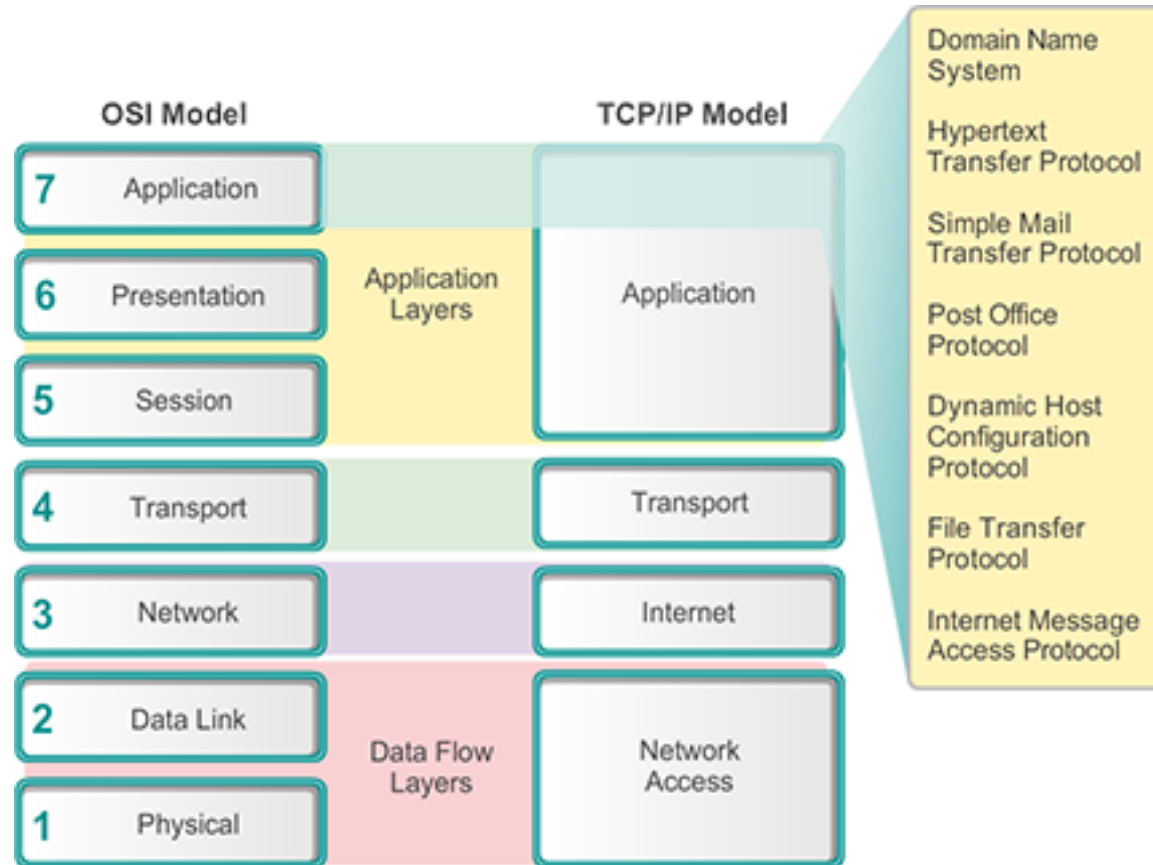
UNIT 5.1

TCP/IP PROTOCOL SUITE



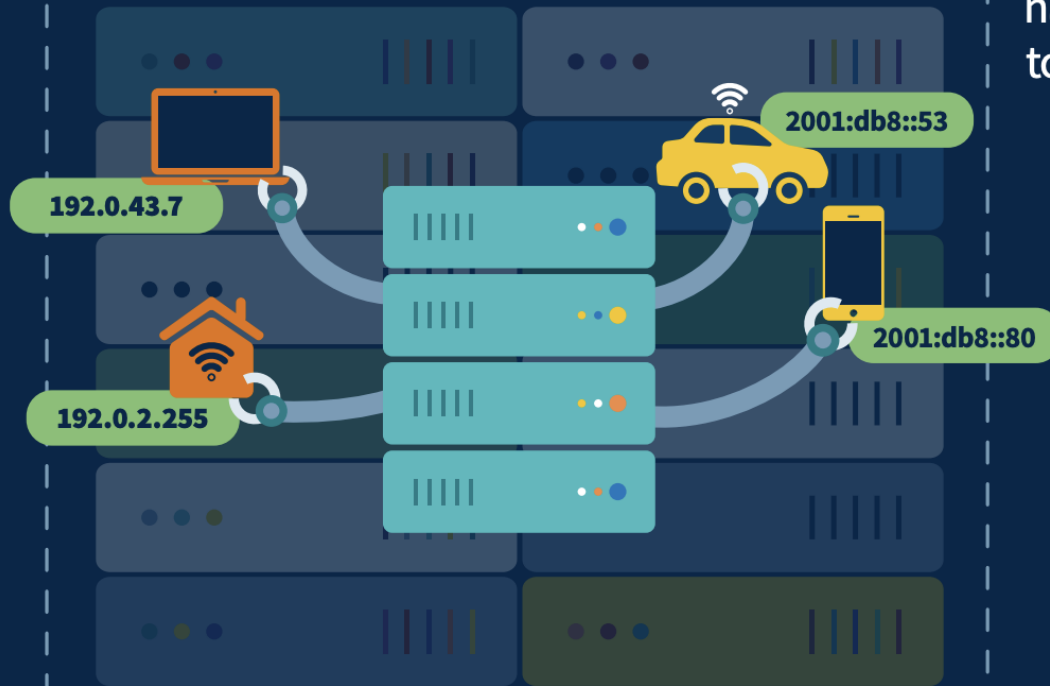
- The application layer is responsible for providing services to the user.
- It provides user interfaces and support for many services.

APPLICATION LAYER



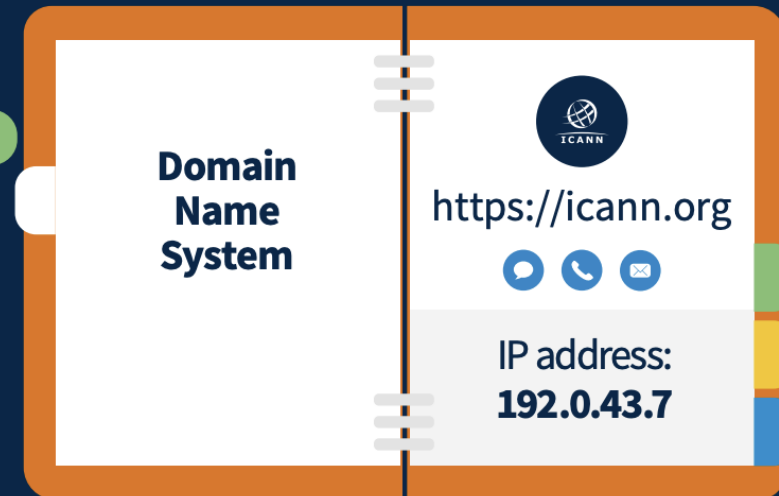
How do billions of connected devices find each other?

- Each device or website on the Internet has a unique address – like a telephone number.



This address is a series of numbers and letters, called an **IP address**.
IP stands for **Internet Protocol**.

- The **Domain Name System (DNS)** makes navigating the Internet easier by allowing users to type in familiar letters – the **domain name** – instead of the **IP address**.

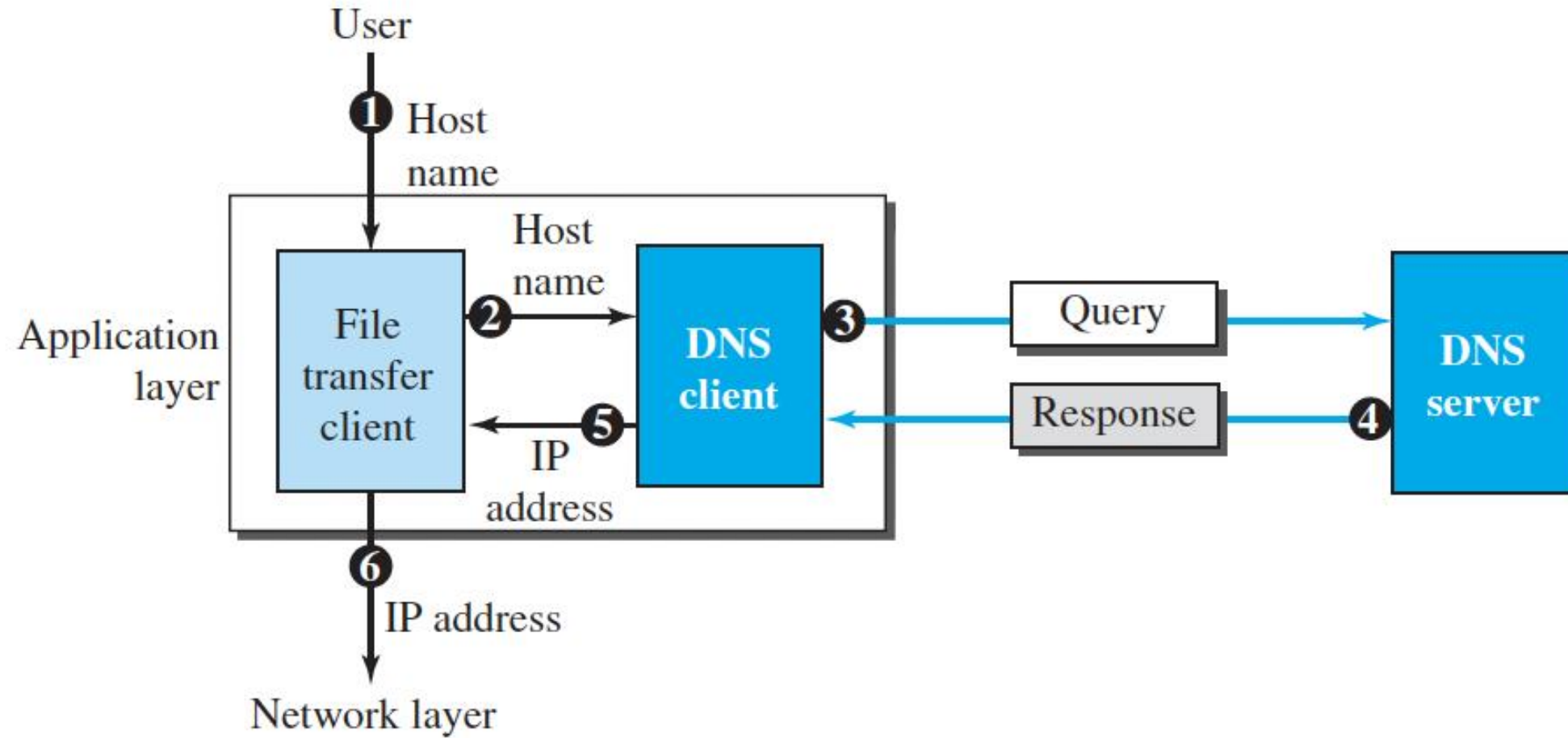


For example, you only need to type in **https://icann.org** to reach ICANN's website, instead of its **IP address - 192.0.43.7**

DNS

- A client/server application that provides name services for other applications.
- It enables the use of application layer address such as email address, URL instead of network layer logical addresses.

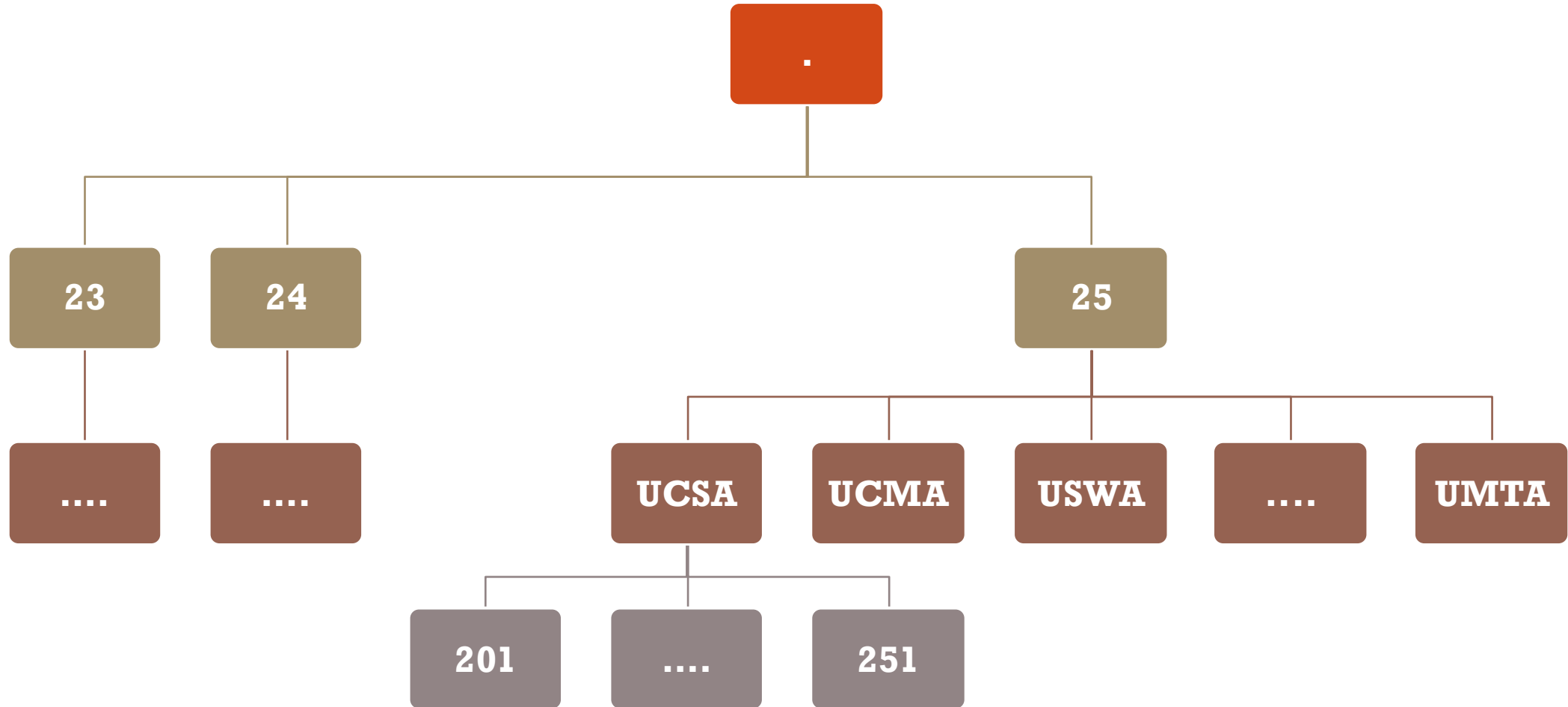
DNS



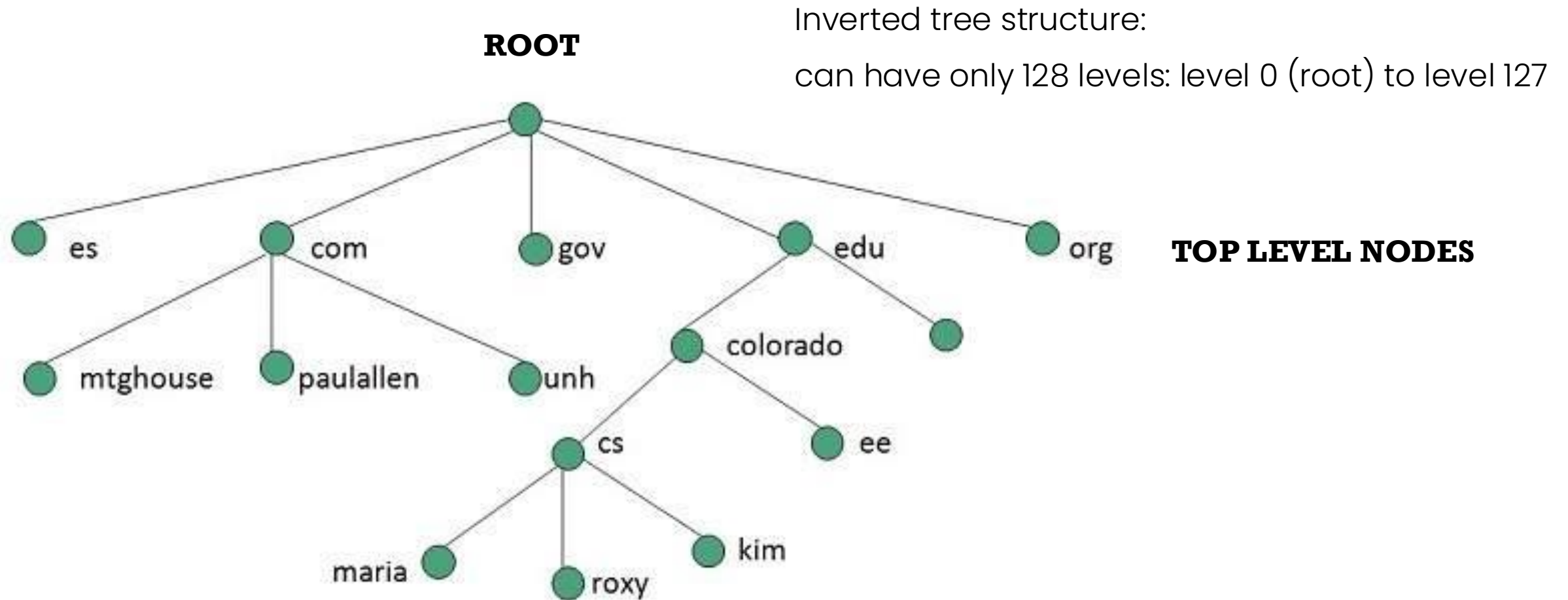
NAME SPACE

- Names must be unique because addresses are unique.
- Name space that maps each address to a unique name can be organized in 2 ways:
 - Flat
 - Hierarchical

NAME SPACE

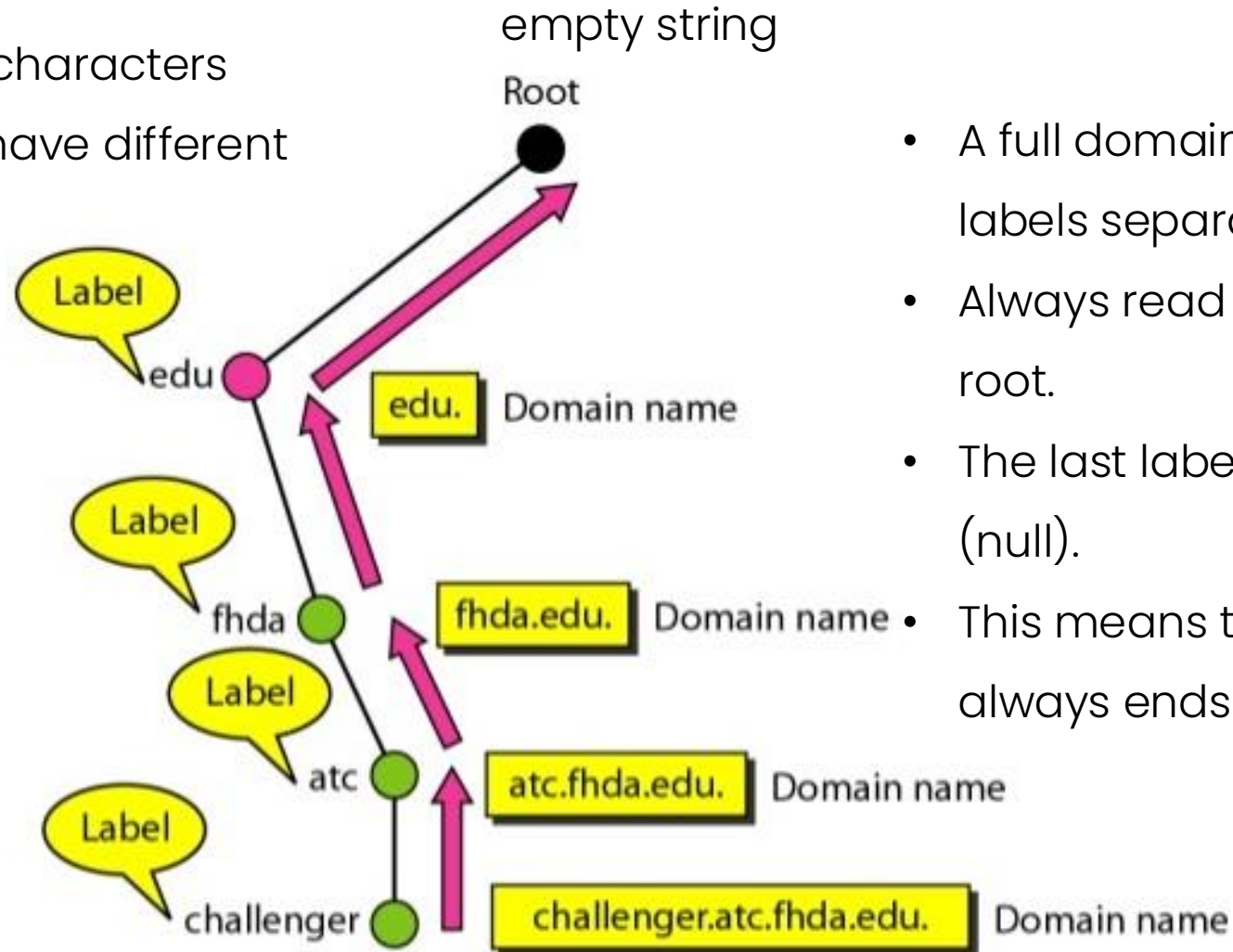


DOMAIN NAME SPACE



LABELS AND DOMAIN NAMES

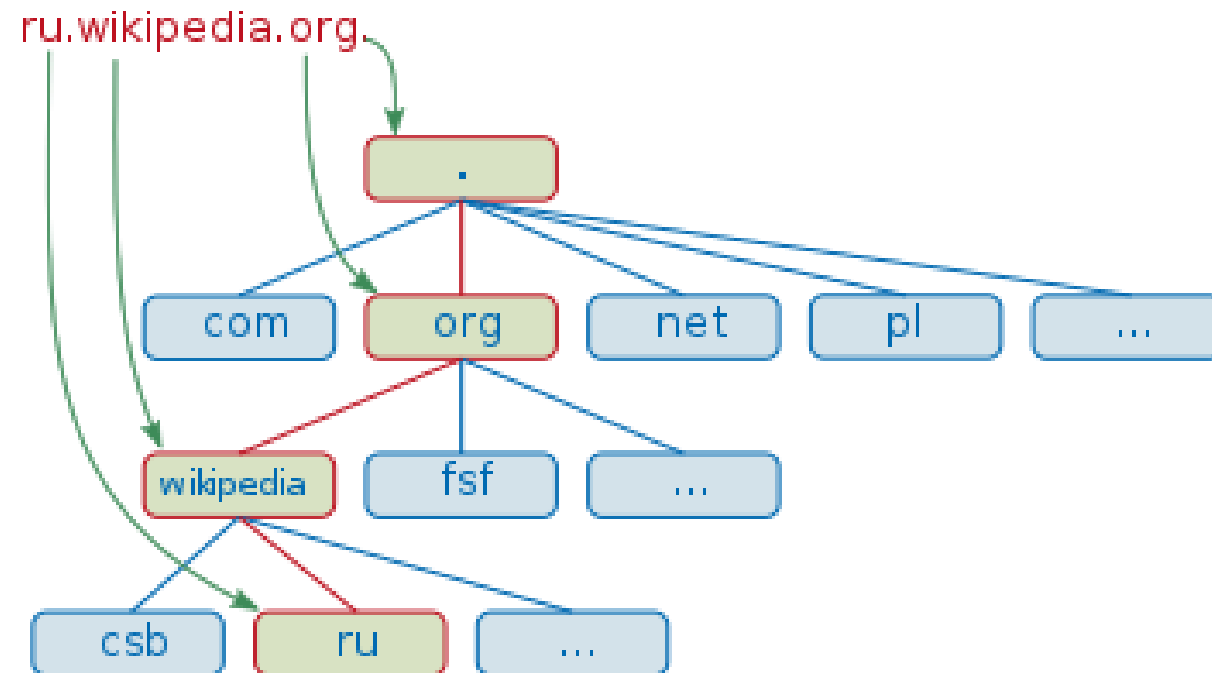
- a maximum of 63 characters
- children of nodes have different labels



- A full domain name is a sequence of labels separated by dots (.).
- Always read from the node up to the root.
- The last label is the label of the root (null).
- This means that a full domain name always ends in a null label

FULLY QUALIFIED DOMAIN NAME (FQDN)

- If a label is terminated by a null string
- Domain name that contains the full name of the host



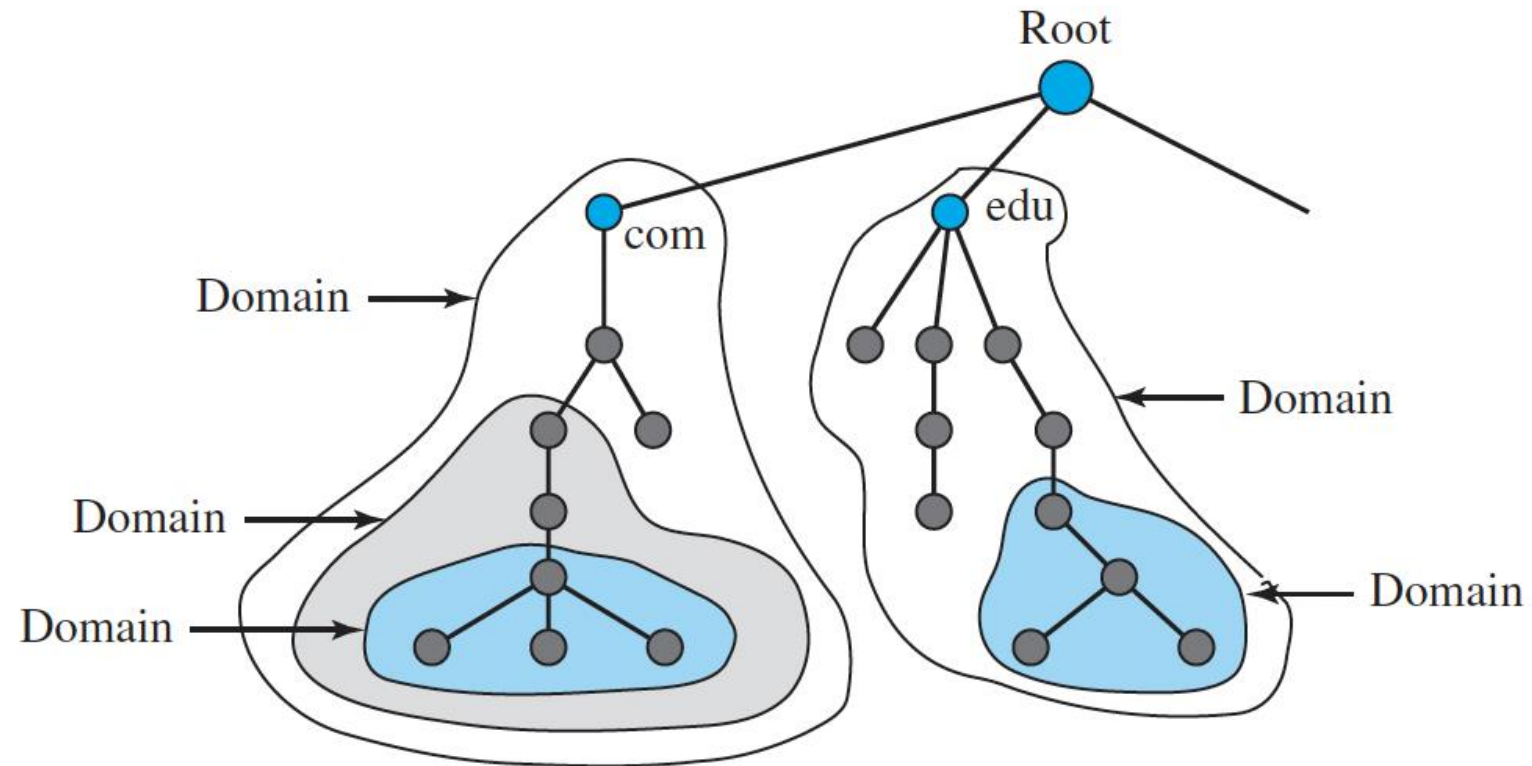
PARTIALLY QUALIFIED DOMAIN NAME (PQDN)

- Does not end with null string i.e., does not reach the root
- It is used when the name to be resolved belongs to the same site as the client
- Resolver can supply the missing part called suffix to create FQDN



DOMAIN

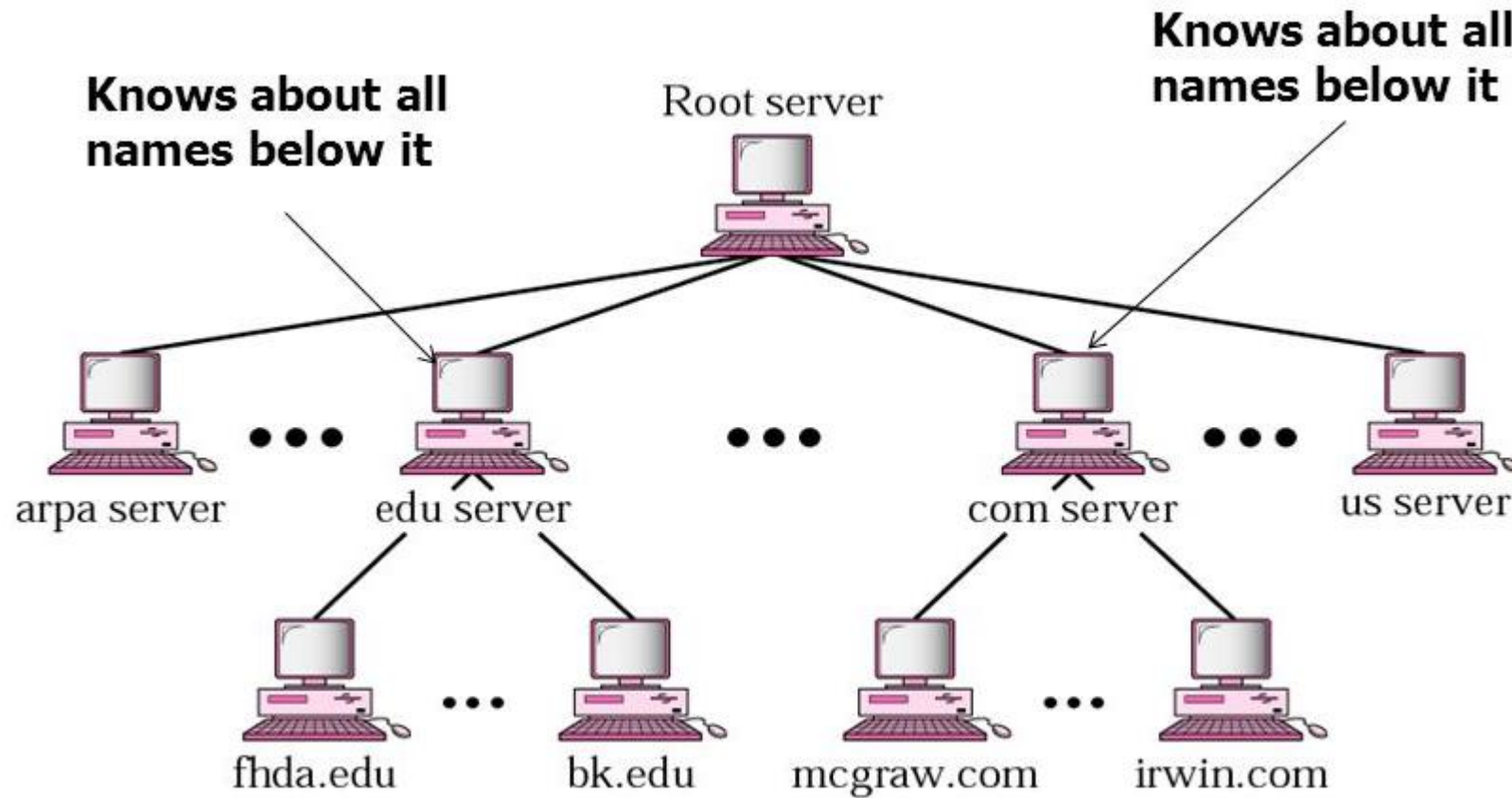
- Subtree of domain name space
- Name of the domain - name of node at top of the subtree



DISTRIBUTION OF NAME SPACE

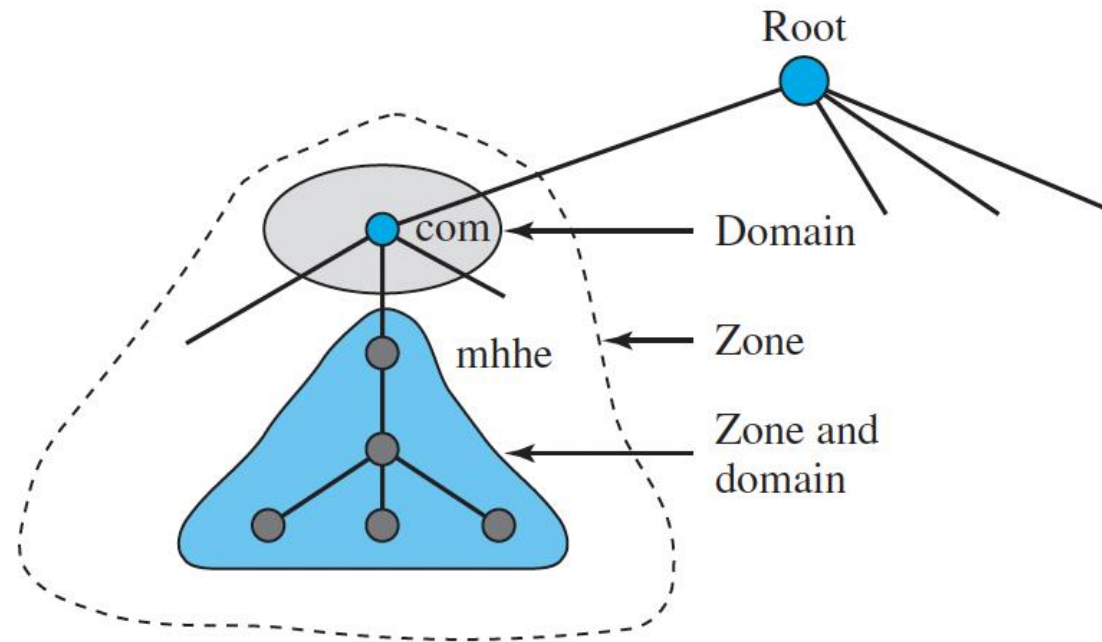
- It is inefficient and unreliable to have just one computer store huge amount of information.
- Solution is to distribute the information among many computer called DNS servers.
- Divide the whole space into many domains based on the first levels and domains to be divided further into smaller domains (subdomains)
- Each server can be responsible (authoritative) for either a large or small domain.
- Hierarchy of servers just like hierarchy of names

DISTRIBUTION OF NAME SPACE



ZONE

- What a server is responsible for or has authority over.



- Domain and zone are same if they are not divided into smaller domains

ZONE

- Server makes a database called zone file and keeps all the information for every node under that domain.
- If server divides its domain into subdomains, part of its authority is delegated to other servers - domain and zone becomes different.
- Original server keeps reference to the servers at lower levels, wherein these servers hold information about the nodes.

ROOT SERVER

- The zone of root server consists of the whole tree.
- Delegates its authority to other servers having their references.
- There are several root servers all over the world covering whole domain name space.

There are 12 independent **root server operators** that manage 13 root identities across the globe. The ICANN organization runs one of these root identities – the L-Root. These identities represent over 1,000 individual servers, each providing identical information from the root zone to resolvers all over the world.



What is the root zone?

The root zone holds referral information for the TLDs, which points to their DNS servers to help **resolve** your device's request.

PRIMARY SERVER

- Stores the zone file about that zone it has authority on its local disk.
- It is responsible for creating, maintaining, and updating the zone file.

SECONDARY SERVER

- Loads all information from the primary server.
- When secondary downloads information from the primary, it is called zone transfer.
- It can also load information from another secondary server.
- The secondary server neither creates nor updates the zone files.

SECONDARY SERVER

- If updating is required, it must be done by the primary server, which sends the updated version to the secondary.
- Both primary and secondary can be authoritative for the zones they serve.
- The idea is to create redundancy for the data so that if one server fails, the other can continue serving clients.

AUTHORITATIVE SERVER

- Has the official DNS data for a particular zone.
- It can be:
 - a **primary authoritative server**
 - a **secondary authoritative server**
- Technically a root server is an **authoritative name server**, but not every authoritative server is called a root server.

https://lookup.icann.org/en/lookup

Registration data lookup tool

Enter a domain name or an Internet number resource (IP Network or ASN)

[Frequently Asked Questions \(FAQ\)](#)

www.stellamariscollege.edu.in

Lookup

Domain Information

Name: stellamariscollege.edu.in

Internationalized Domain Name: stellamariscollege.edu.in

Registry Domain ID: D2887486-IN

Domain Status:

active

Nameservers:

ns3.hostek.com

ns4.hostek.com

Dates

Registry Expiration: 2029-03-19 06:05:19 UTC

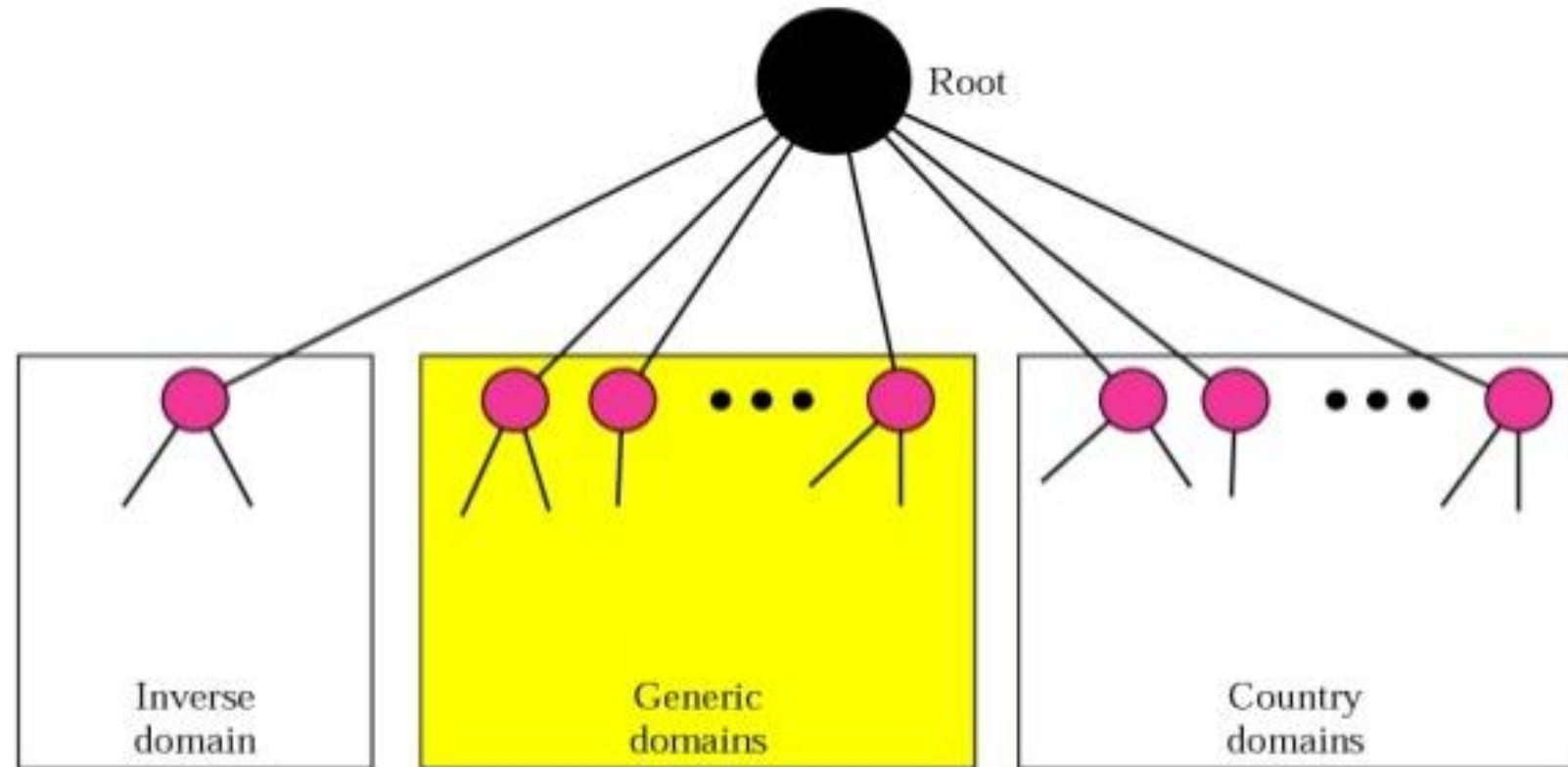
Updated: 2020-04-05 06:00:38 UTC

Created: 2008-03-19 06:05:19 UTC

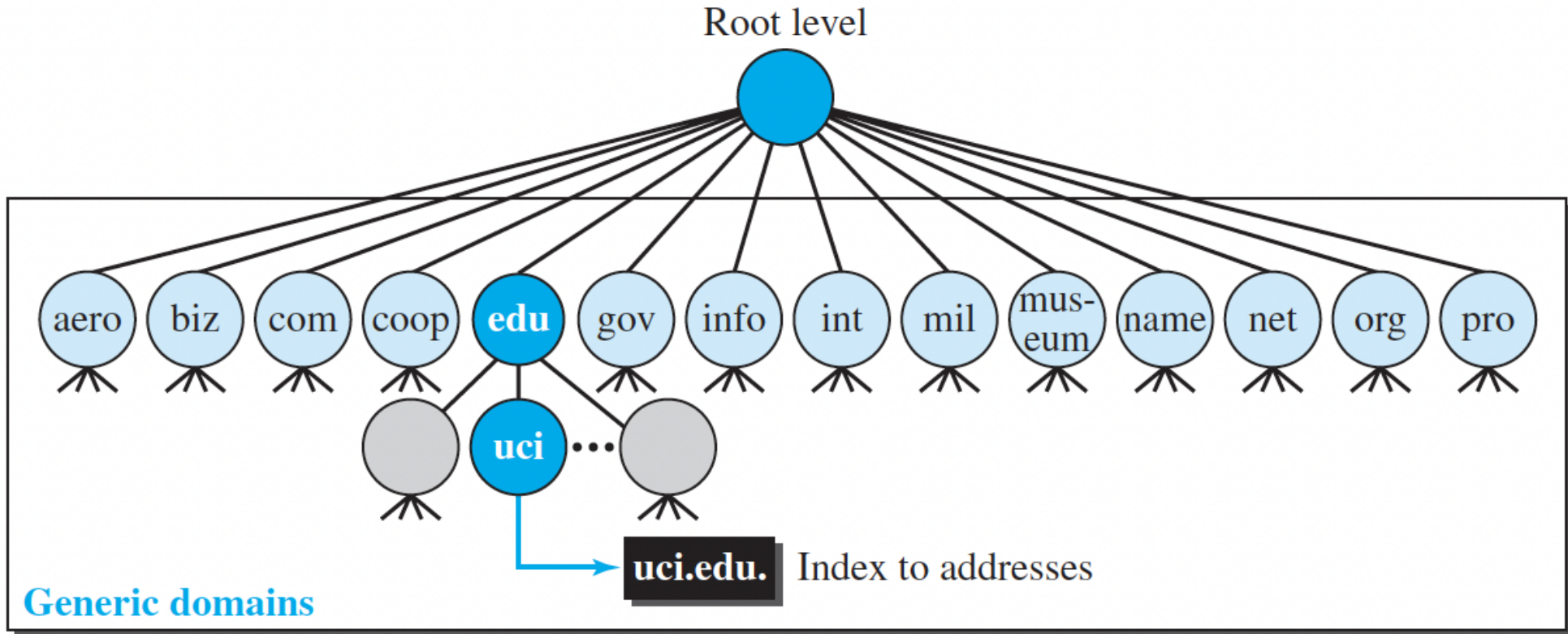
ns3.hostek.com is an authoritative name server for the stellamariscollege.edu.in zone, which contains the DNS record for www.stellamariscollege.edu.in

DNS IN THE INTERNET

- 3 Different Sections



GENERIC DOMAINS

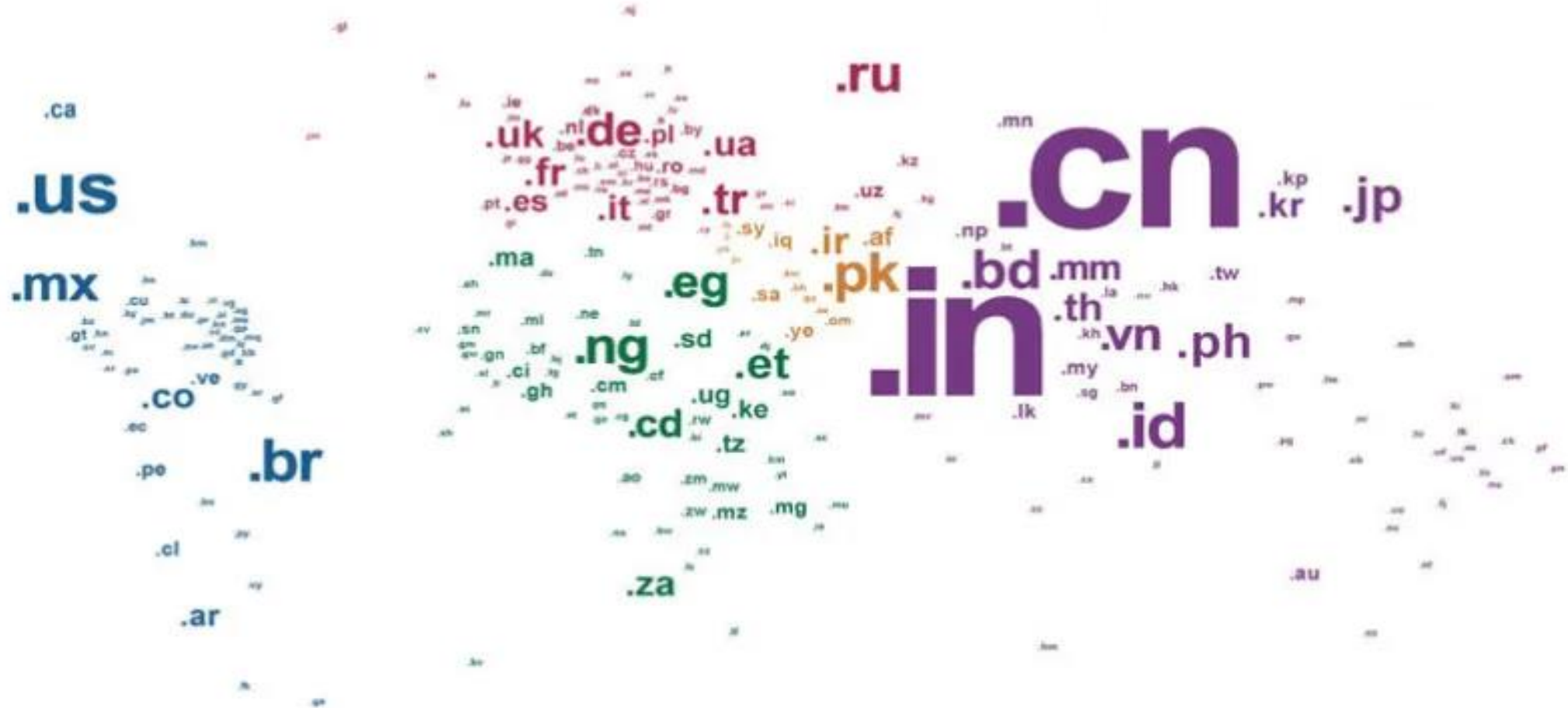


GENERIC DOMAINS

- Define registered hosts according to their generic behaviour

<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to “com”)
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

COUNTRY DOMAINS



RESOLUTION

- Mapping name to an address or an address to a name



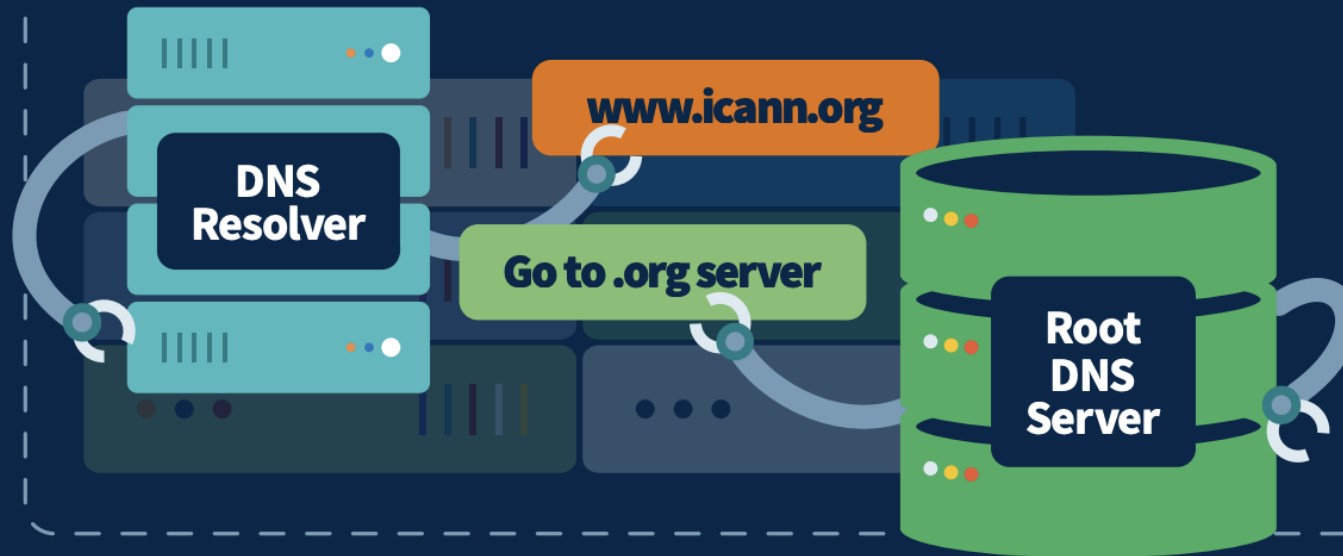
What happens when you type in that domain name?

- **STEP 1:** Your laptop asks the **DNS resolver**, “Where is `www.icann.org`?”



Image Source: <https://www.icann.org/en/system/files/files/root-server-system-infographic-23dec24-en.pdf>

- **STEP 2:** The DNS resolver asks a **root DNS server**, “Where is **www.icann.org**?”



Root DNS servers only know information about **top-level domain (TLD)* names**, so it tells the DNS resolver to “try .org” and gives the DNS resolver referral information to a **.org server**.

Image Source: <https://www.icann.org/en/system/files/files/root-server-system-infographic-23dec24-en.pdf>

- **STEP 3:** The DNS resolver asks the **.org DNS server**, “Where is **www.icann.org**?”

The **.org** server only knows about **.org** domains, so it tells the DNS resolver to “try **www.icann.org**” and gives the DNS resolver referral information to a **www.icann.org DNS server**.

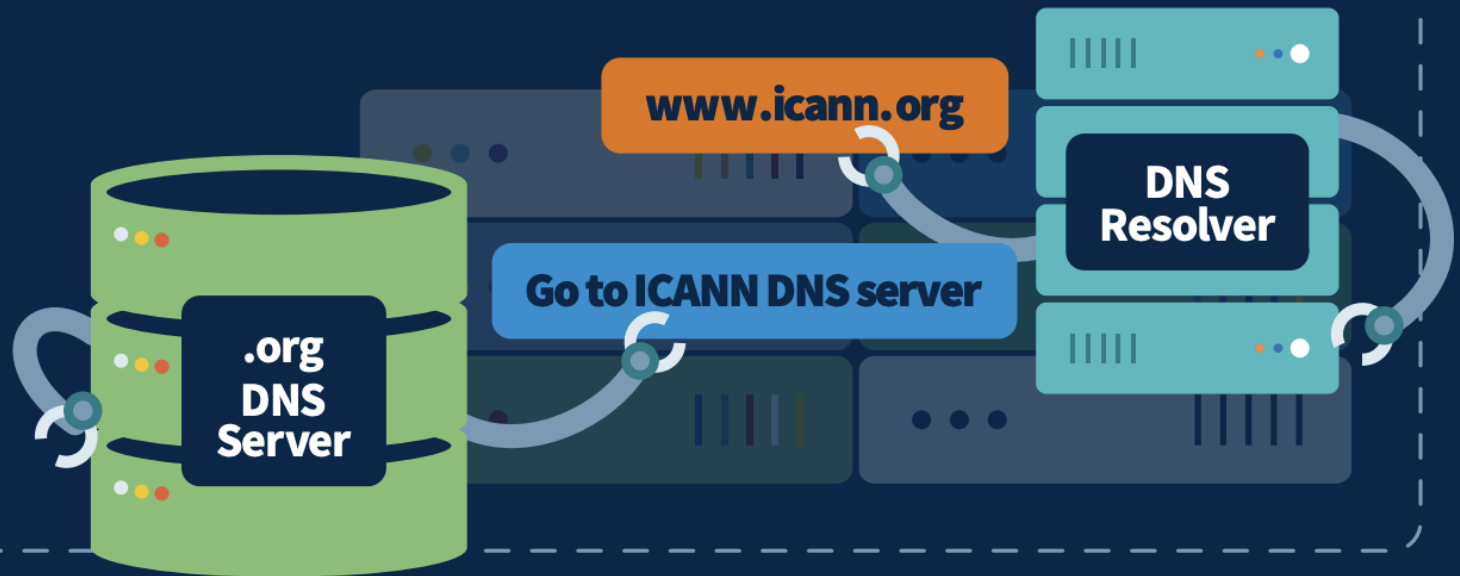
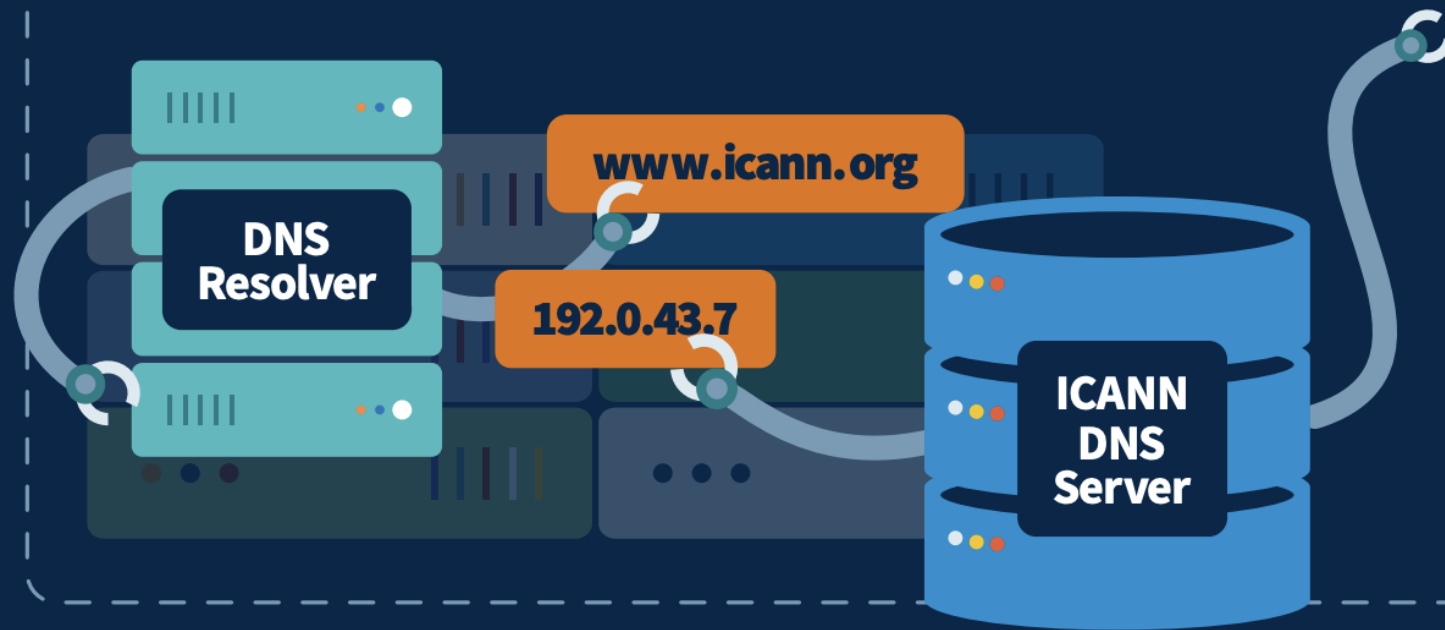


Image Source: <https://www.icann.org/en/system/files/files/root-server-system-infographic-23dec24-en.pdf>

- **STEP 4:** The DNS resolver asks the **www.icann.org DNS server**, “Where is **www.icann.org**?”



Since the **www.icann.org** DNS server knows about the entire website address, not just the TLD, it tells the DNS resolver that “**www.icann.org** is at IP address **192.0.43.7**”

Image Source: <https://www.icann.org/en/system/files/files/root-server-system-infographic-23dec24-en.pdf>

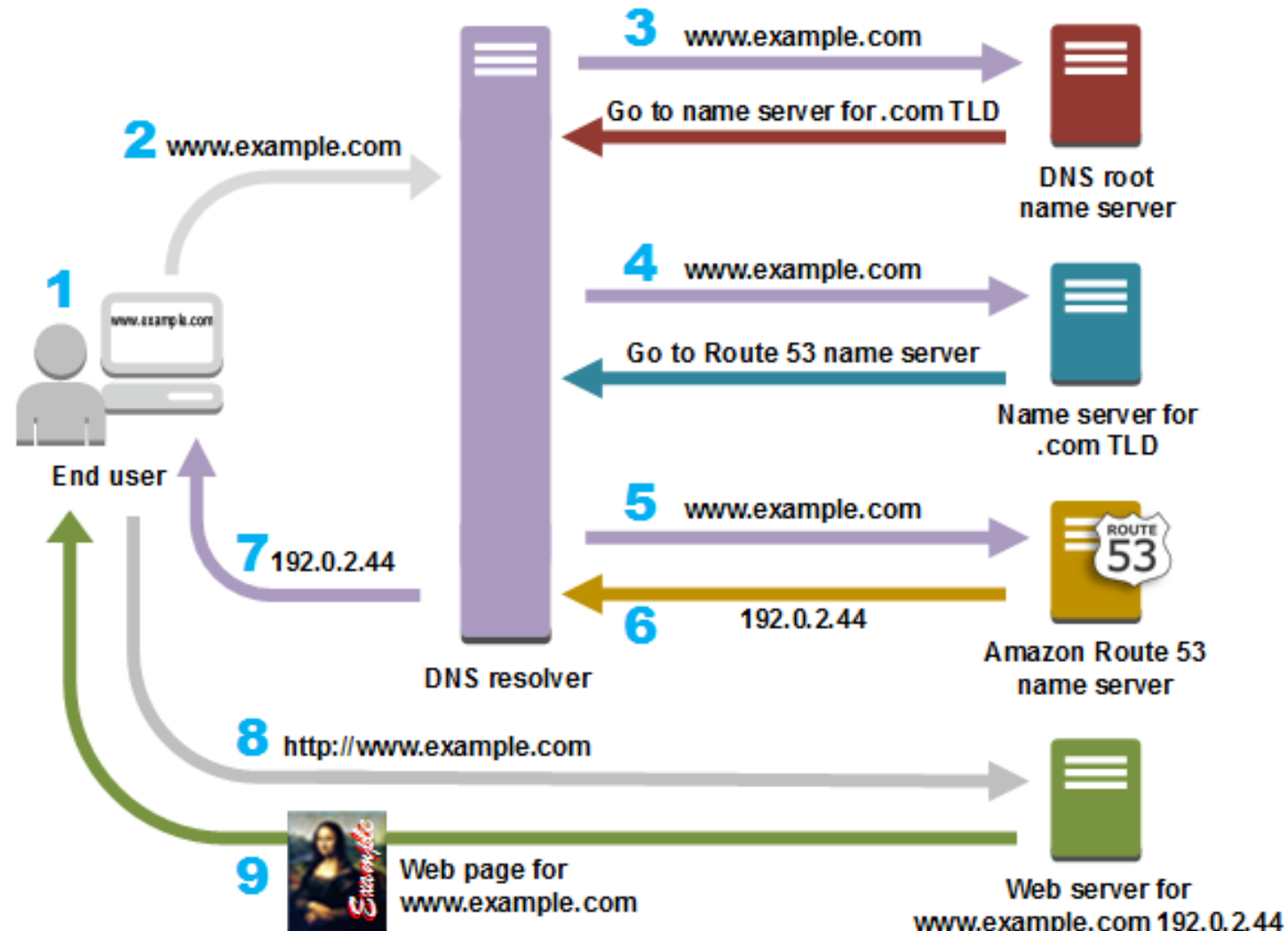
- **STEP 5:** The DNS resolver tells your laptop that “**www.icann.org** is at **192.0.43.7**”

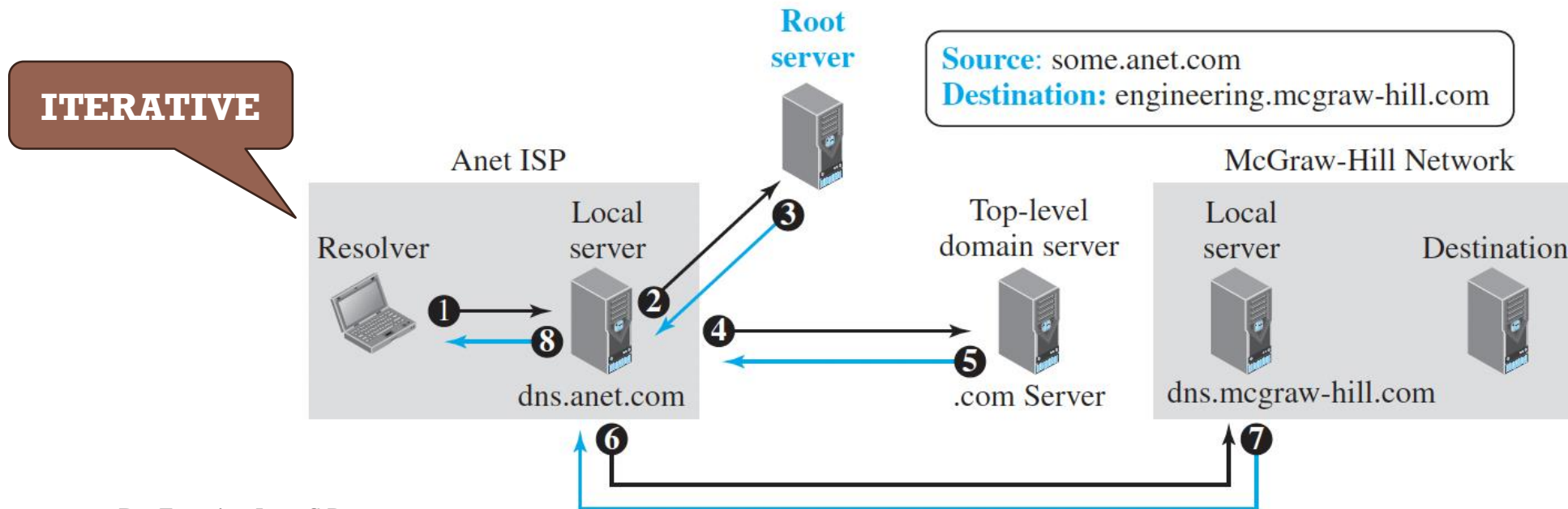
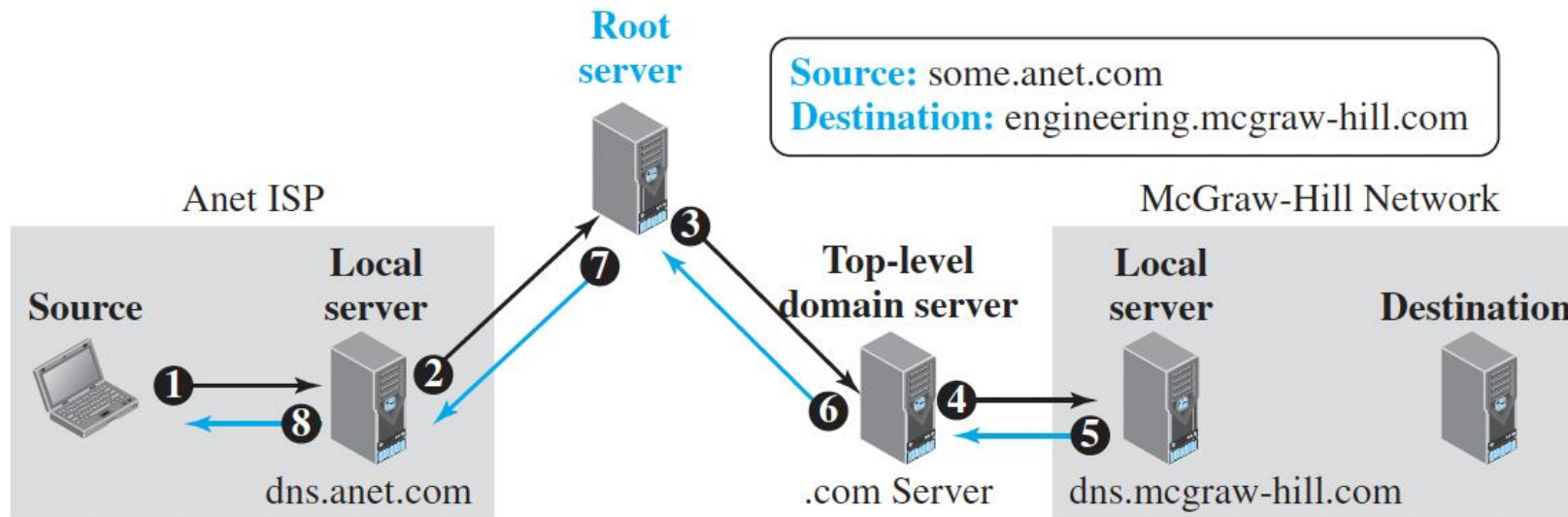


Success!

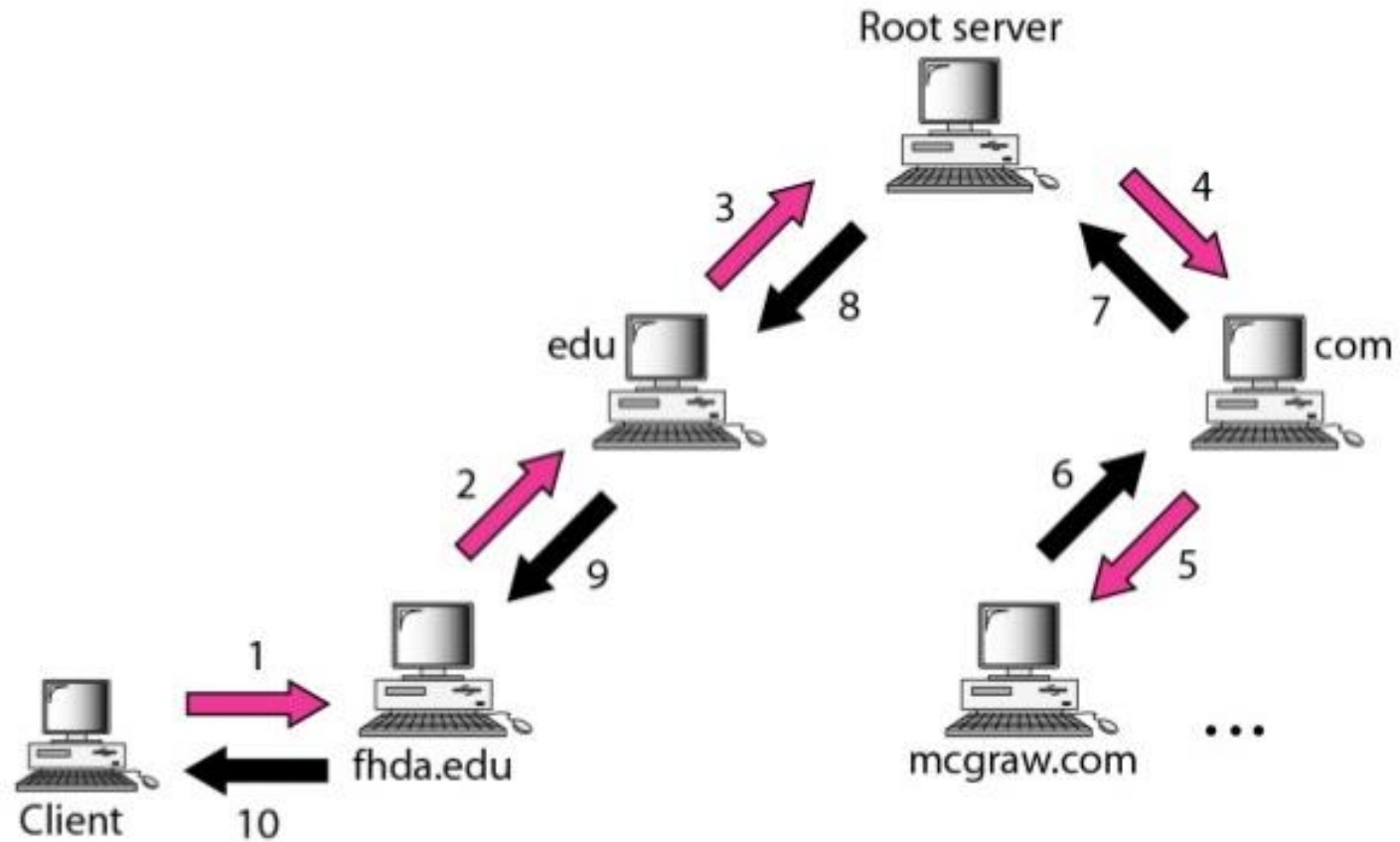
Image Source: <https://www.icann.org/en/system/files/files/root-server-system-infographic-23dec24-en.pdf>

RESOLVER - MAPPING NAMES TO ADDRESSES

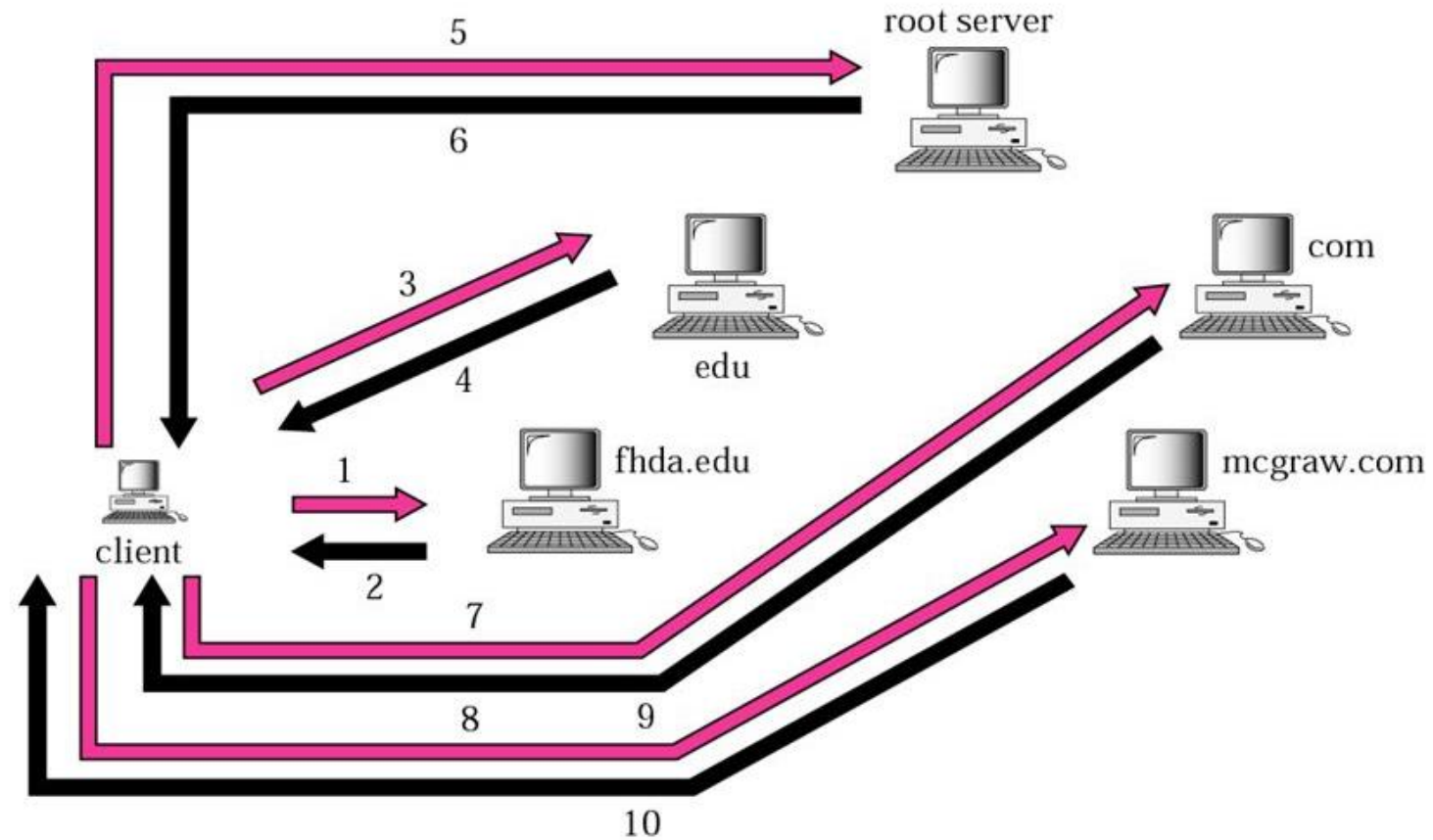




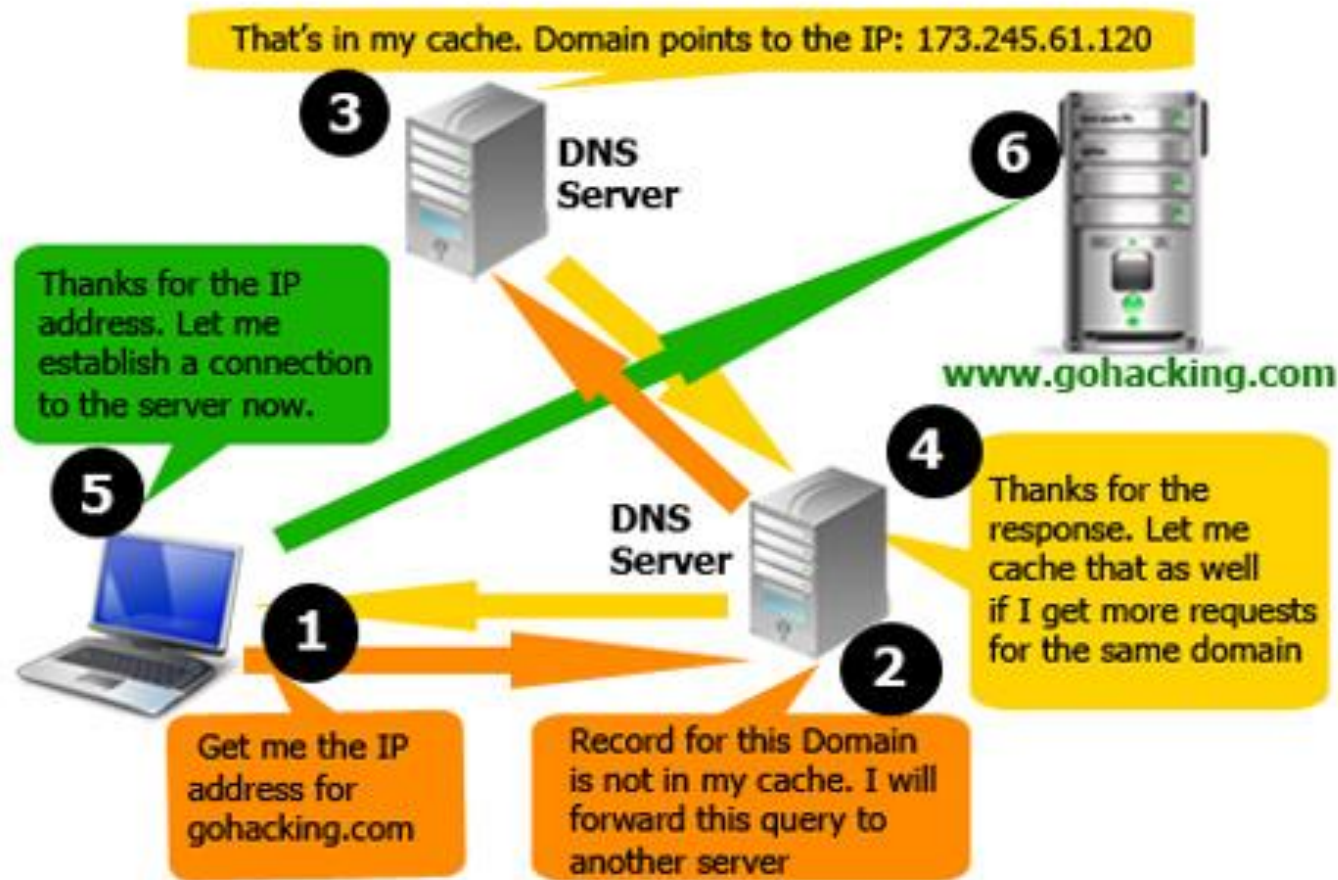
RECURSIVE RESOLUTION



ITERATIVE RESOLUTION



MAPPING ADDRESSES TO NAMES - CACHING



- If a server caches a mapping for a long time, it may send an outdated mapping to the client.
- Time to live (TTL). It defines the time in seconds that the receiving server can cache the information
- TTL counter to purge expired cache

CACHING

ptr:42.111.139.254

Find Problems

Type	IP Address
PTR	42.111.139.254 Vodafone Essar Ltd., Telecommunication - Value Added Services (AS38266)

Domain Name	TTL
42-111-139-254.live.vodafone.in	24 hrs

RESOURCE RECORDS

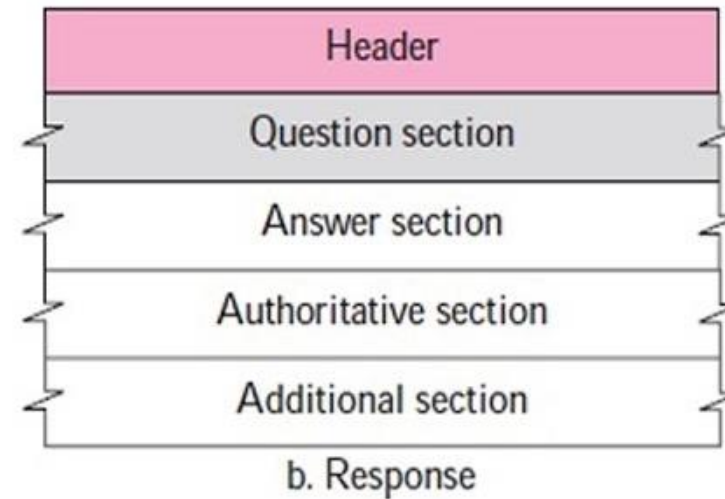
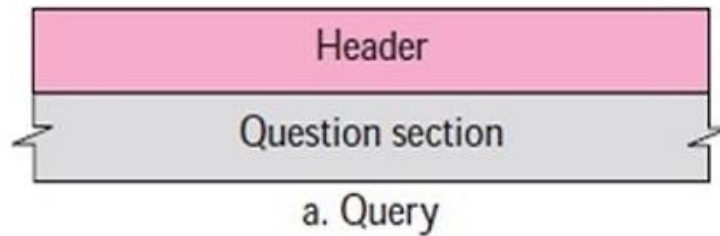
- The zone information associated with a server is implemented as a set of resource records.
- Class is IN for Internet

(Domain Name, Type, Class, TTL, Value)

<i>Type</i>	<i>Interpretation of value</i>
A	A 32-bit IPv4 address (see Chapter 18)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 22)

DNS MESSAGES - 2 TYPES

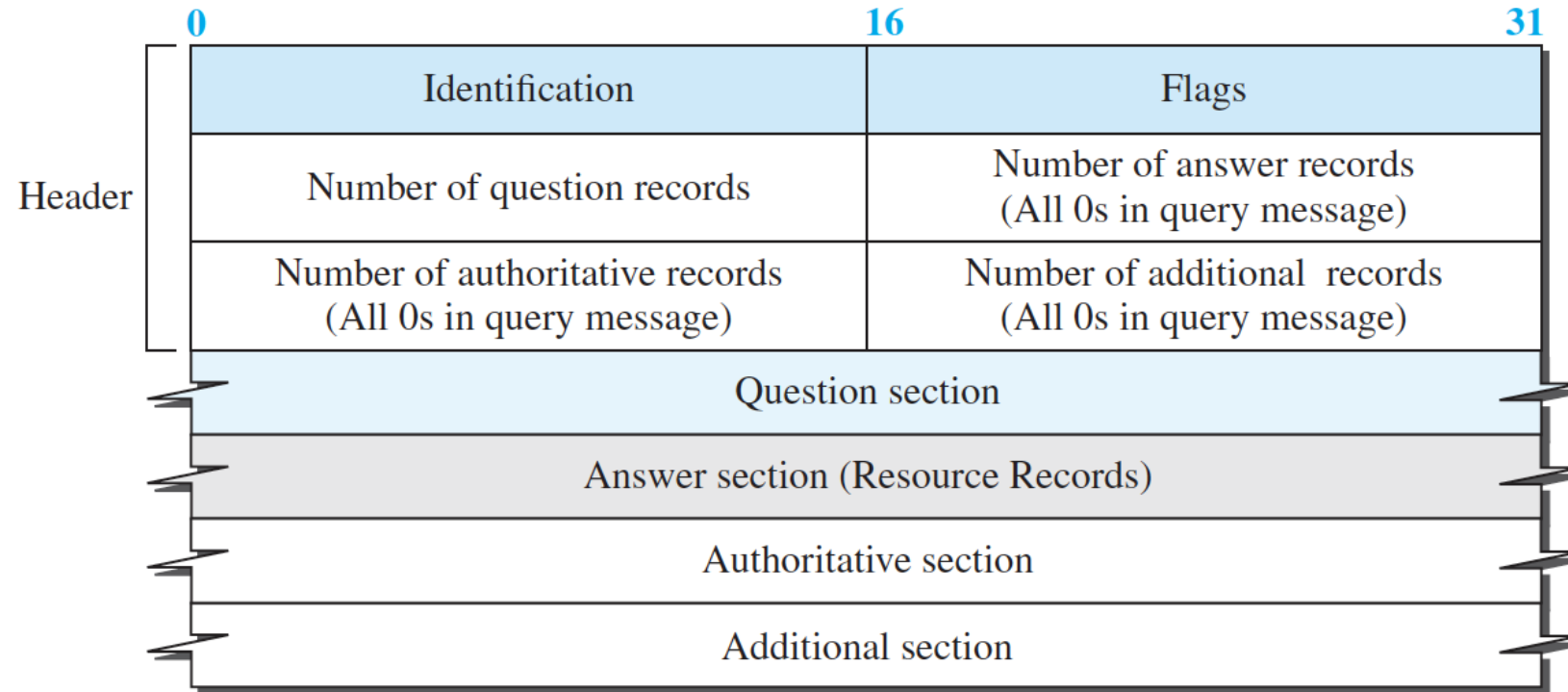
- Query
- Response



DNS MESSAGE SECTIONS

Section	Meaning/Use
Section 1	Message Header
Section 2	The DNS question being asked (aka Question Section)
Section 3	The Resource Record(s) which answer the question (aka Answer Section)
Section 4	The Resource Record(s) which point to the domain authority (aka Authority Section)
Section 5	The Resource Record(s) which may hold additional information (aka Additional Section)

DNS HEADER



Note:

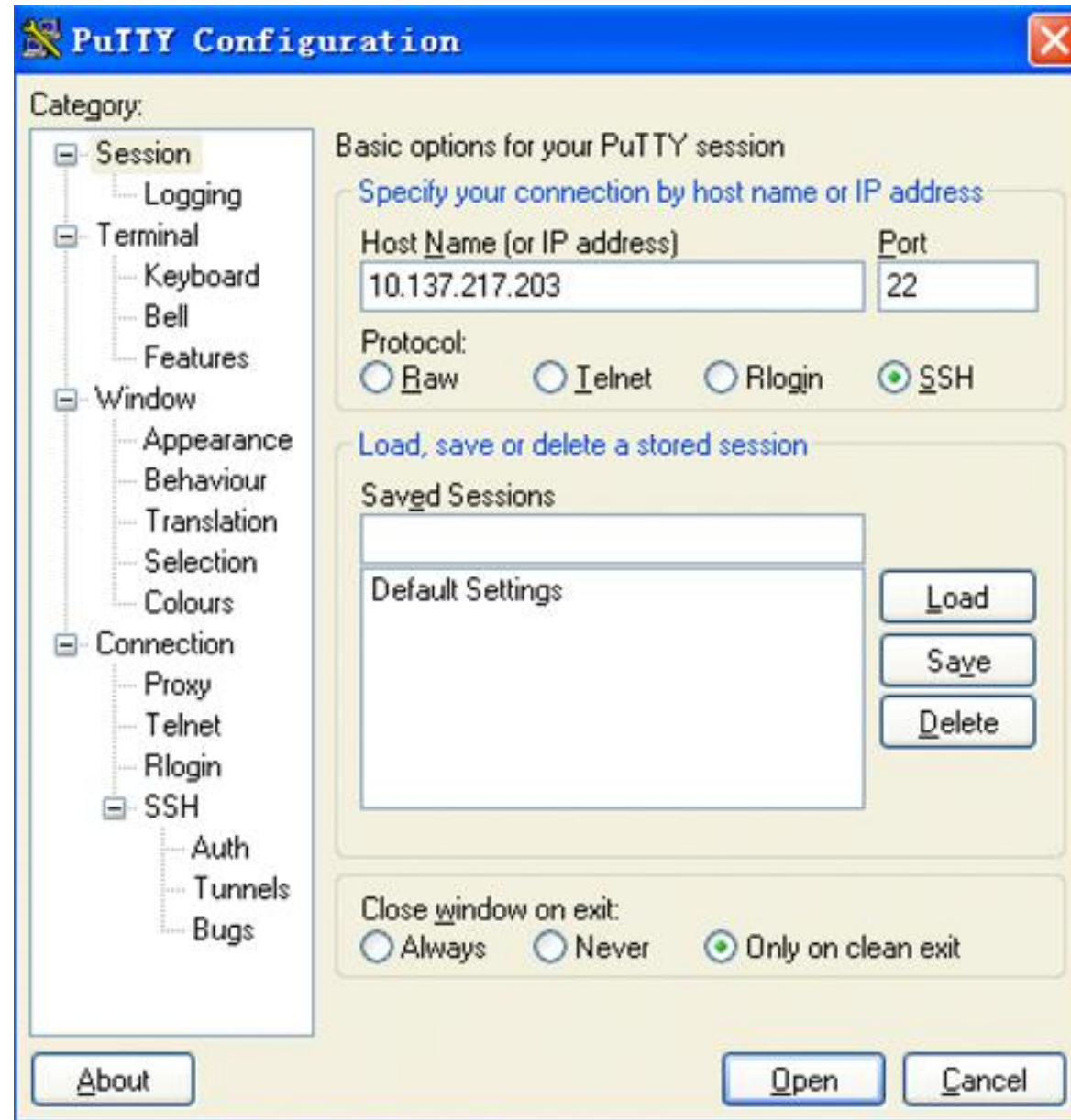
The query message contains only the question section.
The response message includes the question section,
the answer section, and possibly two other sections.



REMOTE LOGGING

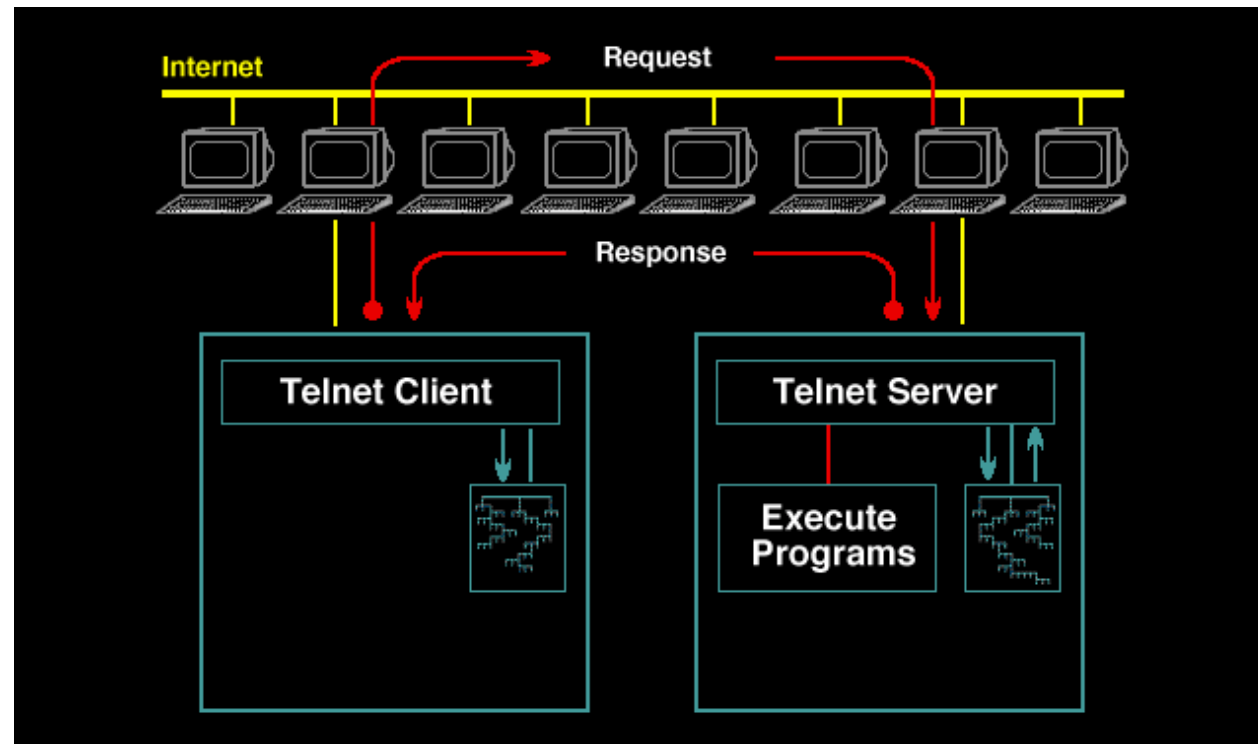
REMOTE LOGGING

- Users may want to run application programs at a remote site and create results that can be transferred to their local site.
- Not possible to write specific client/server program for each demand.
- General purpose program to log on to the remote computer.

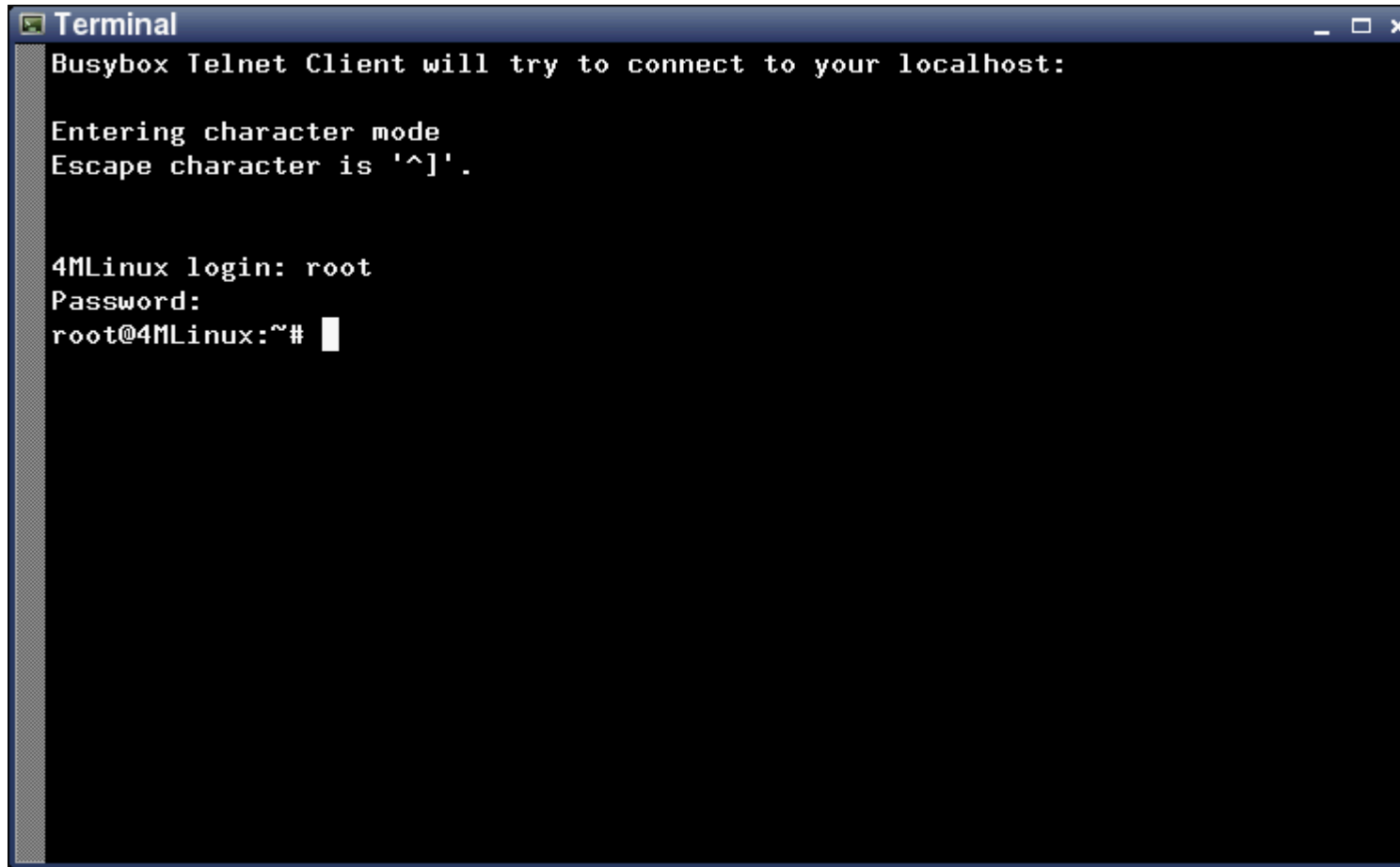


TELNET

- TErminaL NETwork
- Proposed by ISO



EXAMPLE TELNET CLIENT

A terminal window titled "Terminal" with standard window controls. The text inside shows a Busybox Telnet Client session. It starts with a message to connect to localhost, then enters character mode with an escape character. The user logs in as root on 4MLinux, enters a password, and the prompt changes to root@4MLinux:~#. The terminal has a dark background and white text.

```
Terminal
Busybox Telnet Client will try to connect to your localhost:

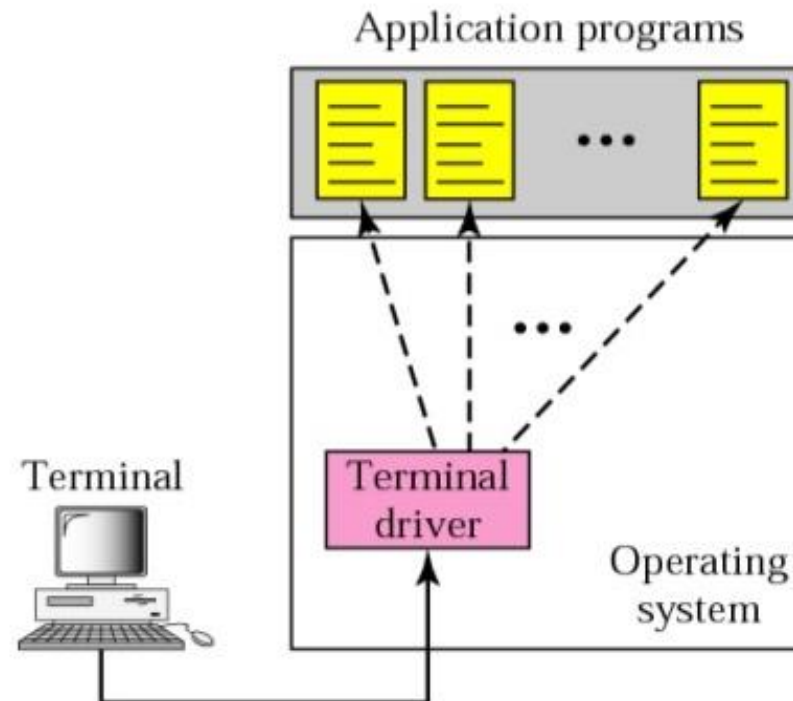
Entering character mode
Escape character is '^]'.

4MLinux login: root
Password:
root@4MLinux:~#
```

- It is vulnerable to hacking because it sends all data including the password in plaintext (not encrypted).
- Favoured protocol – Secure Shell (SSH)

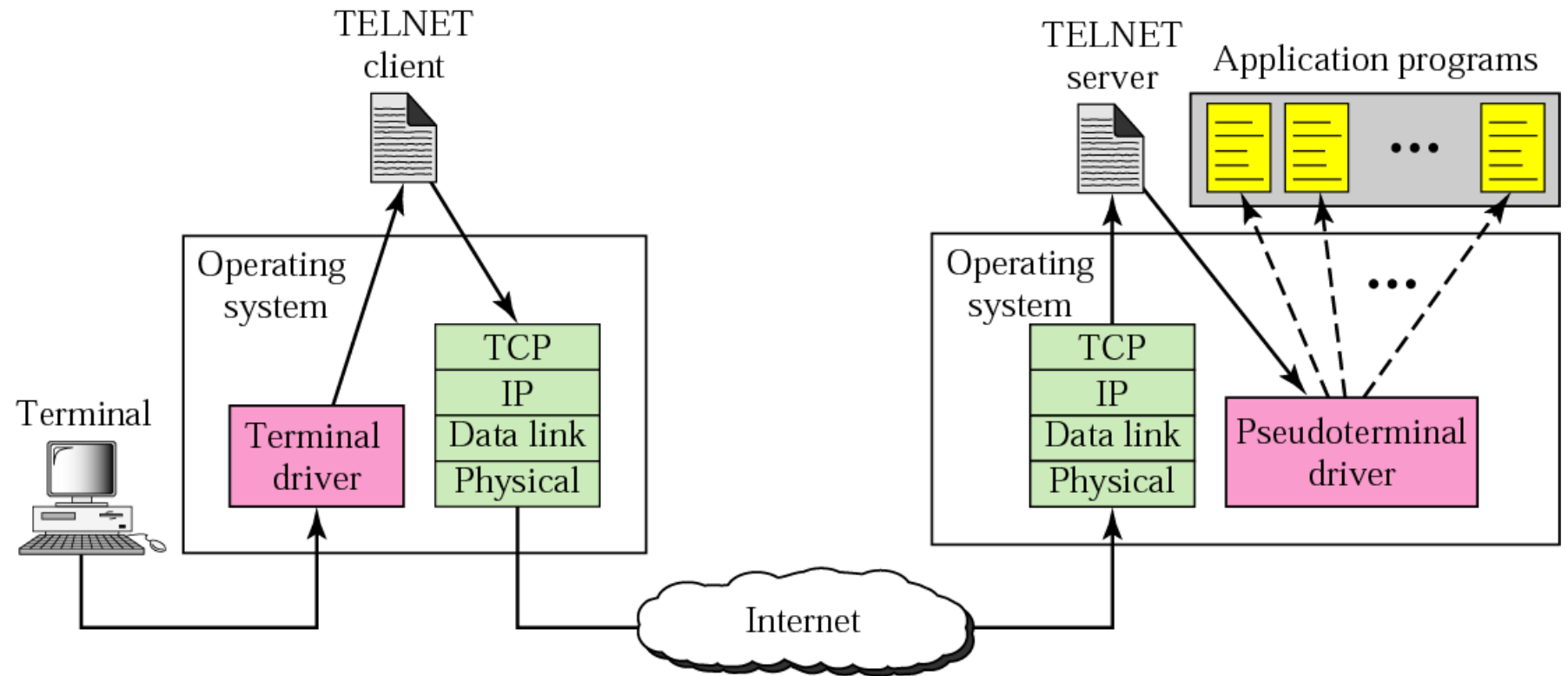
LOGGING

- Local log-in



LOGGING

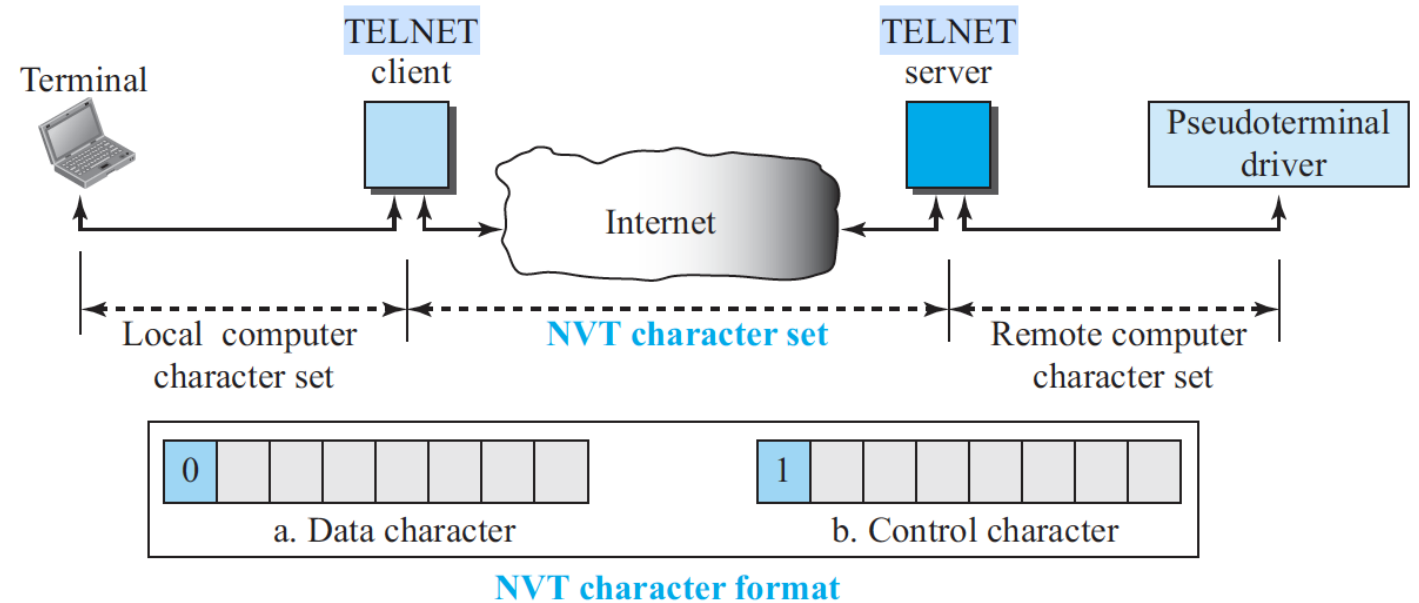
- Remote log-in



- Client transforms keystrokes to Network Virtual Terminal (NVT) characters

NVT

- To deal with heterogeneous systems
- Universal interface called NVT character set
- 2 sets of characters – for data and control
- 8-bits each
- Data – highest order bit 0, remaining ASCII
- Control – highest order bit 1



NVT CONTROL CHARACTERS

<i>Character</i>	<i>Decimal</i>	<i>Binary</i>	<i>Meaning</i>
EOF	236	11101100	End of file
EOR	239	11101111	End of record
SE	240	11110000	Suboption end
NOP	241	11110001	No operation
DM	242	11110010	Data mark
BRK	243	11110011	Break
IP	244	11110100	Interrupt process
AO	245	11110101	Abort output
AYT	246	11110110	Are you there?
EC	247	11110111	Erase character
EL	248	11111000	Erase line
GA	249	11111001	Go ahead
SB	250	11111010	Suboption begin
WILL	251	11111011	Agreement to enable option
WONT	252	11111100	Refusal to enable option
DO	253	11111101	Approval to option request
DONT	254	11111110	Denial of option request
IAC	255	11111111	Interpret (the next character) as control

TELNET INTERFACE COMMANDS

<i>Command</i>	<i>Meaning</i>	<i>Command</i>	<i>Meaning</i>
open	Connect to a remote computer	set	Set the operating parameters
close	Close the connection	status	Display the status information
display	Show the operating parameters	send	Send special characters
mode	Change to line or character mode	quit	Exit TELNET

56

FILE TRANSFER PROTOCOL

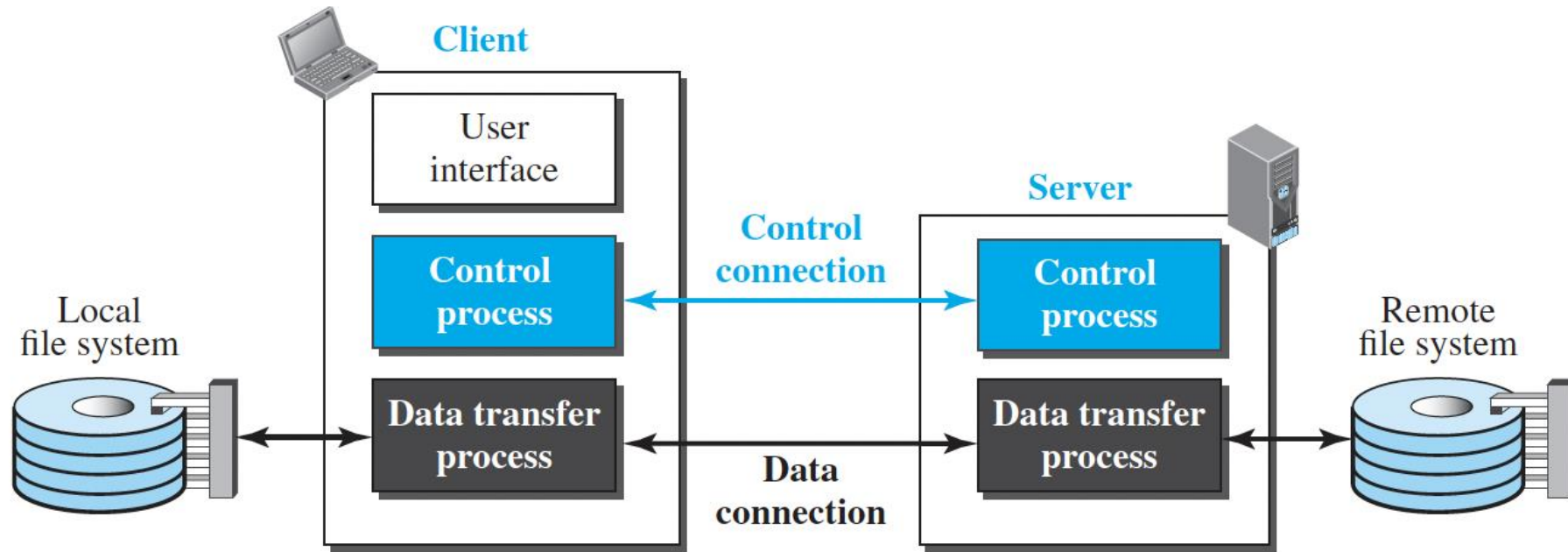
FILE TRANSFER PROTOCOL (FTP)

- Standard protocol provided by TCP/IP for copying a file from one host to another.
- Why FTP?
 - Two systems may use different file name conventions.
 - Two systems may have different ways to represent data.
 - Two systems may have different directory structures.
 - Can be done using HTTP too, but FTP is efficient.

BASIC MODEL OF FTP

- The client has three components:
 - the user interface
 - the client control process
 - the client data transfer process
- The server has two components:
 - the server control process
 - the server data transfer process

FTP - LIFETIMES



- Control connection remains connected during the entire interactive FTP session.
- Data connection is opened and then closed for each file transfer activity.

CONTROL CONNECTION

- Uses the NVT ASCII character set as used by TELNET.
- Communication is achieved through commands and responses.
- We will be sending only one command (or response) at a time.
- Each line is terminated with a two-character (carriage return and line feed) end-of-line token.
- Commands, which are sent from the FTP client control process, are in the form of ASCII uppercase
 - It may or may not be followed by an argument.

FTP COMMANDS

<i>Command</i>	<i>Argument(s)</i>	<i>Description</i>
ABOR		Abort the previous command
CDUP		Change to parent directory
CWD	Directory name	Change to another directory
DELE	File name	Delete a file
LIST	Directory name	List subdirectories or files
MKD	Directory name	Create a new directory
PASS	User password	Password
PASV		Server chooses a port
PORT	Port identifier	Client chooses a port
PWD		Display name of current directory
QUIT		Log out of the system
RETR	File name(s)	Retrieve files; files are transferred from server to client
RMD	Directory name	Delete a directory
RNFR	File name (old)	Identify a file to be renamed
RNTO	File name (new)	Rename the file
STOR	File name(s)	Store files; file(s) are transferred from client to server
STRU	F , R , or P	Define data organization (F : file, R : record, or P : page)
TYPE	A , E , I	Default file type (A : ASCII, E : EBCDIC, I : image)
USER	User ID	User information
MODE	S , B , or C	Define transmission mode (S : stream, B : block, or C : compressed)

FTP RESPONSES

- Every FTP command generates at least one response.
- A response has two parts: 1) a three-digit number 2) text.
- The numeric part defines the code
 - The first digit defines the status of the command.
 - The second digit defines the area in which the status applies.
 - The third digit provides additional information.
- The text part defines needed parameters or further explanations.

FTP RESPONSES

Digit	Category	Meaning
1xx	Positive Preliminary reply	Command accepted, but another reply will follow (e.g. opening a connection).
2xx	Positive Completion reply	Command completed successfully.
3xx	Positive Intermediate reply	Command accepted, need more information (e.g. waiting for password).
4xx	Transient Negative reply	Command failed, but the error is temporary (try again later).
5xx	Permanent Negative reply	Command failed, and retrying won't help (invalid command, wrong user, etc.).

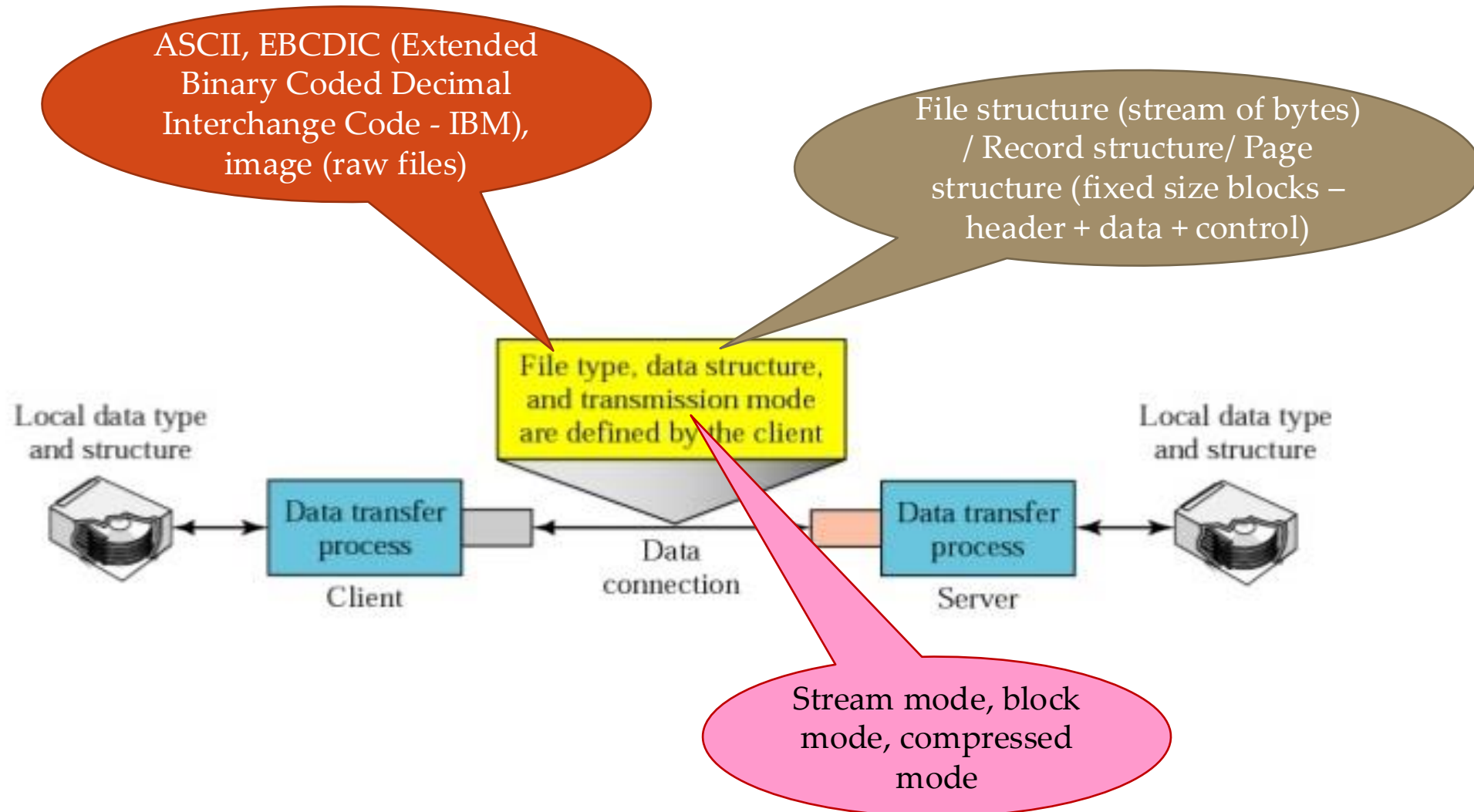
FTP RESPONSES

<i>Code</i>	<i>Description</i>	<i>Code</i>	<i>Description</i>
125	Data connection open	250	Request file action OK
150	File status OK	331	User name OK; password is needed
200	Command OK	425	Cannot open data connection
220	Service ready	450	File action not taken; file not available
221	Service closing	452	Action aborted; insufficient storage
225	Data connection open	500	Syntax error; unrecognized command
226	Closing data connection	501	Syntax error in parameters or arguments
230	User login OK	530	User not logged in

DATA CONNECTION

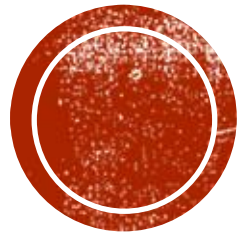
- The data connection uses the well-known port 20 at the server site.
- However, the creation of a data connection is different from the control connection.
 1. The client, not the server, issues a passive open using an ephemeral port.
 - This must be done by the client because it is the client that issues the commands for transferring files.
 2. Using the PORT command the client sends this port number to the server.
 3. The server receives the port number and issues an active open using the well-known port 20 and the received ephemeral port number.

COMMUNICATION OVER DATA CONNECTION



FILE TRANSFER

- File transfer occurs over the data connection under the control of the commands sent over the control connection.
- File transfer in FTP means one of three things:
 - retrieving/downloading a file (server to client) (RETR command)
 - storing a file (client to server) (STOR command)
 - directory listing (server to client) (LIST command)



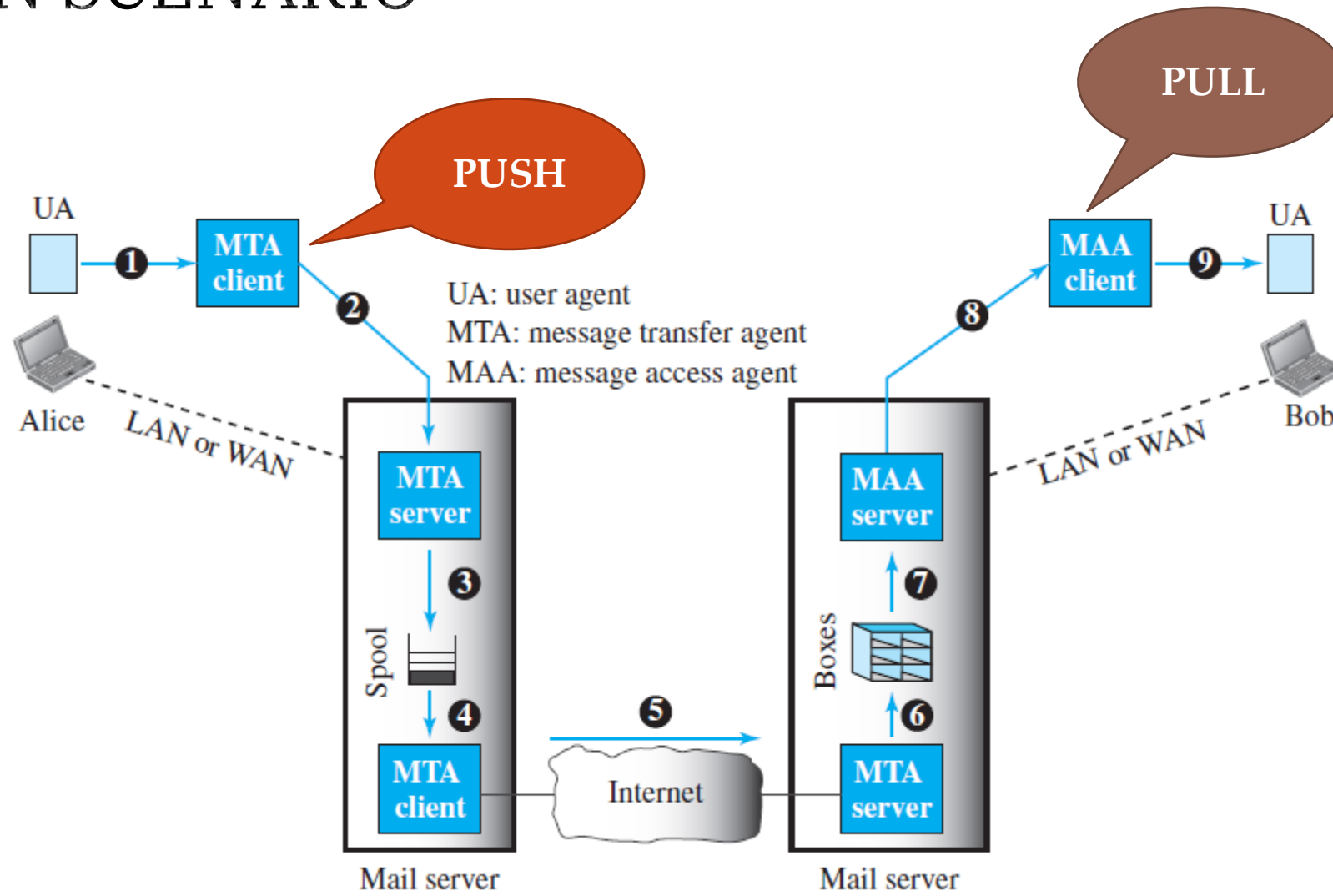
ELECTRONIC MAIL



3 MAJOR COMPONENTS

- User Agent
- Message Transfer Agent
- Message Access Agent

COMMON SCENARIO



MAILBOXES

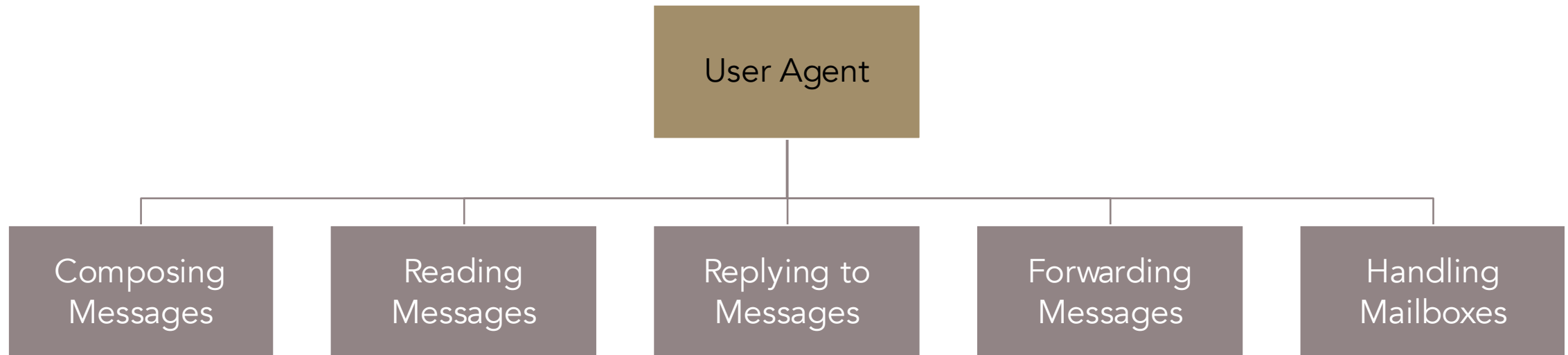


Inbox: Incoming, received mail.

Outbox: Outgoing, waiting-to-be-sent mail.

USER AGENT

- Software package that provides services to help users send and receive messages easier.



<http://whatsmyuseragent.org/>

✓ WHAT'S MY USER AGENT?

Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/64.0.3282.140 Safari/537.36 Edge/17.17134

My IP Address: 171.61.229.2

Mail

Calendar

Contacts

Deleted Items

Drafts

Inbox (1)

Junk Email

Sent Items

[Click to view all folders](#)

[Manage Folders...](#)

New Message | Move | Delete | Junk | |

	From	Subject	Received	Size
			4/19/2018 2:21 PM	80 KB
	test01		4/19/2018 11:02 AM	57 KB
	test01	Accepted: test	3/22/2018 9:18 PM	24 KB
	test01		3/18/2018 4:09 PM	417 KB
	Microsoft Outloo...		3/13/2018 3:19 PM	84 KB
	Microsoft Outloo...		3/13/2018 3:14 PM	84 KB
	test01	test	3/7/2018 5:11 PM	21 KB
	test01	Test	3/7/2018 4:48 PM	17 KB
	test01	Test	3/7/2018 4:30 PM	122 KB
	test01	You're invited to share this cal...	3/6/2018 7:28 PM	34 KB
	test01	I've removed permissions on your...	3/6/2018 7:18 PM	19 KB
	test01	You're invited to share this cal...	3/6/2018 7:16 PM	35 KB
	test01	You're invited to share this cal...	3/6/2018 7:14 PM	34 KB
	test01	You're invited to share this cal...	3/6/2018 6:58 PM	34 KB
	test01	I've removed permissions on your...	3/6/2018 6:03 PM	19 KB
	test01	You're invited to share this cal...	3/6/2018 6:03 PM	35 KB
	test01	I've removed permissions on your...	3/6/2018 6:02 PM	19 KB
	test 02	ttt	3/6/2018 4:39 PM	32 KB
	test01	Accepted: test	3/6/2018 4:33 PM	28 KB
		Accepted: test	3/6/2018 4:23 PM	25 KB

Move Delete

Page: 1 2

Connected to Microsoft Exchange

USER AGENT TYPES

- Command-driven
- GUI-based

```
xterm
PINE 4.44  MAIN MENU                               Folder: INBOX  84 Messages

?  HELP                -  Get help using Pine
C  COMPOSE MESSAGE     -  Compose and send a message
I  MESSAGE INDEX       -  View messages in current folder
L  FOLDER LIST         -  Select a folder to view
A  ADDRESS BOOK        -  Update address book
S  SETUP               -  Configure Pine Options
Q  QUIT                -  Leave the Pine program

Copyright 1989-2002.  PINE is a trademark

? Help      P PrevCmd
0 OTHER CMDS [ListFldrs] N NextCmd
```

```
File Edit Settings Help
PINE 4.33  COMPOSE MESSAGE                               Folder: INBOX  34 Messages

To      : someone@someisp.net
Cc      :
Attachmnt:
Subject : How are you?
----- Message Text -----

Hello,

I was just thinking about you today and wanted to send you
a quick email to see how you are doing.

I hope all is going well.

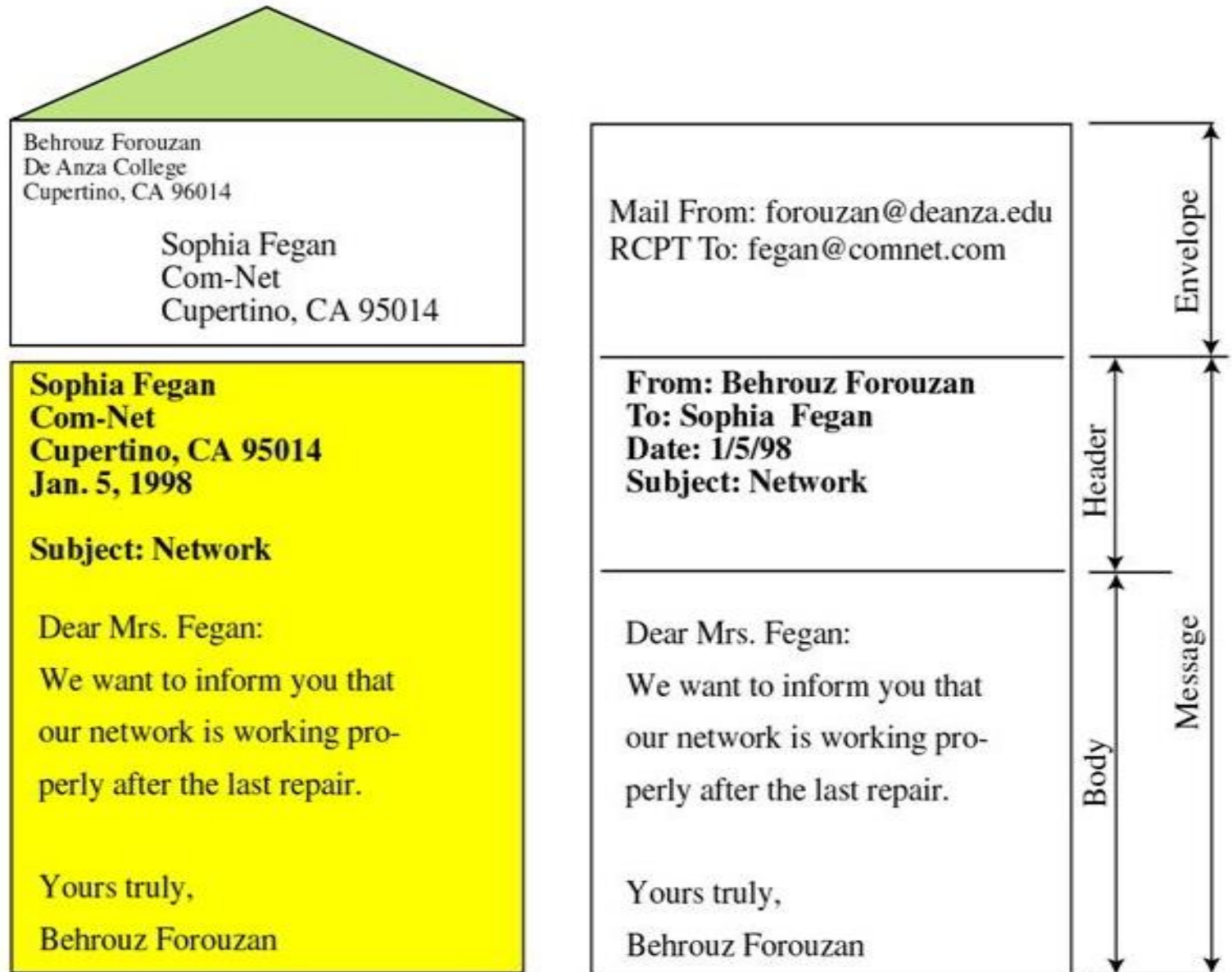
Take care and I hope to hear from you soon.

Best regards,
Old College Friend

^G Get Help  ^X Send      ^R Read File  ^V Prev Pg   ^K Cut Text   ^O Postpone
^C Cancel    ^J Justify   ^W Where is  ^V Next Pg   ^U UnCut Text ^T To Spell
```


SENDING MAILS

- Envelope
- Message



RECEIVING MAILS

- The user agent is triggered by the user (or a timer).
- If a user has mail, the UA informs the user with a notice.
- If the user is ready to read the mail, a list is displayed in which each line contains a summary of the information about a particular message in the mailbox.

EMAIL ADDRESSES - PARTS

- Local Part
- Domain Name

It usually identifies the **type of organisation** or **country** where the server is registered.



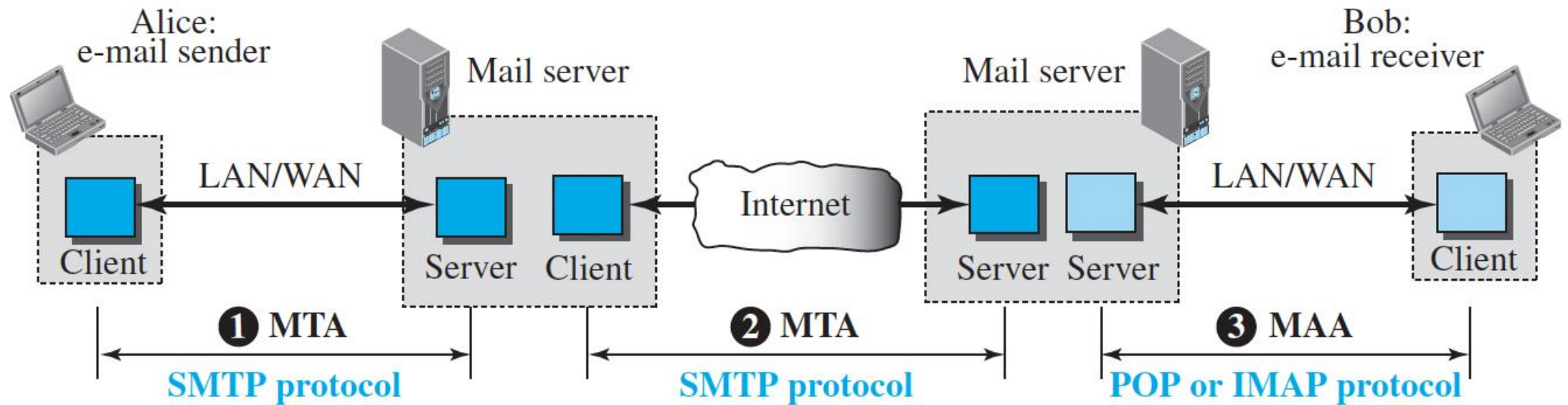
- This is the **unique identifier** chosen by the user (sometimes given by the service provider).
- It tells the mail server **which specific mailbox** to deliver the message to.
- This identifies the **mail server** (the organisation or service provider) where the user's mailbox is hosted.

MAILING LIST

alias

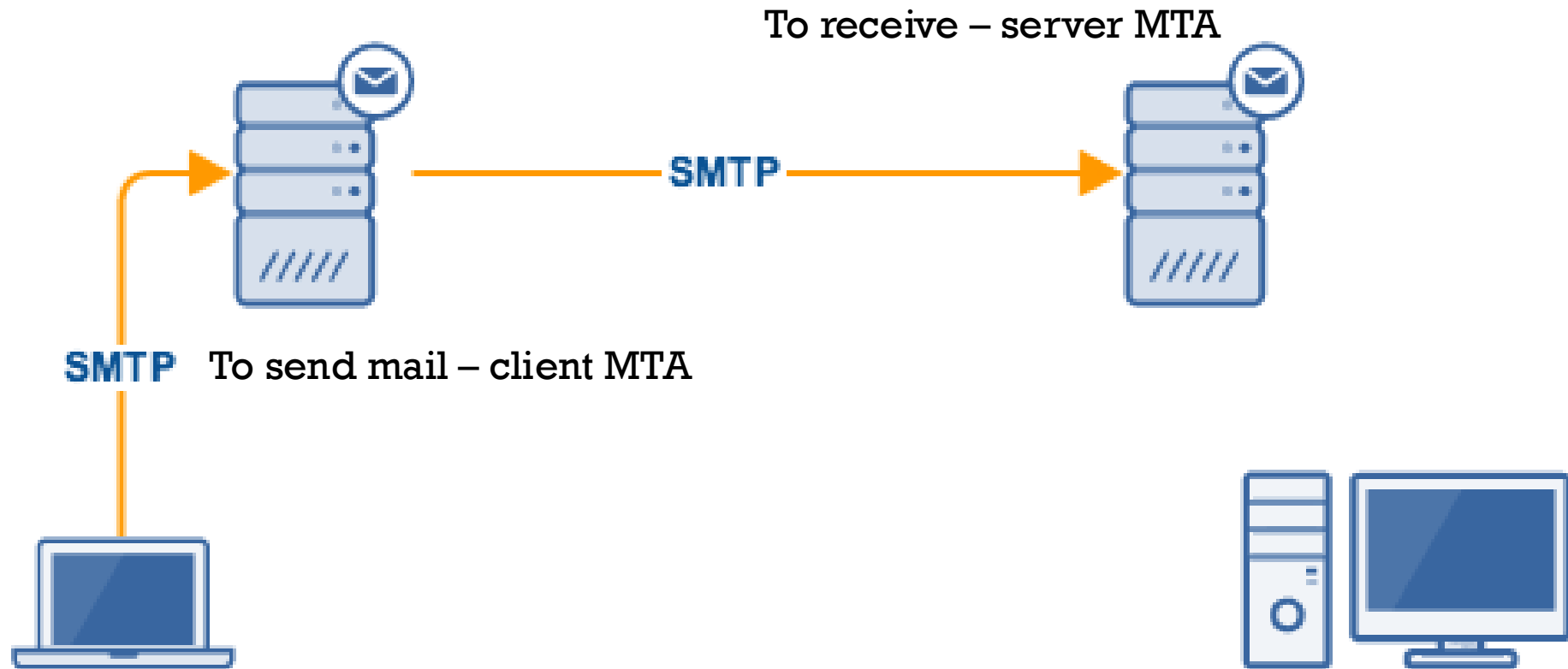
	Mailing List Name (Alias)	Email Address	Number of Members
☐	mailinglist00	mailinglist00@sftaiko.com	1 members
☐	mailinglist1	mailinglist1@sftaiko.com	63 members
☐	mailinglist2	mailinglist2@sftaiko.com	63 members
☐	mailinglist3	mailinglist3@sftaiko.com	55 members
☐	mailinglist4	mailinglist4@sftaiko.com	71 members
☐	mailinglist5	mailinglist5@sftaiko.com	58 members
☐	mailinglist6	mailinglist6@sftaiko.com	69 members
☐	mailinglist7	mailinglist7@sftaiko.com	76 members
☐	mailinglist8	mailinglist8@sftaiko.com	85 members
☐	mailinglistall	mailinglistall@sftaiko.com	41 members

PROTOCOLS USED



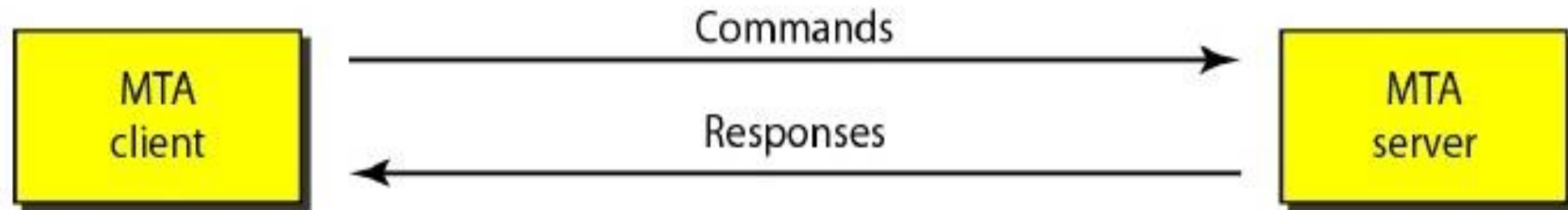
MESSAGE TRANSFER AGENT: SMTP

- Simple Mail Transfer Protocol: Does the actual mail transfer – **push protocol**



COMMANDS AND RESPONSES

- To transfer messages between an MTA client and MTA server
- Each command or response is terminated by a two-character end-of-line token
 - carriage return and line feed



COMMANDS

- Sent from client to server
- Keyword: argument(s) – can take zero or more arguments
- 14 commands: first 5 – mandatory; next 3 – often used and recommended; last 6 – seldom used

<i>Keyword</i>	<i>Argument(s)</i>	<i>Keyword</i>	<i>Argument(s)</i>
HELO	Sender's host name	NOOP	
MAIL FROM	Sender of the message	TURN	
RCPT TO	Intended recipient	EXPN	Mailing list
DATA	Body of the mail	HELP	Command name
QUIT		SEND FROM	Intended recipient
RSET		SMOL FROM	Intended recipient
VRFY	Name of recipient	SMAL FROM	Intended recipient

RESPONSES

- Sent from server to client
- 3-digit code that may be followed by additional textual information

<i>Code</i>	<i>Description</i>
Positive Completion Reply	
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
Positive Intermediate Reply	
354	Start mail input
Transient Negative Completion Reply	
421	Service not available
450	Mailbox not available
451	Command aborted: local error
452	Command aborted: insufficient storage

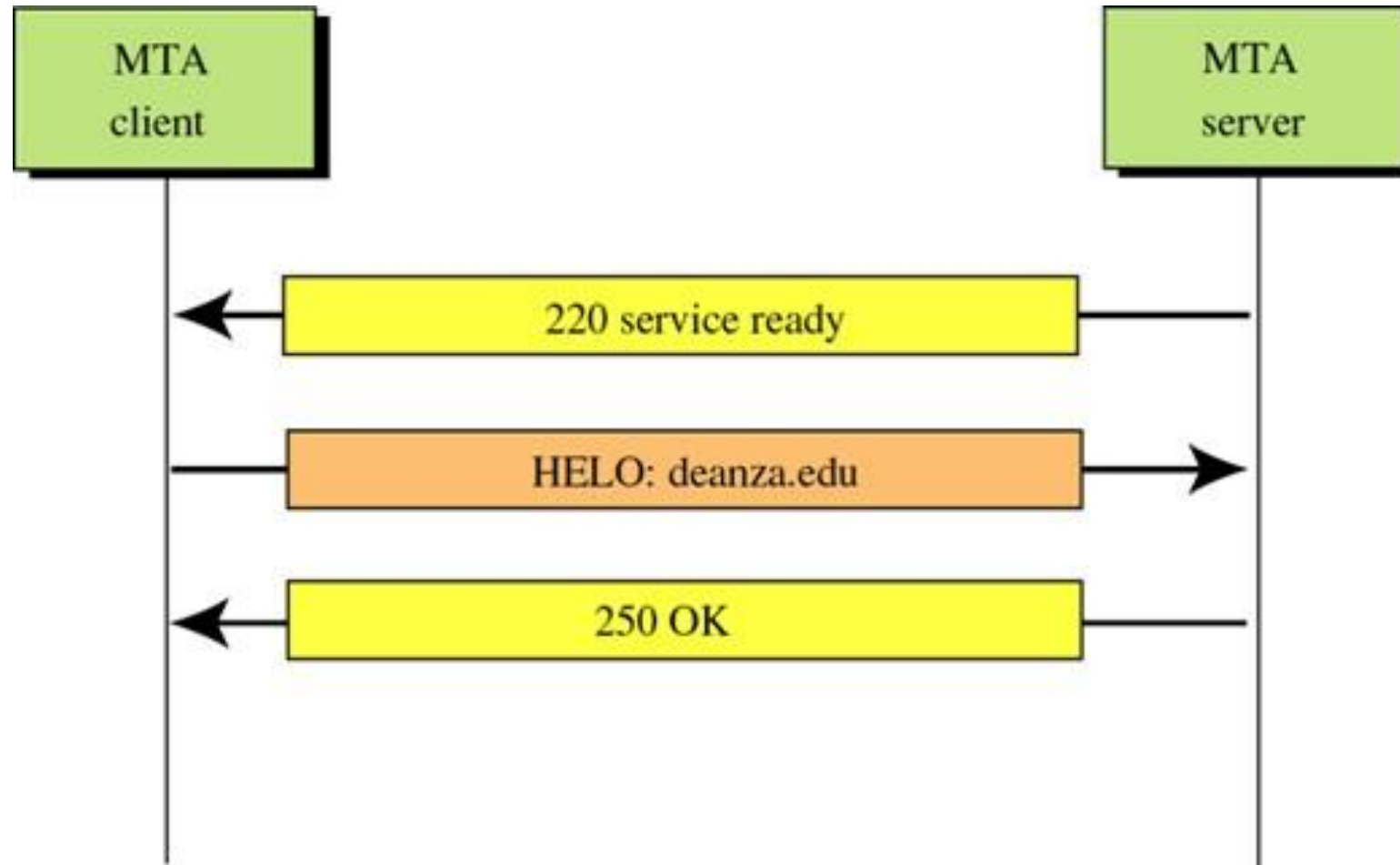
RESPONSES

<i>Code</i>	<i>Description</i>
Permanent Negative Completion Reply	
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

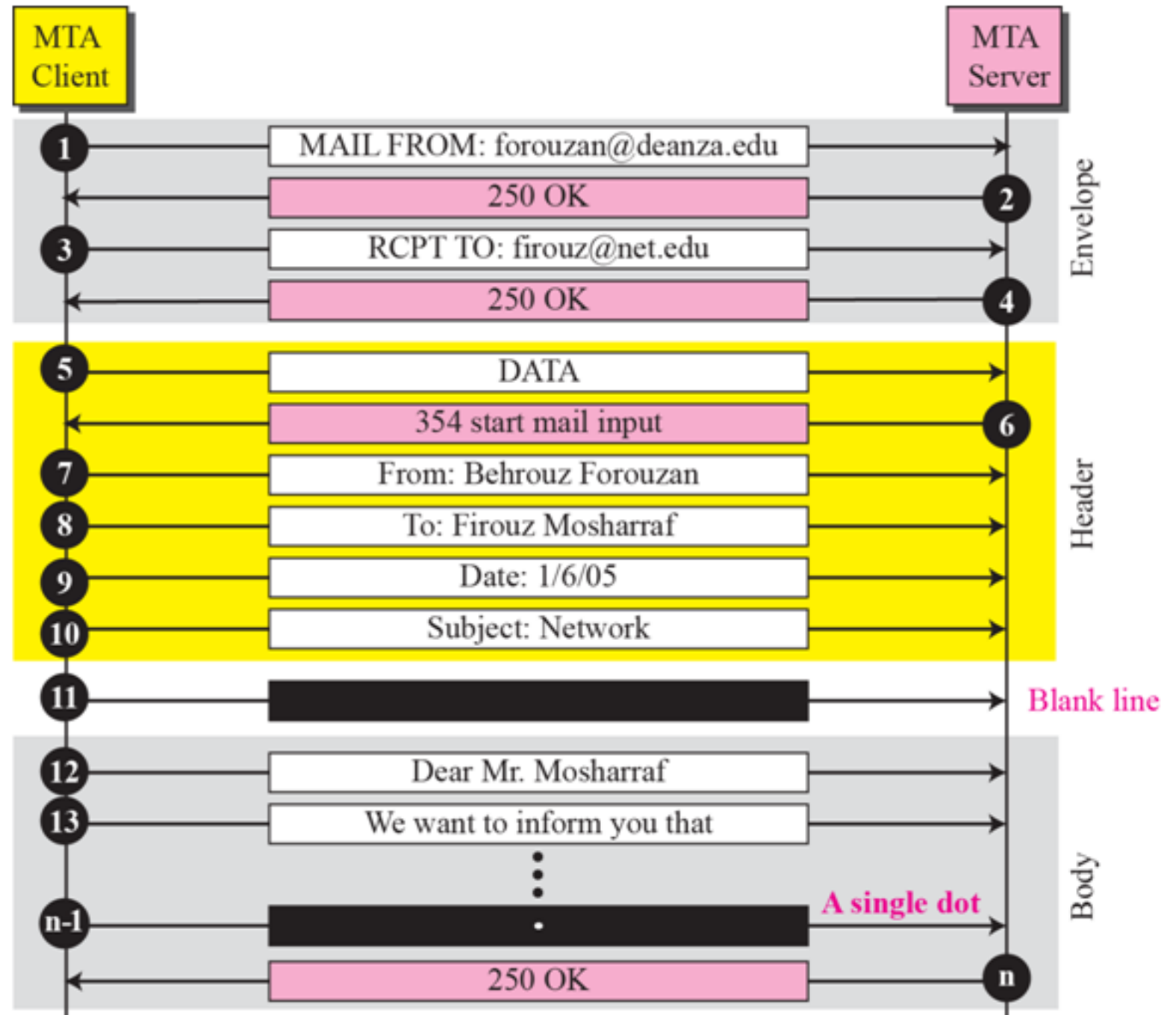
MAIL TRANSFER – THREE PHASES

- Connection Establishment
- Mail Transfer
- Connection Termination

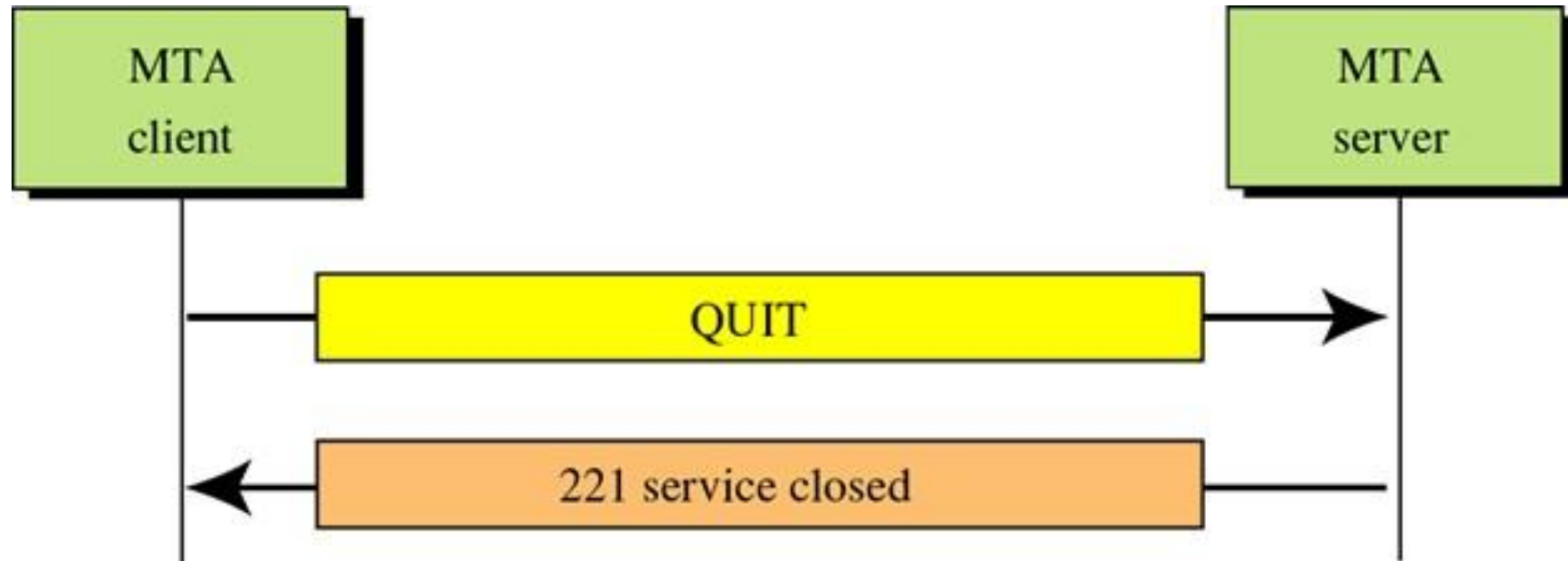
CONNECTION ESTABLISHMENT



MAIL TRANSFER

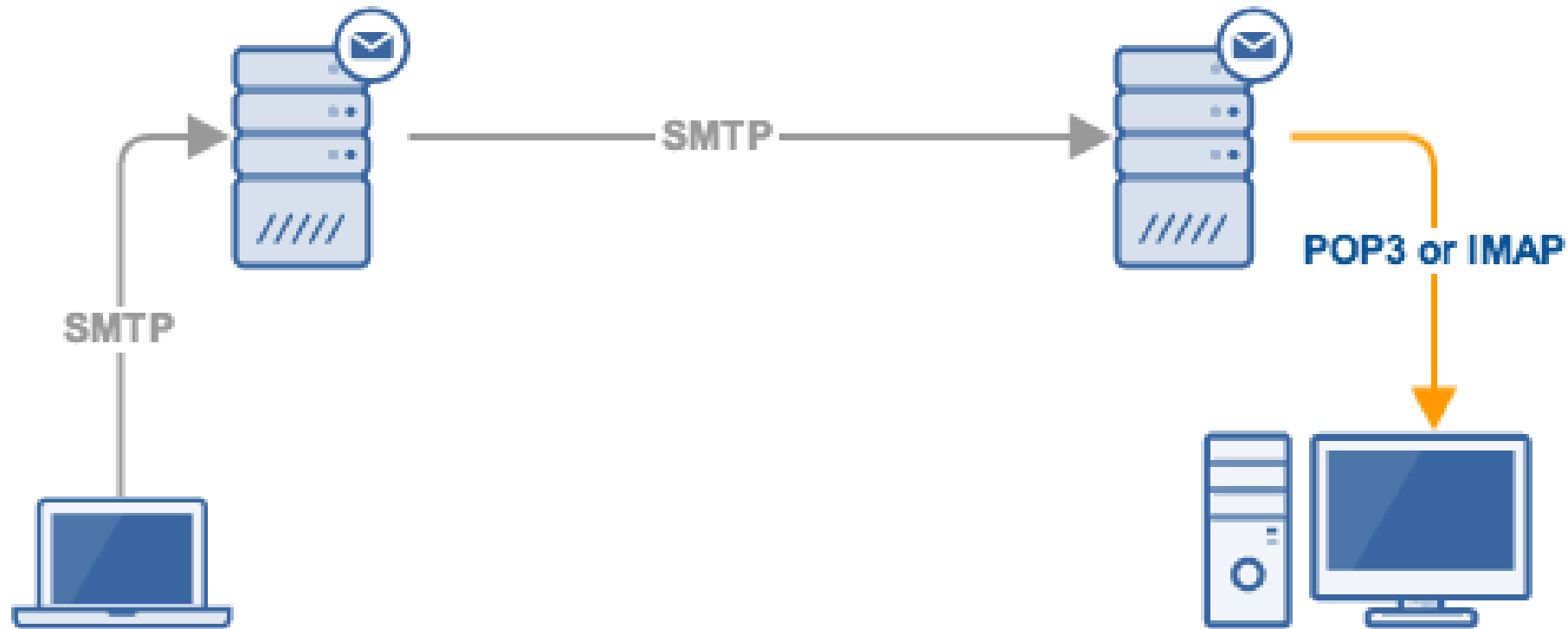


CONNECTION TERMINATION



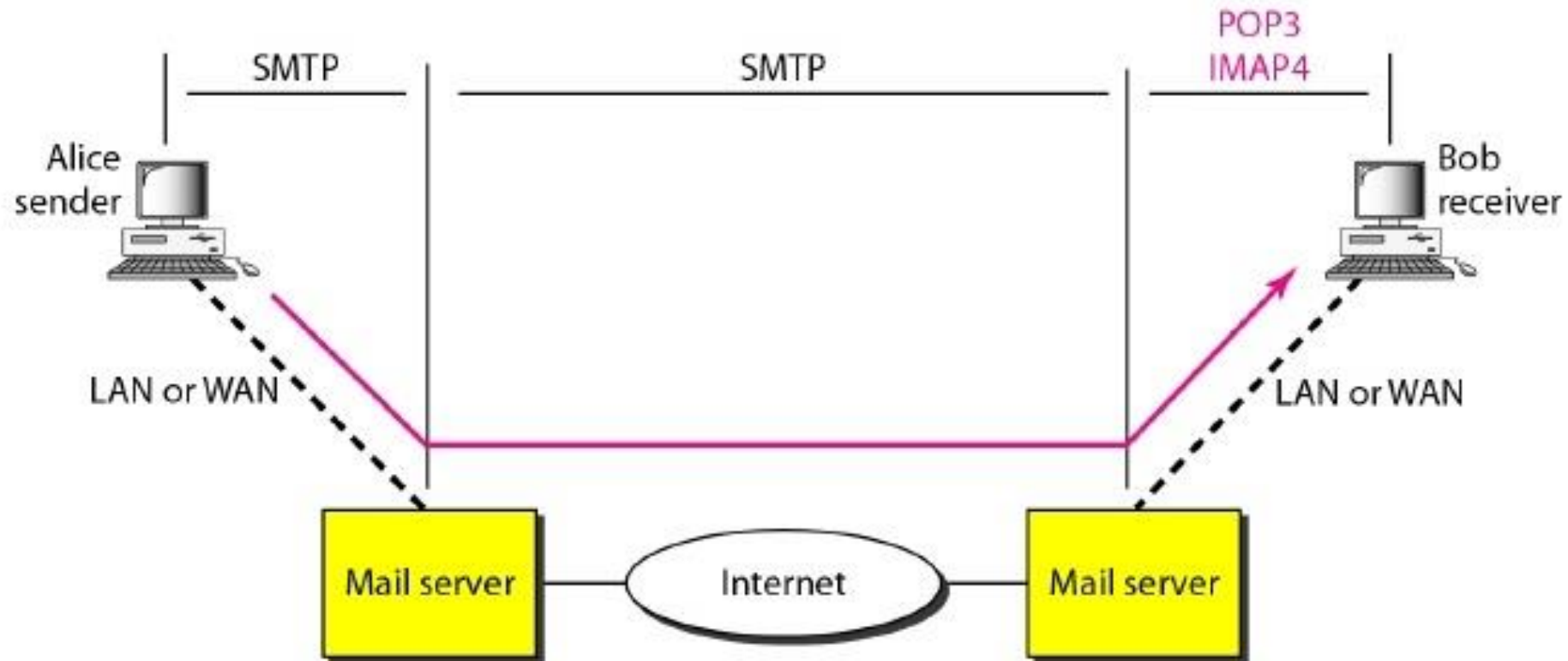
MESSAGE ACCESS AGENT - POP & IMAP

- Post Office Protocol, version 3 (POP3), Internet Mail Access Protocol, version 4 (IMAP4) - **pull protocol**

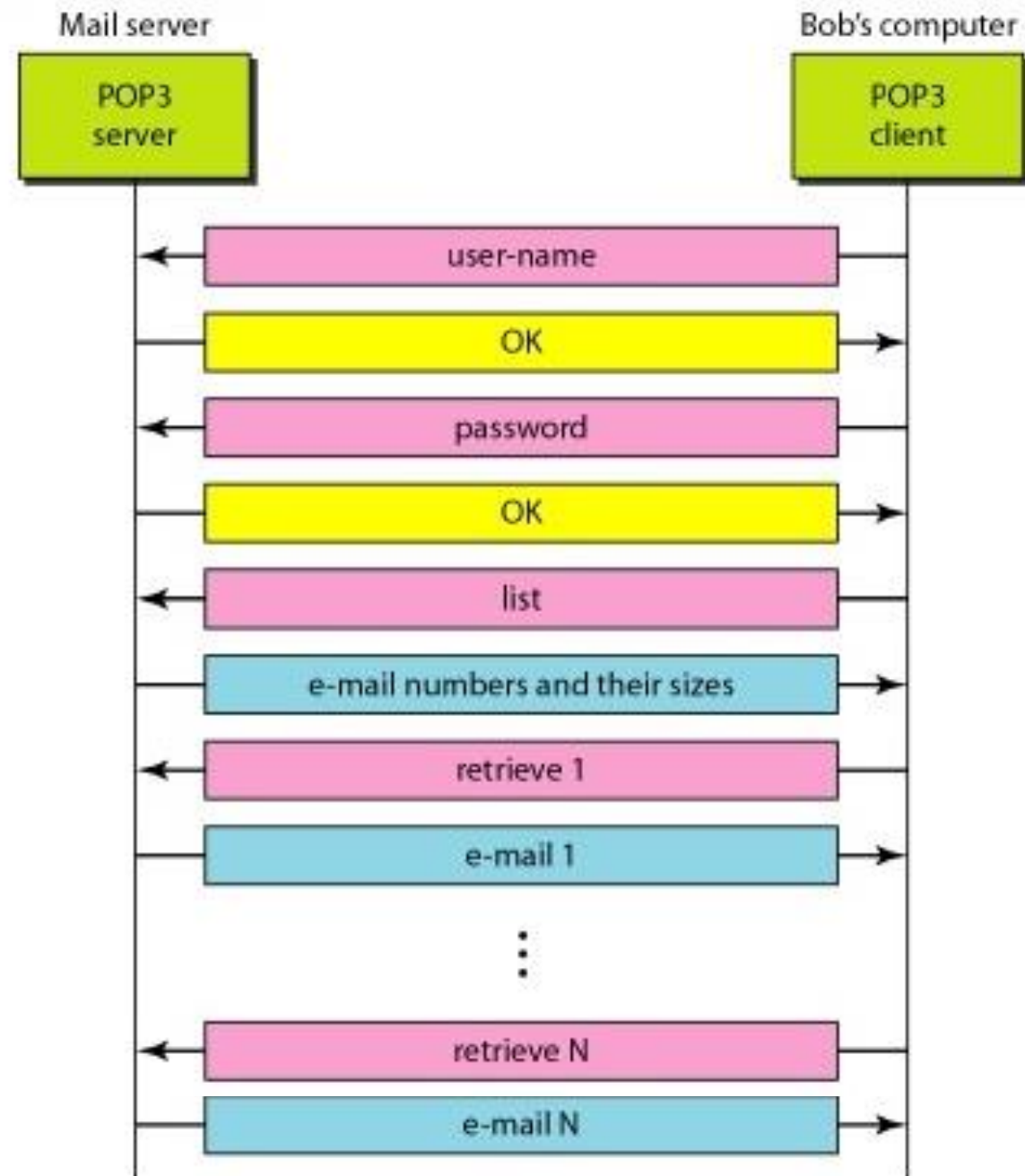


POP3

- Client POP3 software is installed on the recipient computer; server POP3 on mail server



COMMANDS AND RESPONSES IN POP3



POP3 - 2 MODES

- Delete mode
 - When the email client downloads a message from the server, it **removes (deletes) the copy** from the server and it exists **only on the local device**.
 - If you check your mail from another device, you won't see the message again (since it was deleted from the server).
- Keep mode
 - When the email client downloads a message, it **keeps a copy on the server**.
 - Lets you access emails from multiple devices but not fully synchronised

IMAP4

- Effective than POP3, has many features
 - User can check email header prior to downloading
 - Search email contents with specific string prior to downloading
 - Can partially download email
 - Create, delete, rename mailboxes on the mail server
 - Create hierarchy of mail boxes in a folder for email storage

POP3 VS. IMAP4

POP3 DOWNLOADS AN EMAIL FROM THE SERVER TO ONE COMPUTER AND THEN DELETES IT FROM THE SERVER



POP3, YOU CAN'T ORGANISE EMAILS IN THE MAILBOX OF THE MAIL SERVER



IMAP STORES THE EMAIL ON THE SERVER AND SYNCS IT ACROSS SEVERAL DEVICES TO ACCESS OVER MULTIPLE CHANNELS



IMAP, YOU CAN ORGANISE EMAILS IN THE MAILBOX OF THE MAIL SERVER



YOU CAN DOWNLOAD ALL THE EMAILS AT ONCE WITH POP3



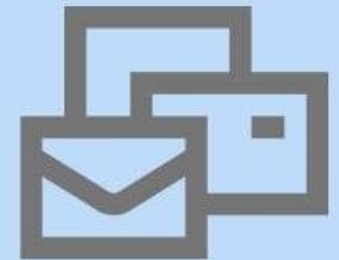
WHILE WITH IMAP, YOU SEE THE MESSAGE HEADER BEFORE YOU DOWNLOAD THE EMAIL



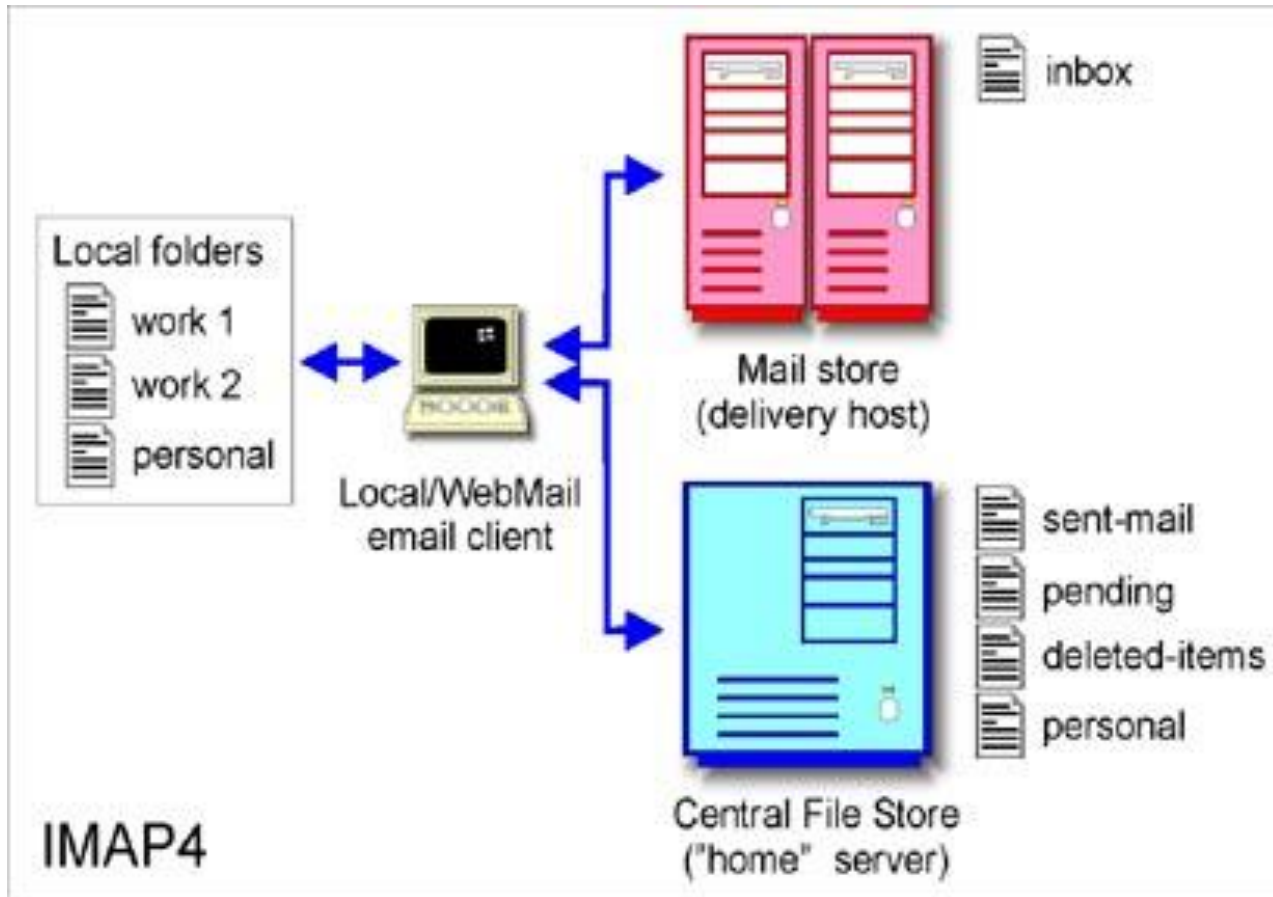
THERE ARE TWO MODES OF POP3 - DELETE MODE AND KEEP MODE



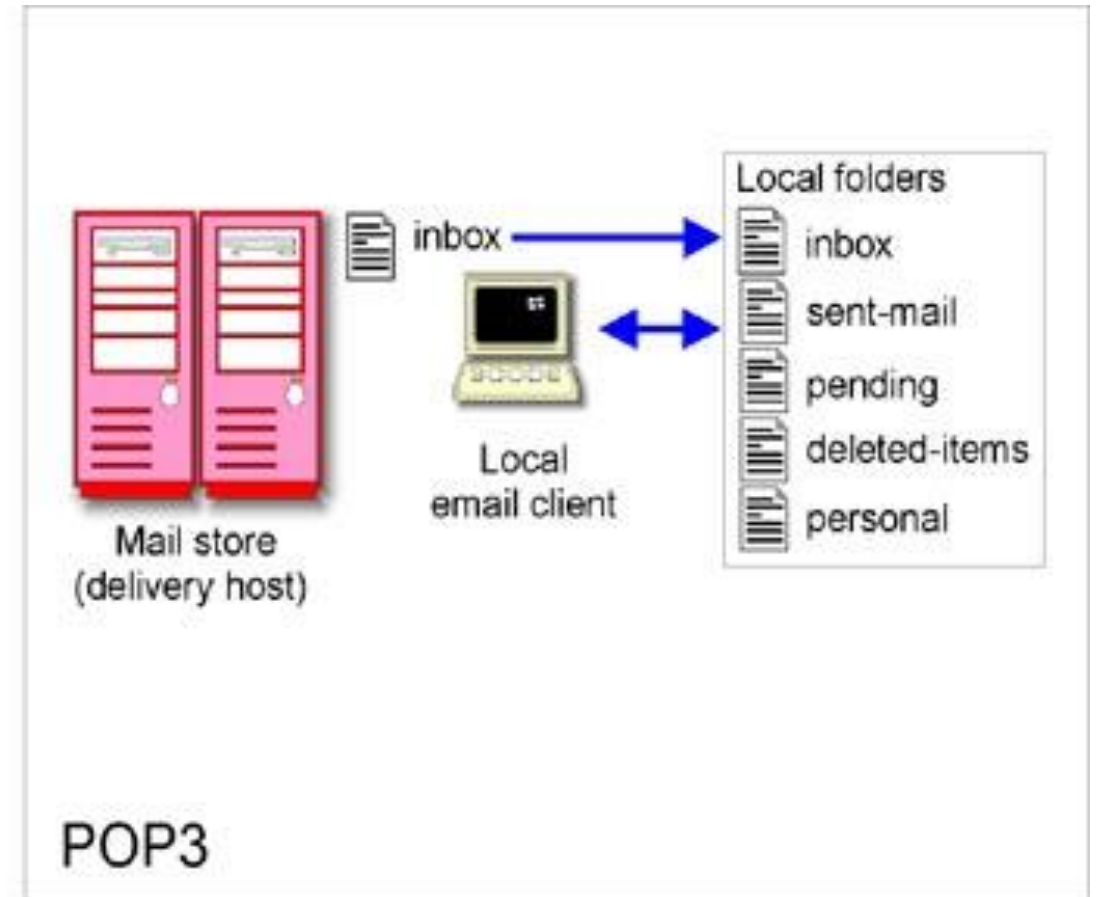
WITH IMAP, THERE ARE SEVERAL COPIES OF THE EMAIL ON THE MAIL SERVER TO RETRIEVE IT EASILY



POP3 VS. IMAP4



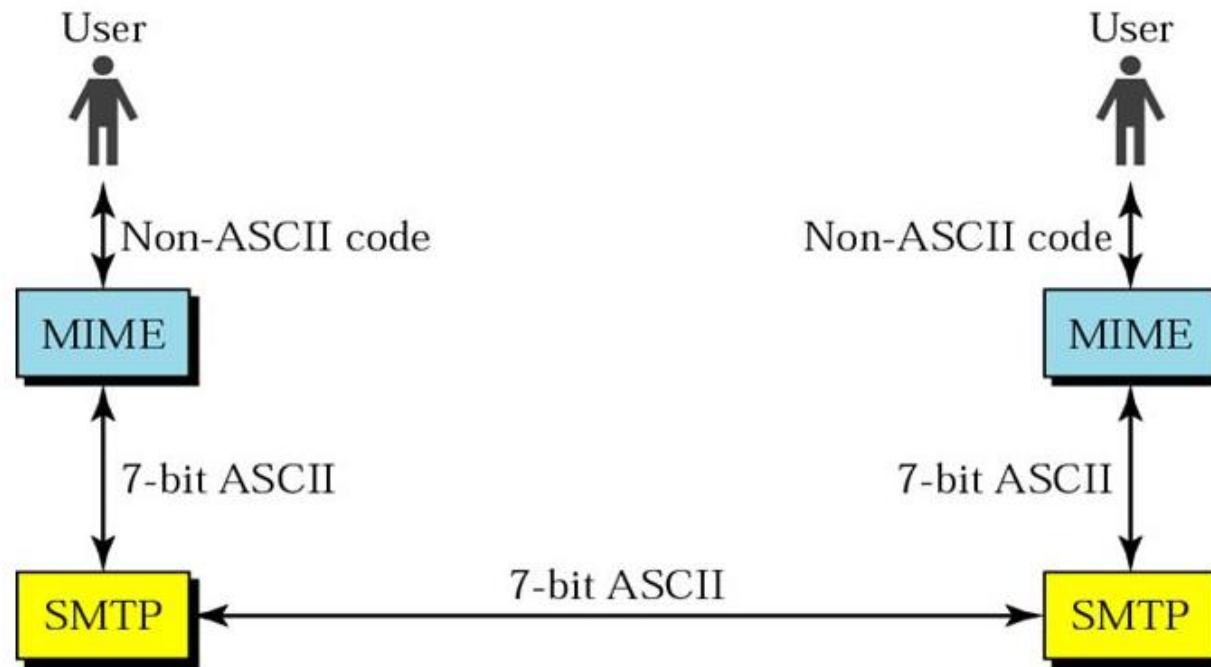
IMAP4 = Mail stays on the server, synced across devices.



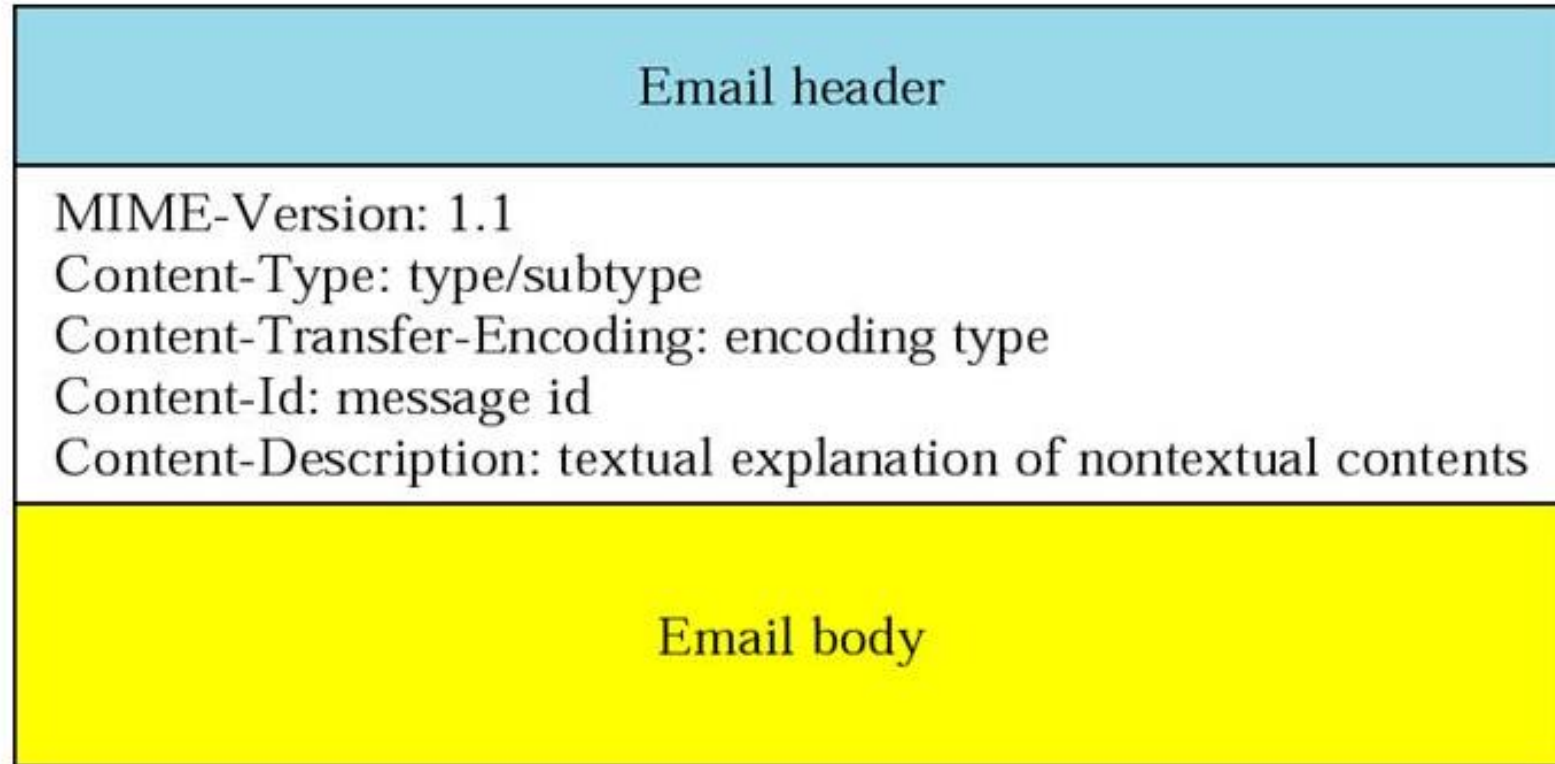
POP3 = Mail downloaded and stored locally, usually removed from server.

MIME

- Multipurpose Internet Mail Extensions (MIME)
- Supplementary protocol that allows non-ASCII data to be sent through email



MIME HEADER



MIME header

CONTENT-TYPE

- Type of data in the message
- Content-Type: <type / subtype; parameters>

Type	Subtype	Description
Text	Plain	Unformatted
	HTML	HTML format
Multipart	Mixed	Body contains ordered parts of different data types
	Parallel	Same as above, but no order
	Digest	Similar to mixed subtypes, but the default is message/RFC822
	Alternative	Parts are different versions of the same message
Message	RFC822	Body is an encapsulated message
	Partial	Body is a fragment of a bigger message
	External-Body	Body is a reference to another message
Image	JPEG	Image is in JPEG format
	GIF	Image is in GIF format
Video	MPEG	Video is in MPEG format
Audio	Basic	Single-channel encoding of voice at 8 kHz
Application	PostScript	Adobe PostScript
	Octet-stream	General binary data (8-bit bytes)

MIME EXAMPLE

```
Received: from 9t2.com ([212.237.25.118]) by [removed] for [removed];  
        Mon, 26 Feb 2018 11:44:34 +0000 (UTC)  
Received: from [69.172.229.5] (unknown [69.172.229.5])  
        by 9t2.com (Postfix) with ESMTPA id C13154335C;  
        Mon, 26 Feb 2018 12:05:48 +0100 (CET)  
MIME-Version: 1.0  
Date: Mon, 26 Feb 2018 02:38:36 -0800  
Subject: FULL OPTION  
Content-Type: multipart/mixed; boundary="=====0574962845=="  
From: "Basil mega" <tpr.sales@intelexpresscargo.com>  
To: Recipients <tpr.sales@intelexpresscargo.com>  
Reply-To: kayali.abdullatif@hotmail.com
```

This is a multi-part message in MIME format.

```
-----0574962845==  
Content-Type: multipart/alternative;  
        boundary="=====0574962845a=="
```

```
-----0574962845a==  
MIME-Version: 1.0  
Content-Transfer-Encoding: quoted-printable  
Content-Type: text/plain; charset="iso-8859-1"  
Content-Description: Mail message body
```

Hello Dear

We are Seamac company limited based in America, and i am company sales manager

CONTENT-TRANSFER-ENCODING

- Method used to encode message into 0s and 1s for transport
- Content-Transfer-Encoding: <type>

Type	Description
7 bit	NVT ASCII characters and short lines.
8 bit	Non-ASCII characters and short lines.
Binary	Non-ASCII characters with unlimited-length lines.
Radix-64	6-blocks of data are encoded into 8-bit ASCII characters using Radix-64 conversion.
Quoted-printable	Non-ASCII characters are encoded as an equal sign followed by an ASCII code.

CONTENT-ID

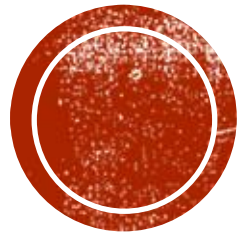
- Uniquely identifies the whole message in a multiple-message environment
- Content-Id: id=<content-id>

CONTENT-DESCRIPTION

- Defines whether the body is a image, audio or video
- Content-Description: <description>

WEB-BASED MAIL

- HTTP for transfer of mail from client to client server
- SMTP for transfer between the sending and receiving servers
- HTTP for transfer of mail from receiving server to receiver



WIRELESS COMMUNICATION

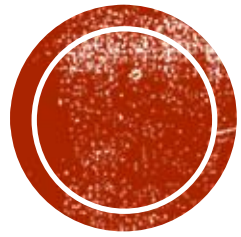
Wireless Technology



INTRODUCTION

- Involves the transmission of information over a distance without the help of wires, cables or any other forms of electrical conductors.
- A broad term that incorporates all procedures and forms of connecting and communicating between two or more devices using a wireless signal through wireless communication technologies and devices.





COMPONENTS OF A WIRELESS COMMUNICATION SYSTEM

COMPONENTS

- Transmitter
- Receiver
- Antenna
- Filters
- Amplifiers
- Mixers

TRANSMITTER

- Modulate, encode several baseband signals onto a high frequency carrier signal
- The signal is then radiated and propagated
- Helps make more efficient use of radio frequency (RF) spectrum

RECEIVER

- Receives the modulated signals
- Reverses the functions of transmitter and recovers the transmitted signals
- Antenna receives electromagnetic waves from many sources over a relatively broad frequency range

ANTENNA

- Transmitter side: Converts electric signals to propagating electromagnetic RF wave
- Receiver side: Converts propagating RF wave to electric signals
- Transceiver: Transmitter and Receiver are co-located for full-duplex communication, so same antenna is shared
- Two types of antennas: Omni-directional and directional

FILTERS

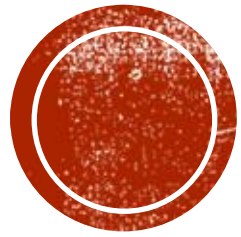
- Present in all wireless transmitters and receivers
- Reject interfering signals lying outside the operating band
- Reject unwanted noise signals generated by amplifier

AMPLIFIER

- Amplifies the strength of the signal
- Specifications include **power gain and noise figure**
- Noise figure: how much noise is added to the amplified signal; this is critical for the front-end of the receiver
- First amplifier to have low noise figure as possible

MIXERS

- Used for frequency conversion at transmitters and receivers
- Required as it is advantageous to transmit signals at a higher frequency
- Achieved by modulating carrier waveform with original baseband frequency
- With this, signals can be easily and effectively radiated and becomes less susceptible to noise and attenuation
- When transmission range increases, signal quality improves

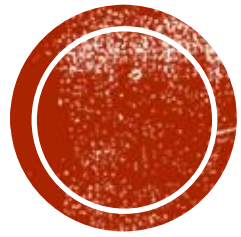


WIRELESS NETWORKING STANDARDS

WIRELESS NETWORKING STANDARDS

- The IEEE 802.11a/b/g/n/ac/ax standards are widely accepted throughout the industry.

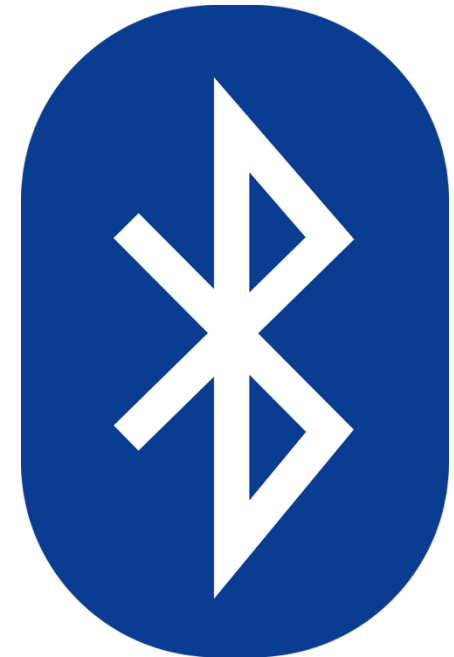
IEEE Standard	WiFi Gen	Year	Frequency	Max PHY Data Rate	Max Range
802.11	-	1997	2.4 GHz	2 Mbps	20m (indoor) 100m (outdoor)
802.11 a	-	1999	5 GHz	54 Mbps	35m (indoor) 120m (outdoor)
802.11 b	-	1999	2.4 GHz	11 Mbps	35m (indoor) 140m (outdoor)
802.11 g	-	2003	2.4 GHz	54 Mbps	38m (indoor) 140m (outdoor)
802.11 n	WiFi 4	2009	2.4/5 GHz	600 Mbps	70m (indoor) 250m (outdoor)
802.11 ac	WiFi 5	2013	5 GHz	6.9 Gbps	35m (indoor)
802.11 ad	-	2012	60 GHz	8.1 Mbps	3.3m (indoor)
802.11 ah	-	2017	Sub 1 GHz	347 Mbps	1km
802.11 ax	WiFi 6	2021	2.4/5/6 GHz	9.6 Gbps	30m (indoor) 120m (outdoor)
802.11 ay	-	2021	60 GHz	303 Gbps	10m (indoor) 100m (outdoor)
802.11 be	WiFi 7	2024	2.4/5/6 GHz	46.1 Gbps	30m (indoor) 120m (outdoor)



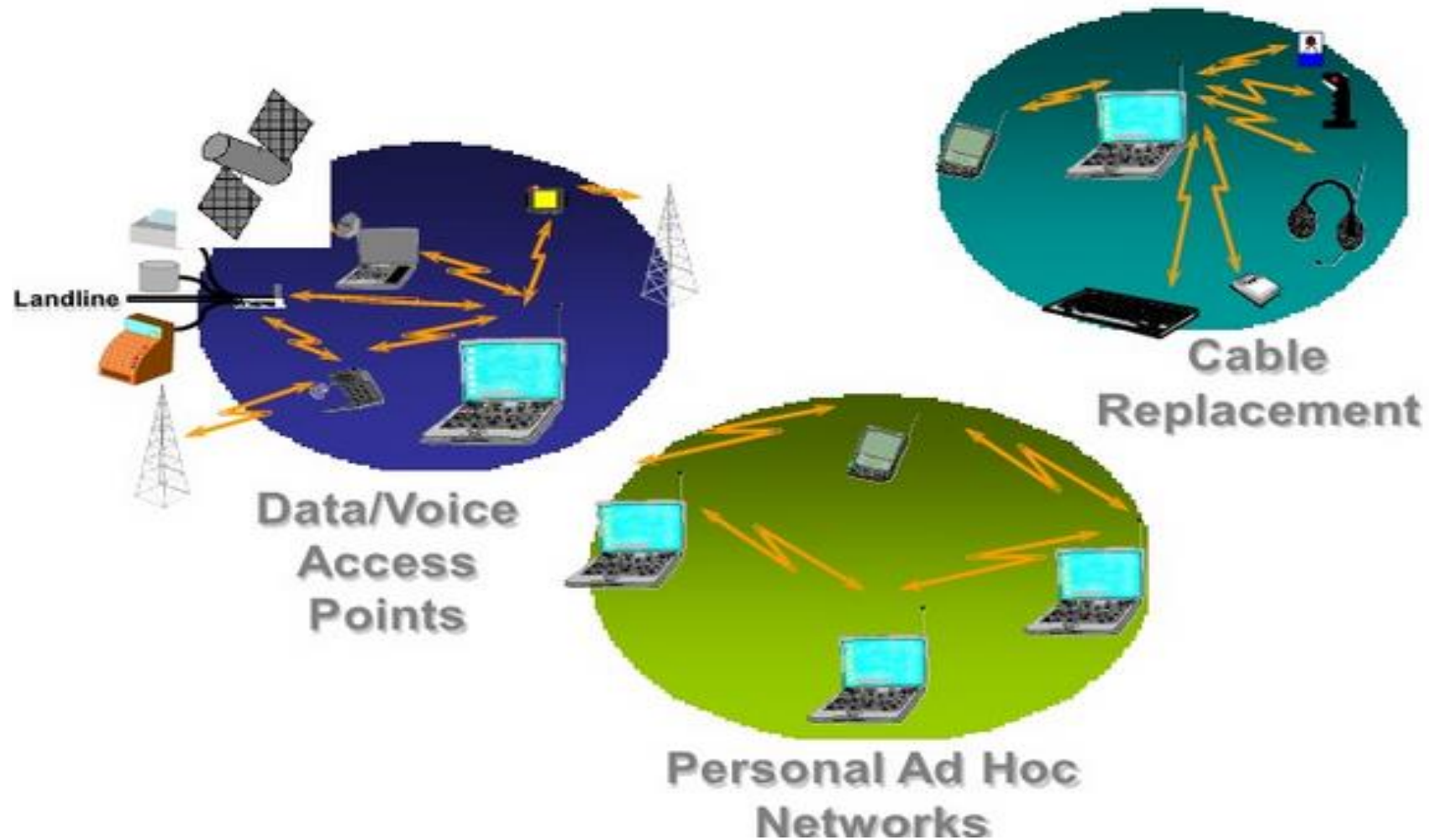
BLUETOOTH

INTRODUCTION

- Implements **PAN** (Personal Area Network)
- PAN helps to interconnect a **set of computerized devices**
- **255 devices within 10 meters**



What Can Bluetooth Do For You?



BLUETOOTH

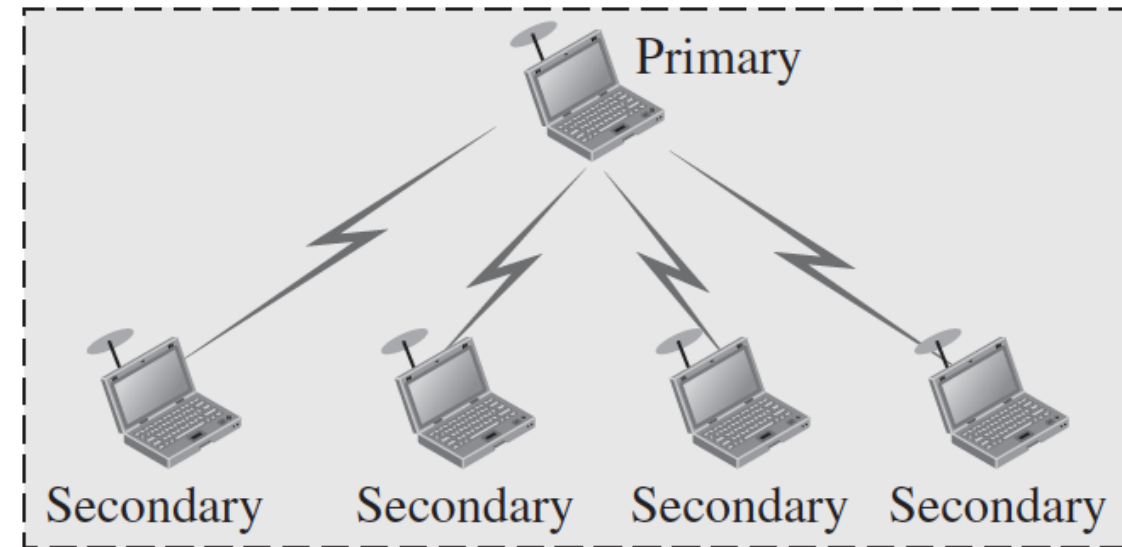
- Two types of Networks:
 - Piconet
 - Scatternet

PICONET

- Bluetooth network is called a piconet, or a small net.
- Can have up to eight stations; one of which is called the primary; the rest are called secondaries.
- It can have only one primary station.
- The communication between the primary and secondary stations can be one-to-one or one-to-many.

Active State and Parked State

Piconet

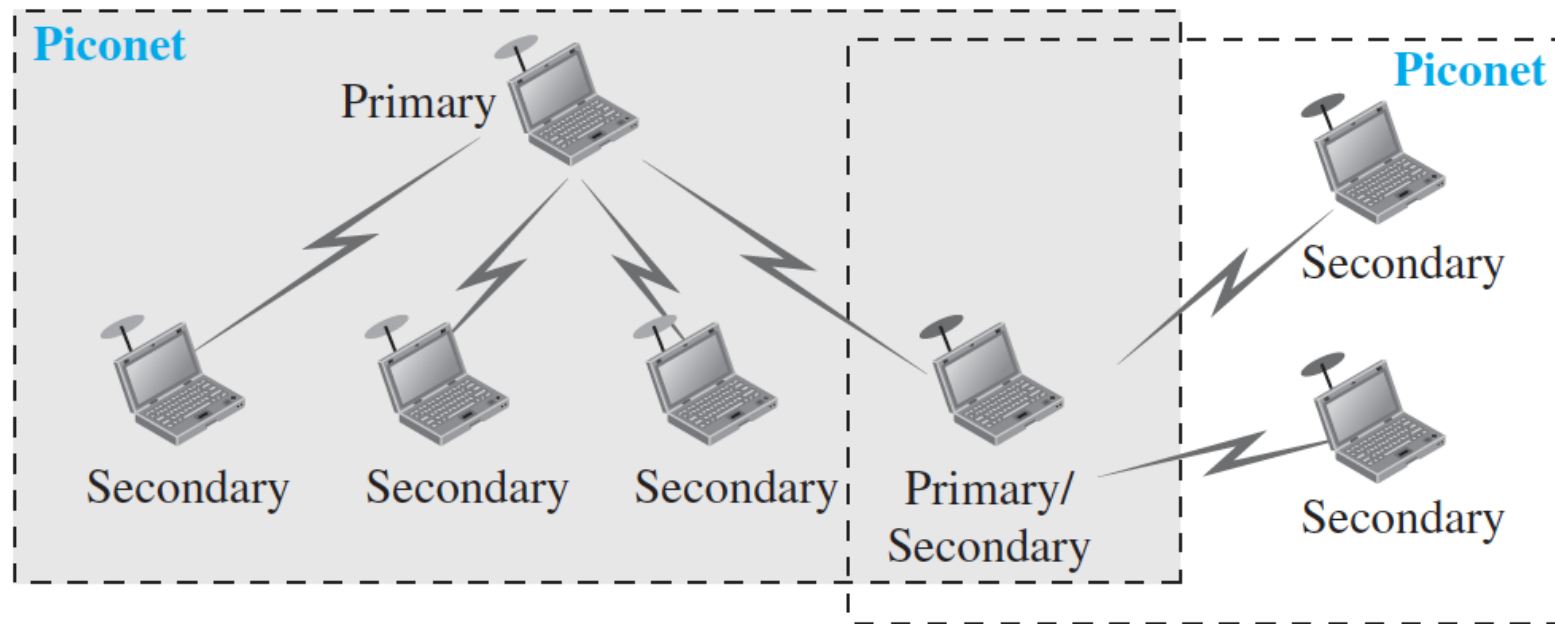


Active State: Devices are actively connected to the master.

Parked State: Devices are still part of the piconet, but not actively communicating.

SCATTERNET

- Piconets can be combined to form what is called a scatternet.
- A secondary station in one piconet can be primary in another piconet.
- A station can be a member of two piconets.



BLUETOOTH PROTOCOL STACK

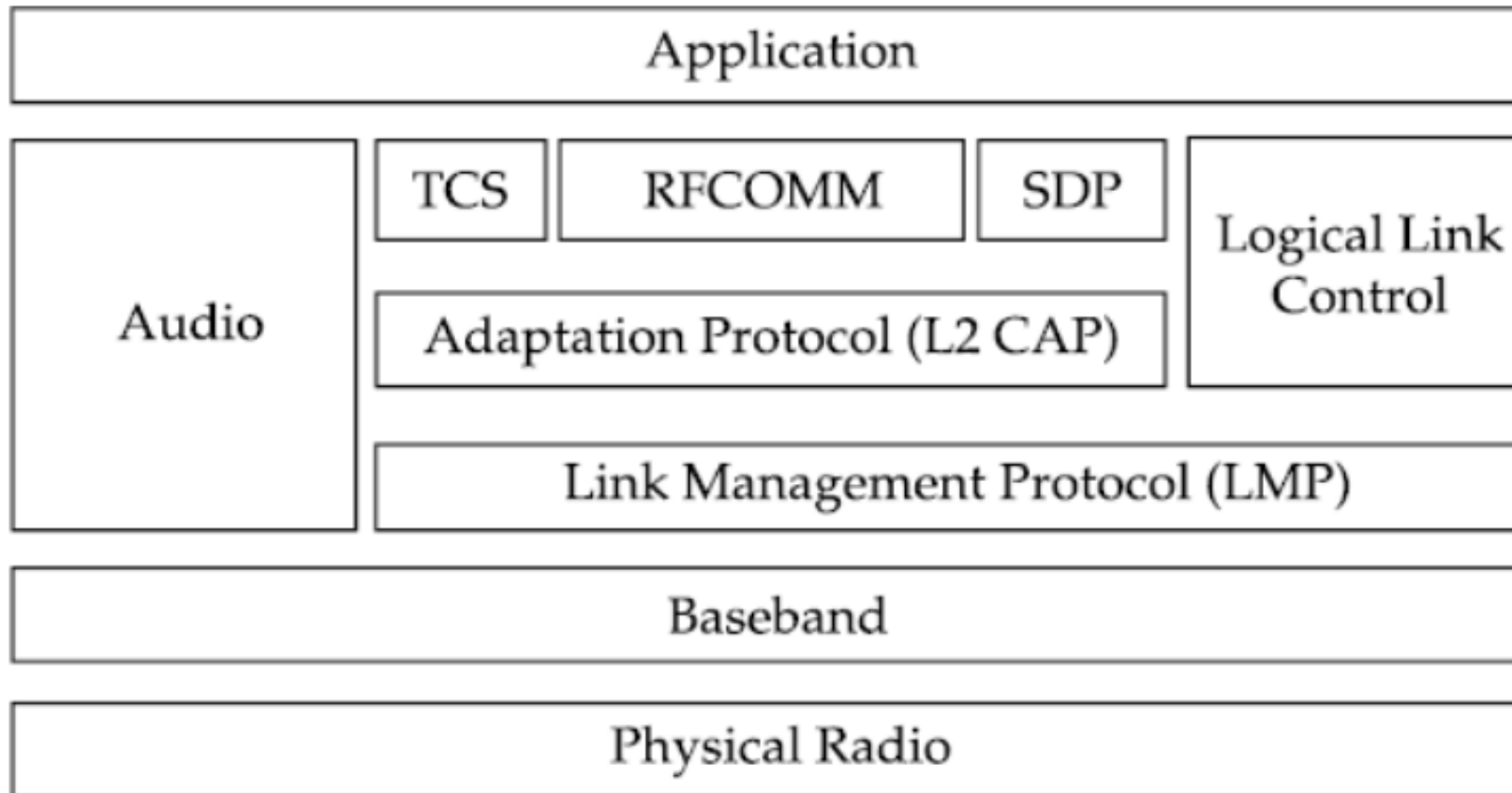


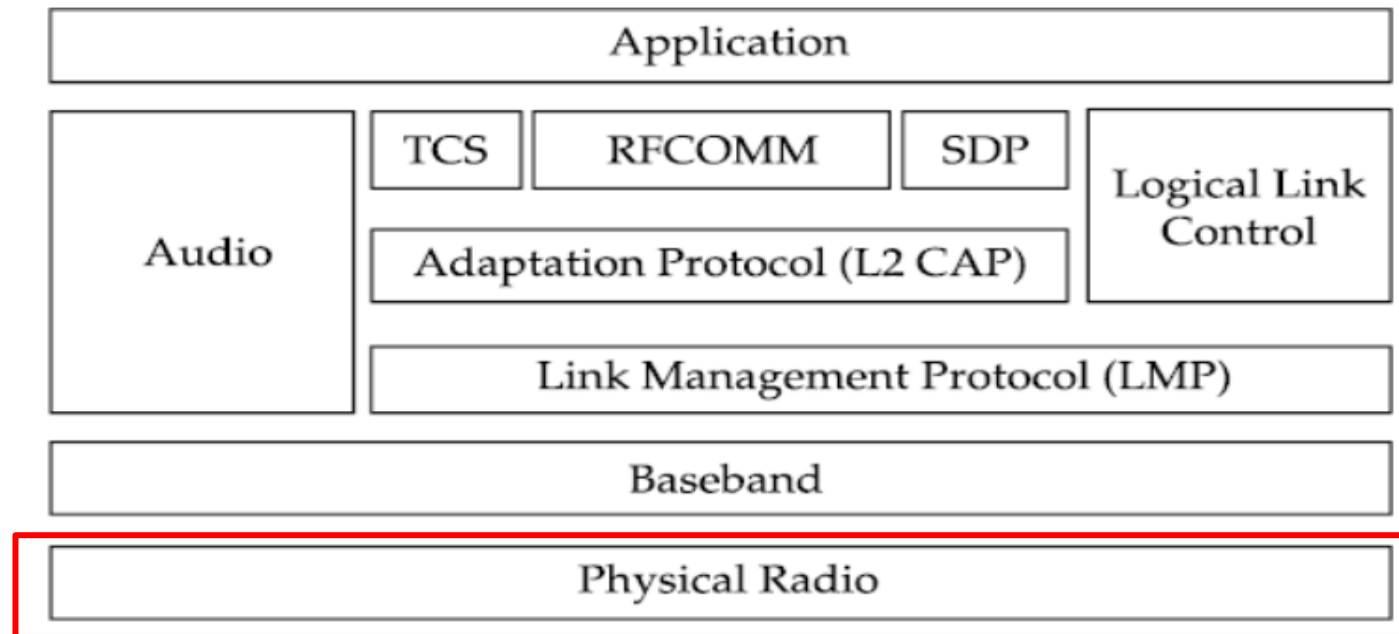
Figure 1.8 *Bluetooth protocol stack.*

BLUETOOTH PROTOCOL STACK

- Radio Layer
- Baseband
- Link Management Protocol (LMP)
- Logical Link Control and Adaptation Protocol (L2CAP)
- Telephony Control Protocol Specification (TCS)
- Radio Frequency Communication (RFCOMM)
- Services Discovery Protocol (SDP)
- Application

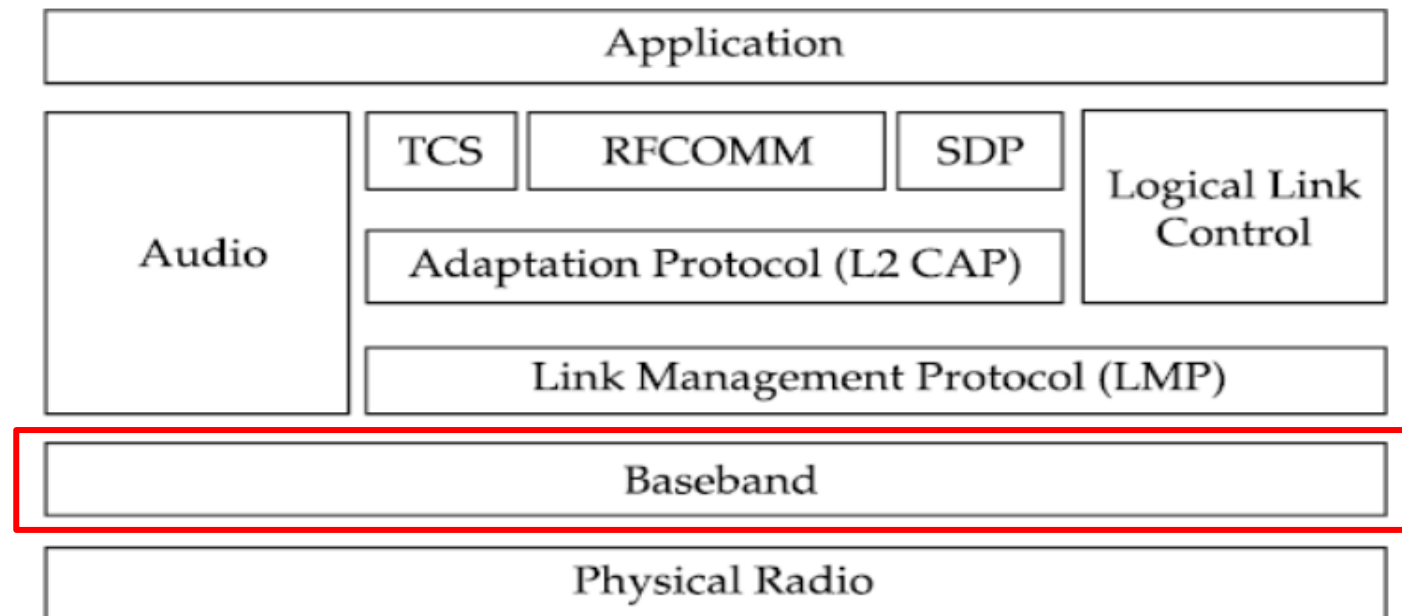
RADIO LAYER

- Modulation / Demodulation of data into RF signals
- Defines the physical characteristics of a Bluetooth transceiver



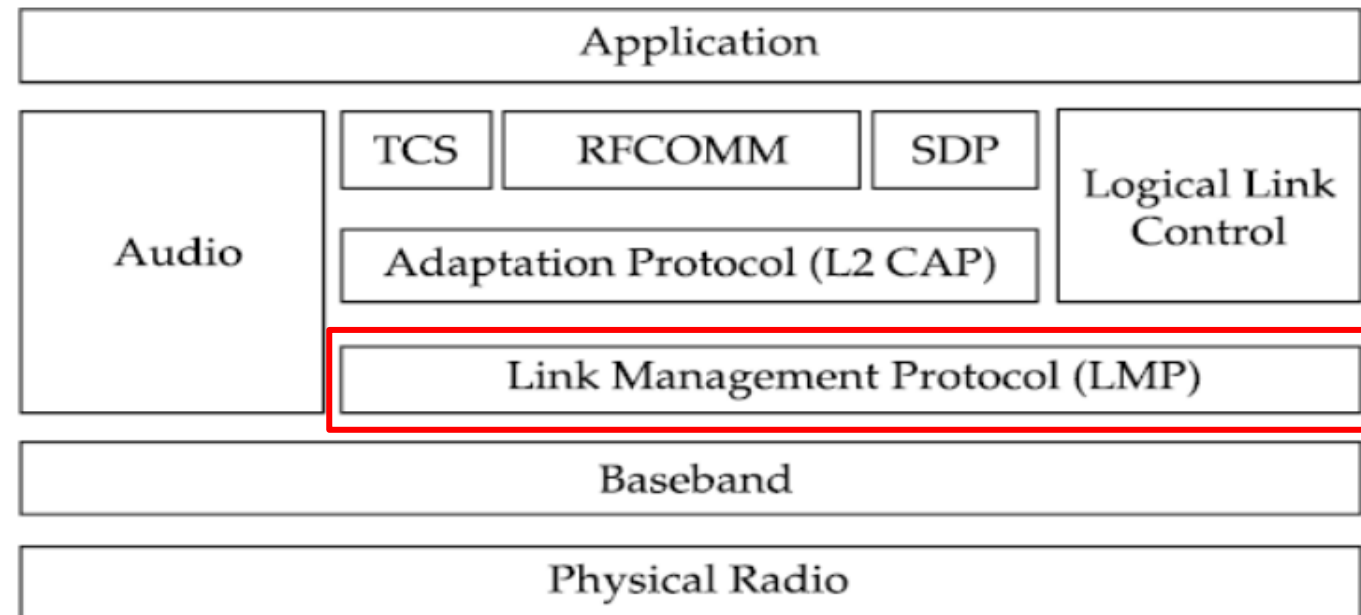
BASEBAND

- Takes those signals and structures them into **packets**
- Performs connection establishment within a piconet



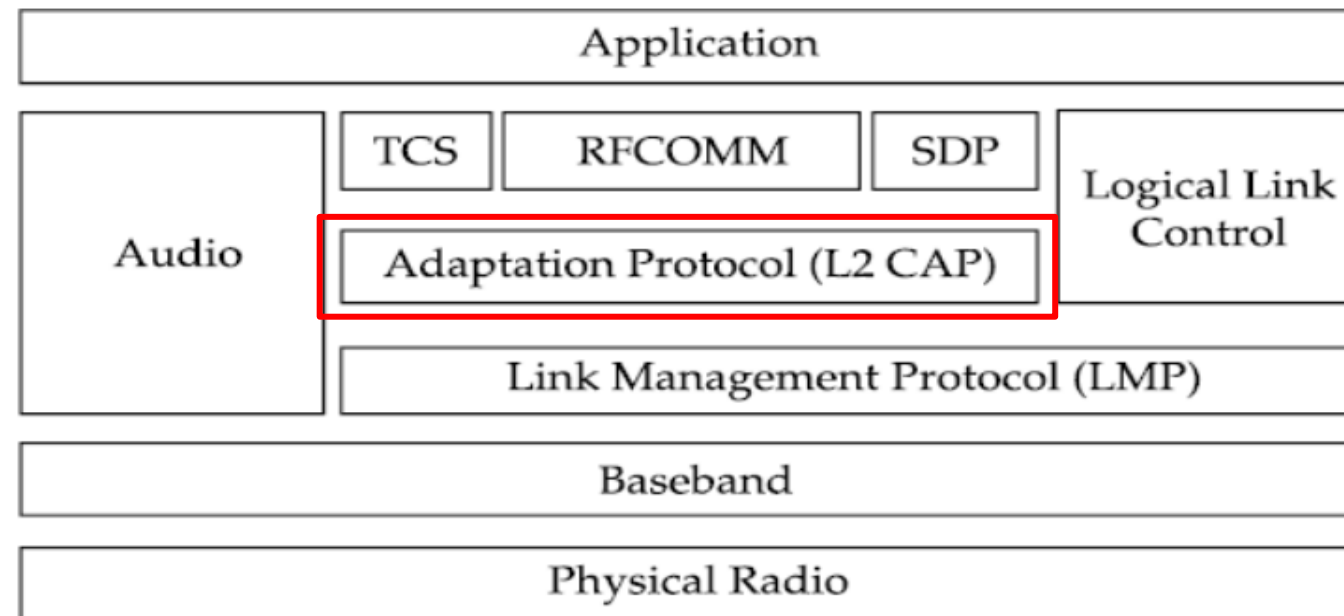
LINK MANAGEMENT PROTOCOL (LMP)

- Manages the already established links
- It includes authentication, encryption, security processes



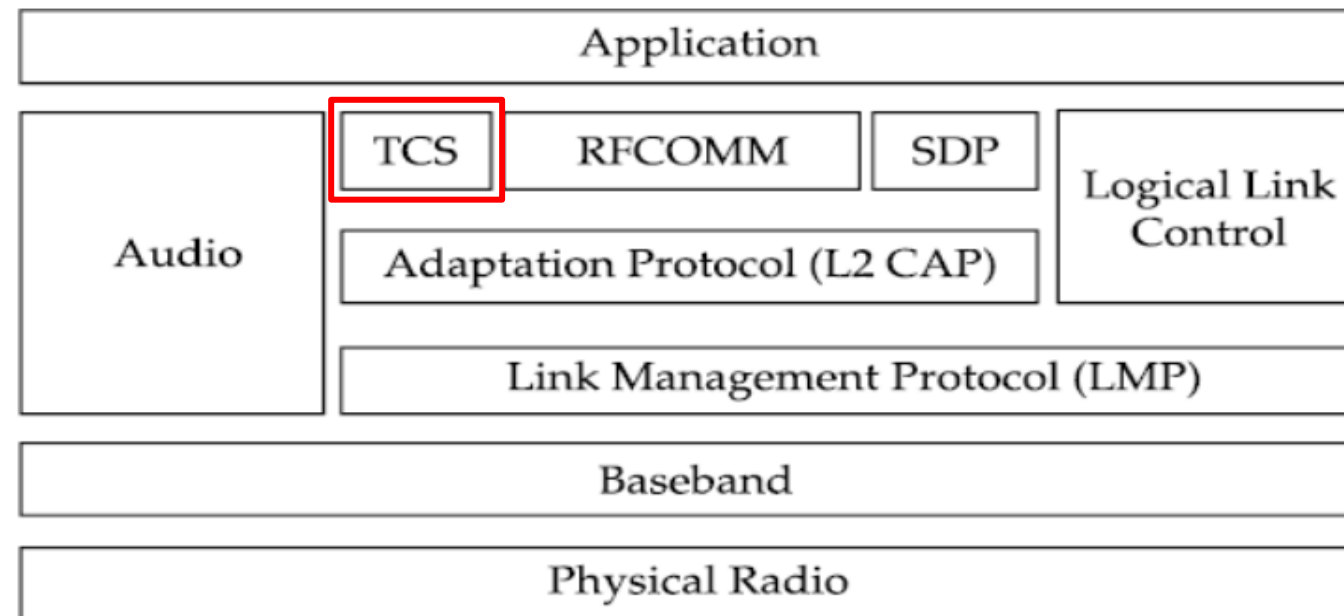
LOGICAL LINK CONTROL AND ADAPTATION PROTOCOL (L2CAP)

- Heart of the Stack
- Allows communication between upper and lower layers
- Packages data packets from upper layers in the form expected by the lower layers; does segmentation and multiplexing



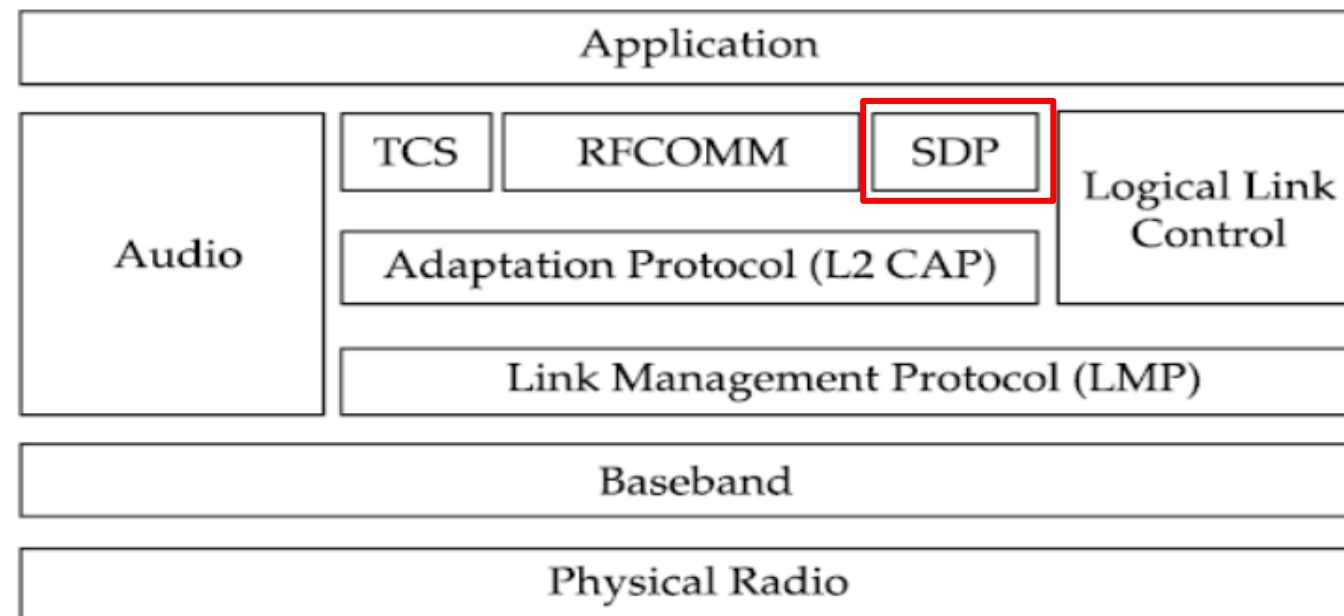
TELEPHONY CONTROL PROTOCOL SPECIFICATION (TCS)

- Provides telephony services - **setting up and managing voice calls** between Bluetooth devices.
- Example: Controlling a **Bluetooth headset** or a **hands-free car kit**.



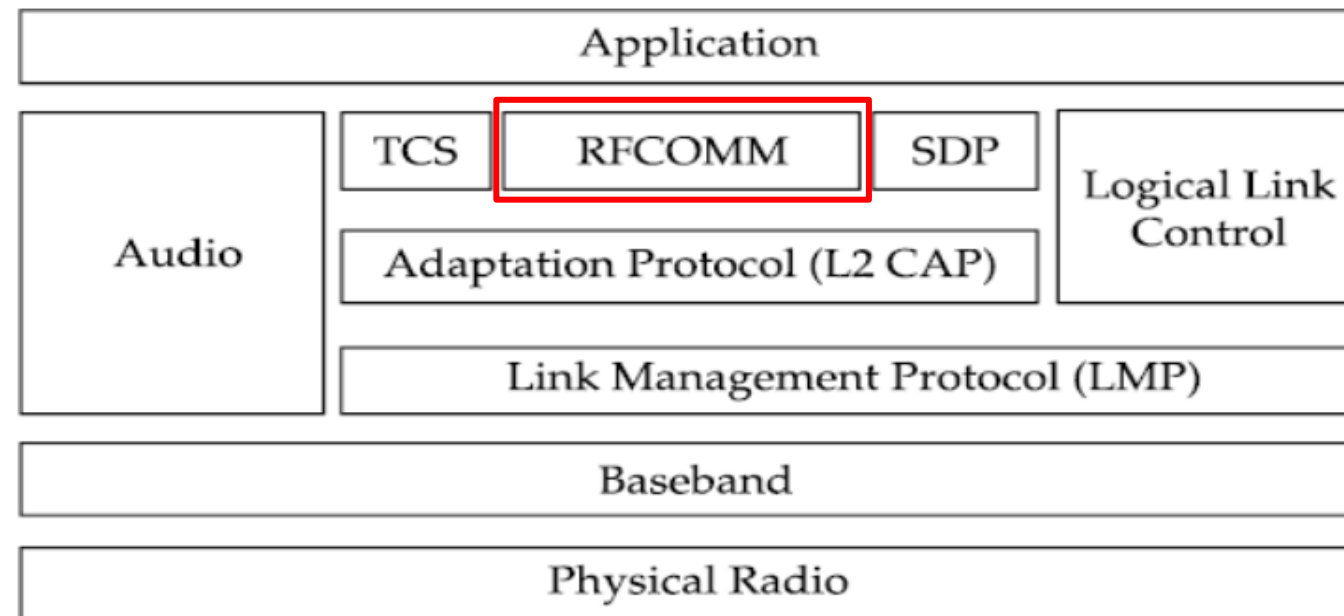
SERVICES DISCOVERY PROTOCOL (SDP)

- Allows to discover services available on other Bluetooth enabled devices
- Example: A phone checking if a headset supports **hands-free audio** etc.



RADIO FREQUENCY COMMUNICATION (RFCOMM)

- A transport protocol running on top of L2CAP
- Provides serial interface with application layer's WAP (Wireless Application Protocol) and OBEX (Object Exchange)

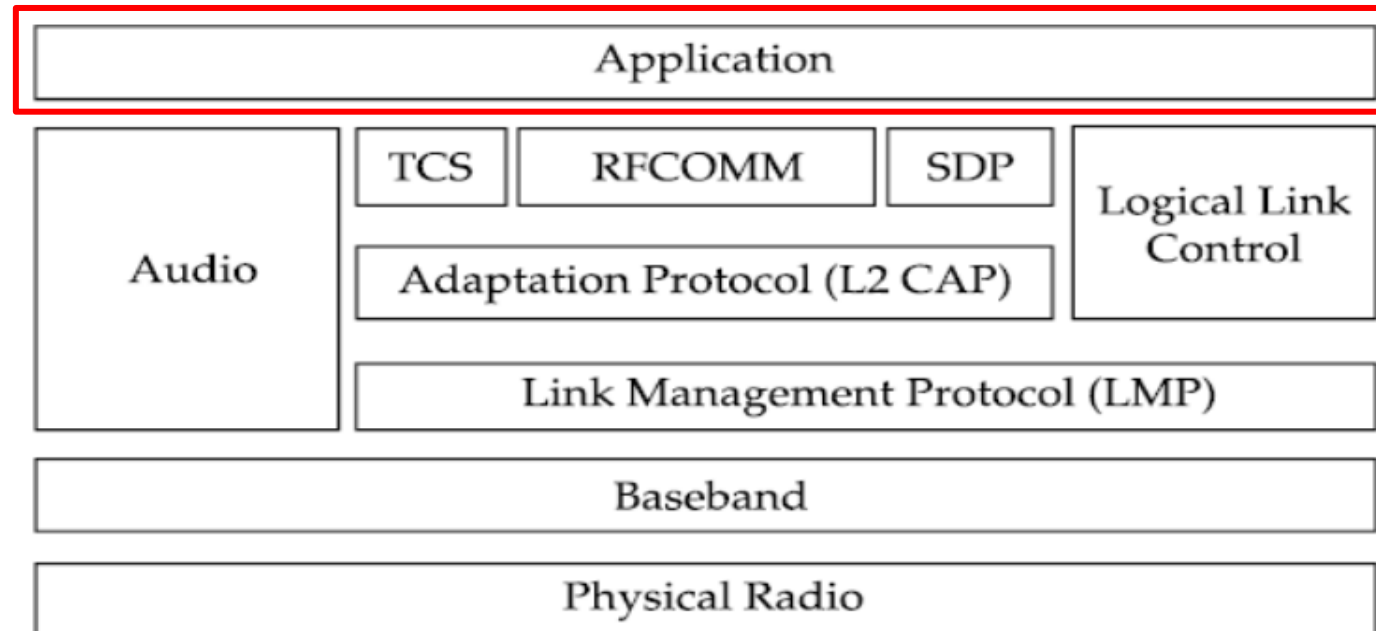


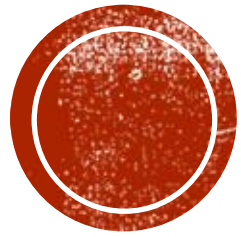
WAP AND OBEX

- WAP (Wireless Application Protocol): A protocol designed to deliver internet content to wireless devices like mobile phones, PDAs, or other handheld devices.
- OBEX (Object Exchange): A communication protocol used to exchange binary objects (files, contacts, calendar entries, etc.) between devices over a short-range wireless link like Bluetooth.

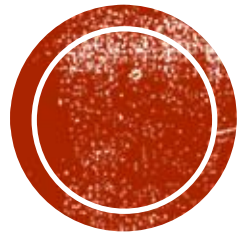
APPLICATION LAYER

- Enables the user to interact with the application





OTHER WIRELESS TECHNOLOGIES



WiMAX

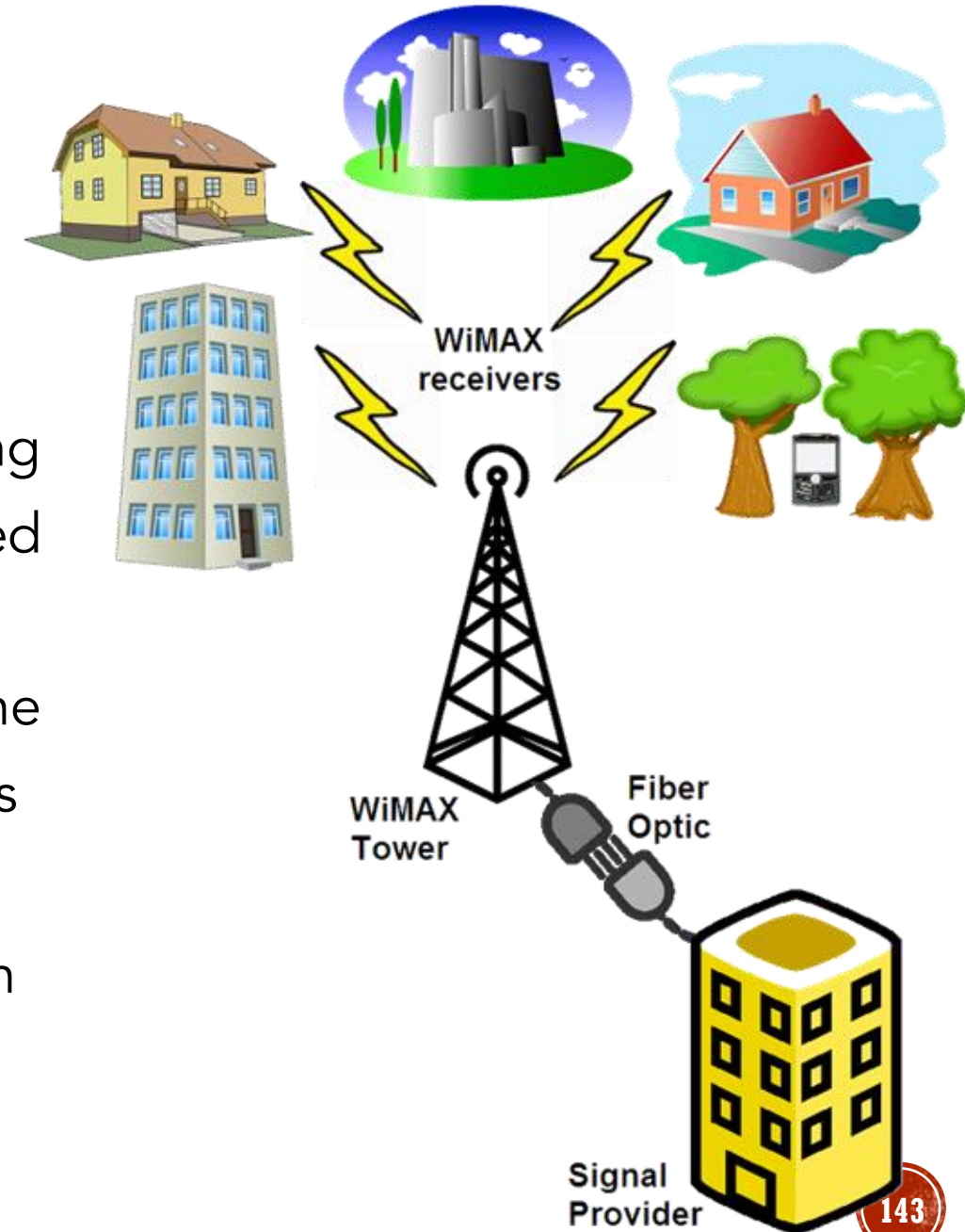
INTRODUCTION

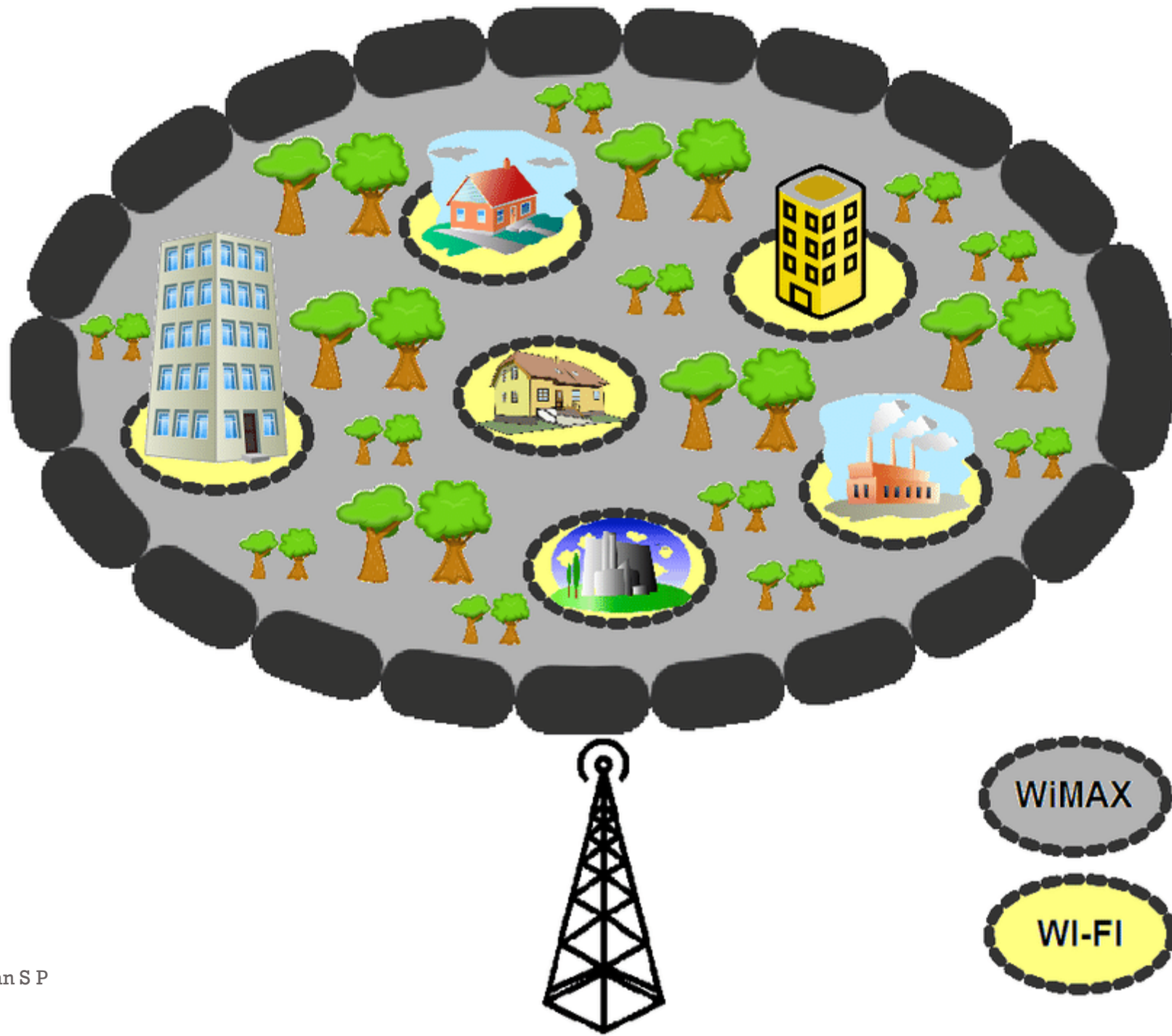
- WiMAX: Worldwide Interoperability for Microwave Access
- It is based on IEEE 802.16 standard, which is called Wireless MAN (Metropolitan Area Network)
- WiMAX is a non-profit organization established to promote the adoption of WiMAX compatible products, services and to ensure a high level of interoperability among them.

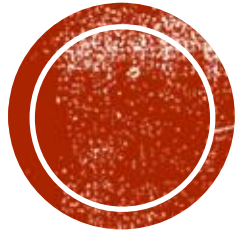


WiMAX COMPONENTS

- Base Station
 - where the signals are broadcasted from.
 - It is connected to the public network using fiber optics, radio links or other high-speed point-to-point.
 - can cover up to 50 kms radius but due the geographic limitations it is just about 10 kms
- Receiver
 - receive the signals from WiMAX base station
 - just like WiFi but allows longer distances







WIRELESS PAN PROTOCOLS

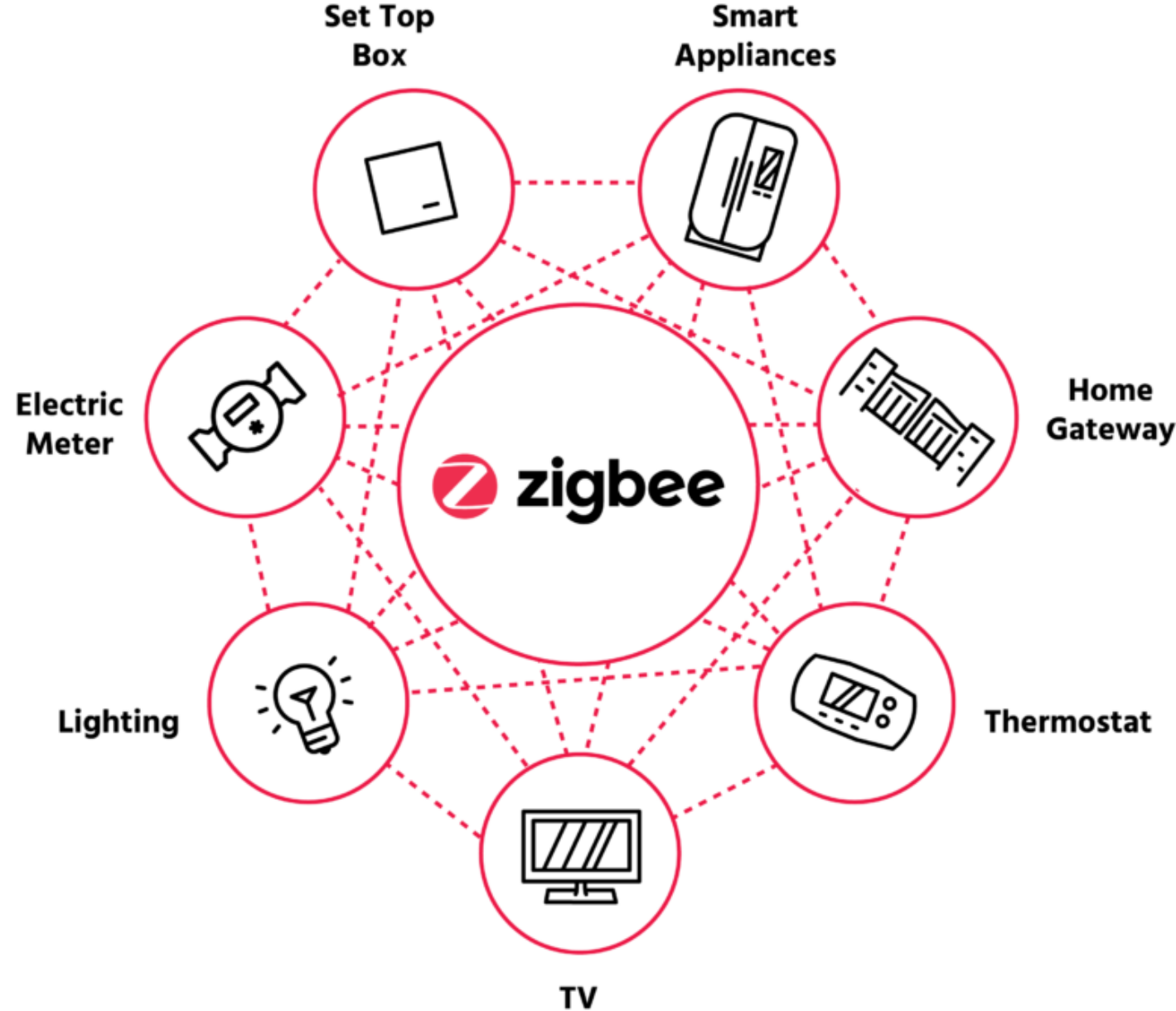
ZigBee, ZWave, Thread, BLE, 6LoWPAN

ZIGBEE

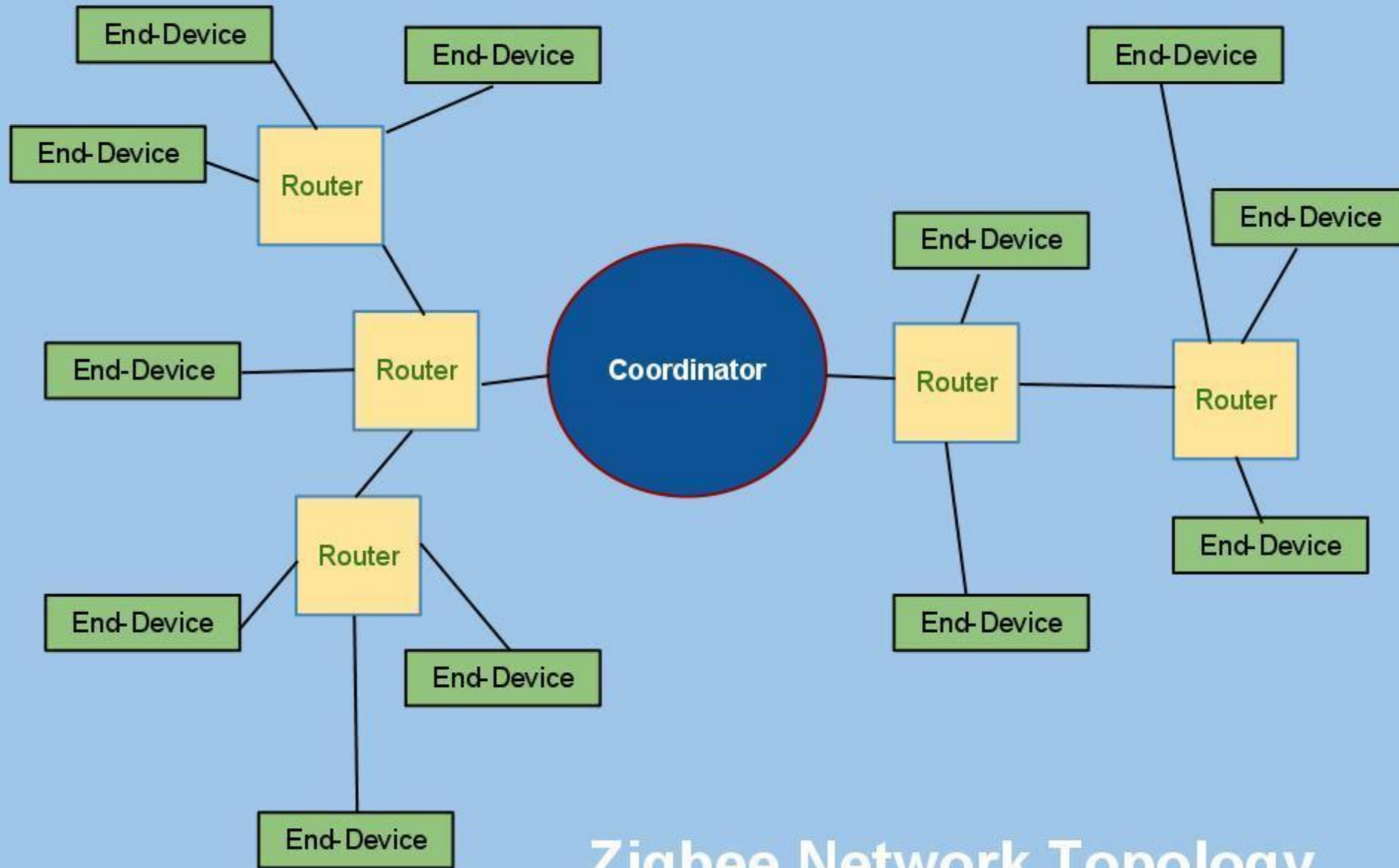
- Zigbee is an IEEE 802.15.4-based specification for a suite of high-level communication protocols used to create personal area networks
- Zigbee is a low-power, low data rate, and close proximity (i.e., personal area) wireless ad hoc network.

ZIGBEE

- Used for small, low-power digital radios needing wireless connection
 - Home automation
 - medical device data collection
 - Other low-power low-bandwidth needs designed for small scale projects



Smart Home



Zigbee Network Topology

PHILIPS

hue

 **SmartThings**

 **Yale** smart Living

Honeywell

 **ZigBee**[®]

 **BOSCH**



HIVE



amazon alexa



ZIGBEE

Pros

- Very low energy use
- Secure transmission
- Common protocol for smart home devices

Cons

- Requires a central hub
- More vulnerable to hackers because it is open protocol

ZWAVE

- A wireless communications protocol used primarily for residential and commercial building automation.
- It is a mesh network using low-energy radio waves to communicate from device to device
- Allows wireless control of smart home devices, such as smart lights, security systems, thermostats, sensors, smart door locks, and garage door openers.





ZWAVE

Pros

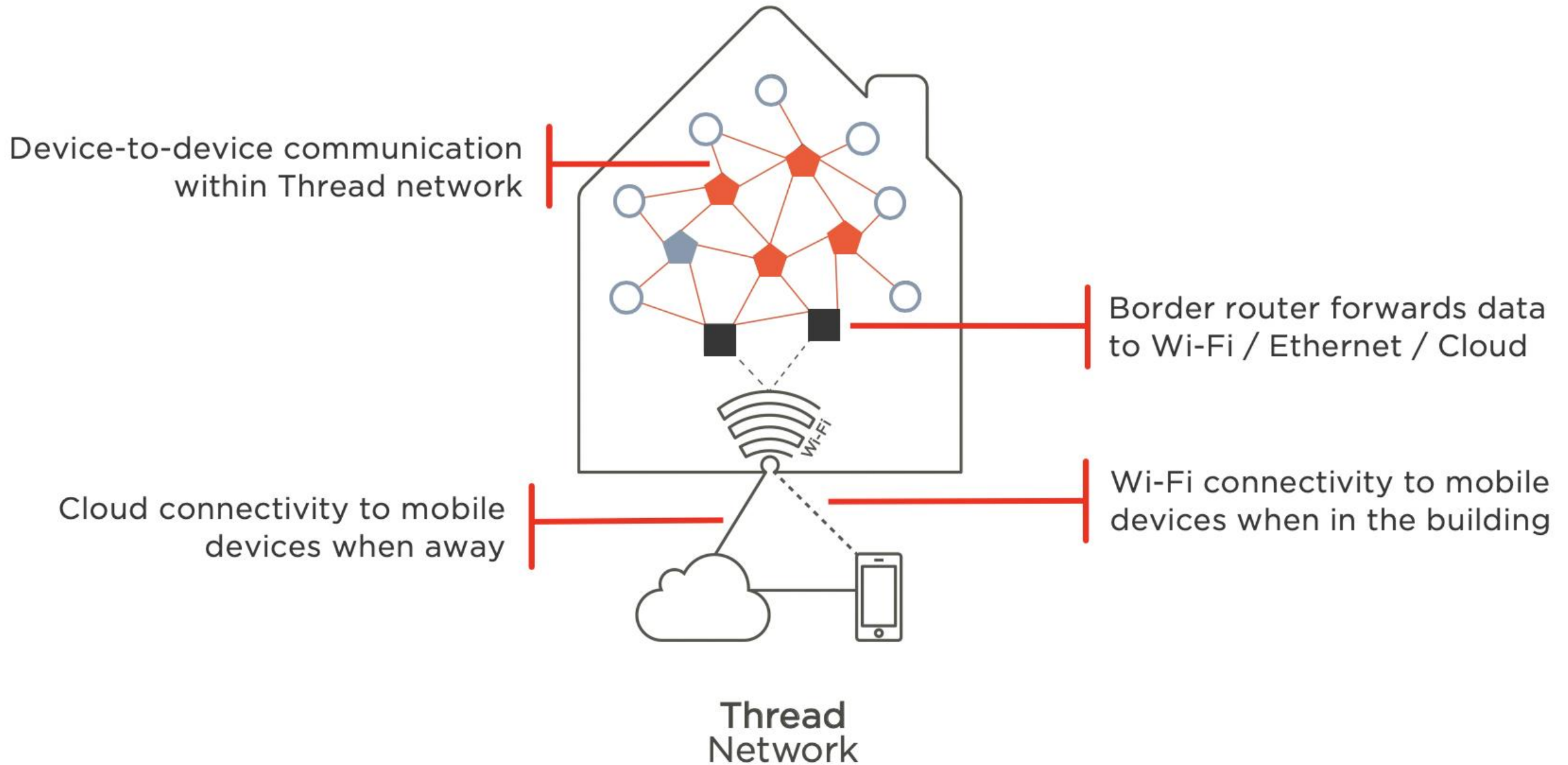
- Secure transmission
- Low interference
- Compatible with a variety of devices

Cons

- Requires a central hub
- There is a maximum amount of devices that can connect to a single hub
- Closed system

THREAD

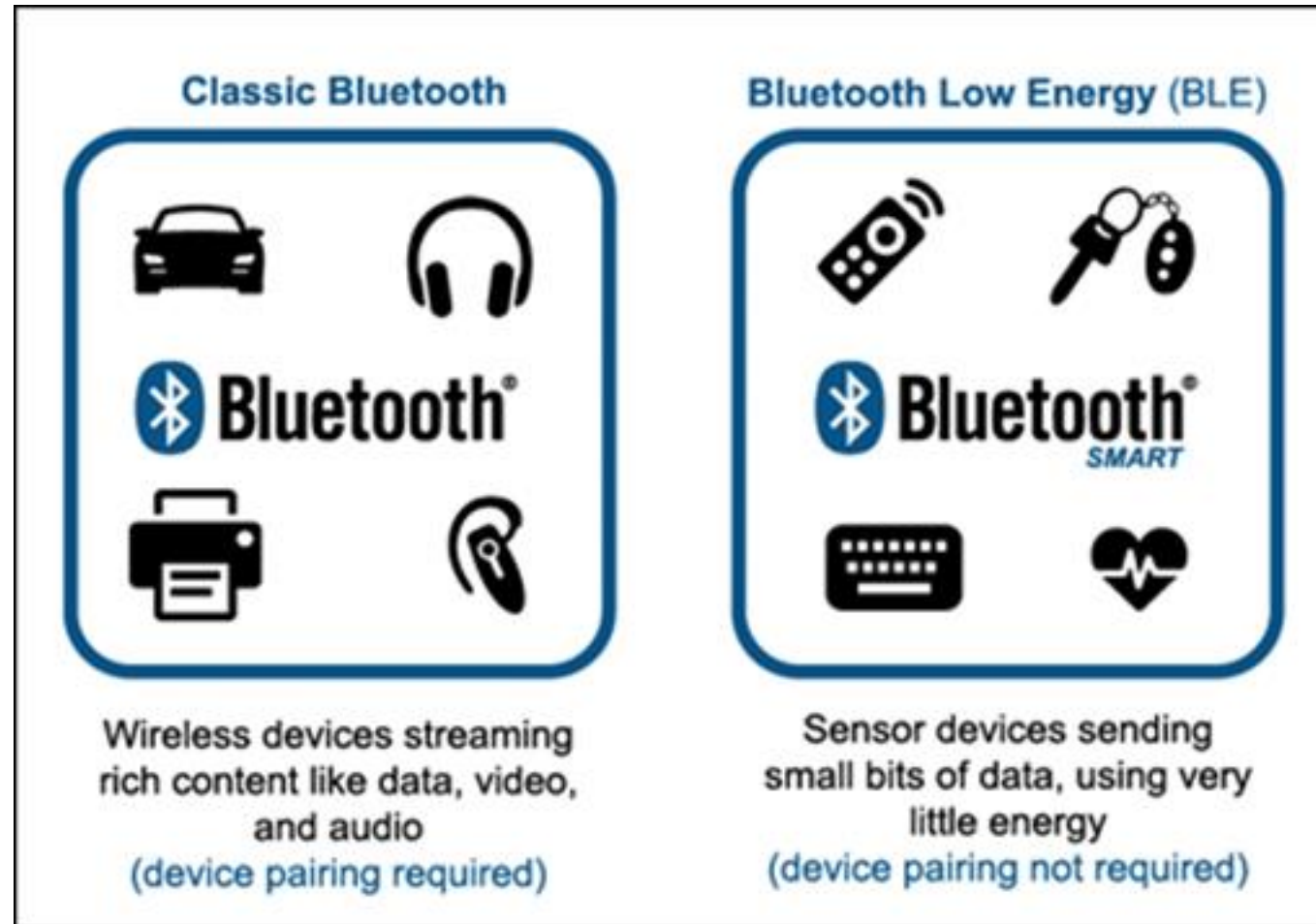
- Established by Google Nest
- Thread uses an IP network, which makes it easier for it to communicate with Wi-Fi networks and cellular networks.



BLUETOOTH LOW ENERGY (BLE)

- A wireless personal area network technology designed and marketed by the Bluetooth Special Interest Group (Bluetooth SIG)
- Aimed at novel applications in the healthcare, fitness, beacons, security, and home entertainment industries.

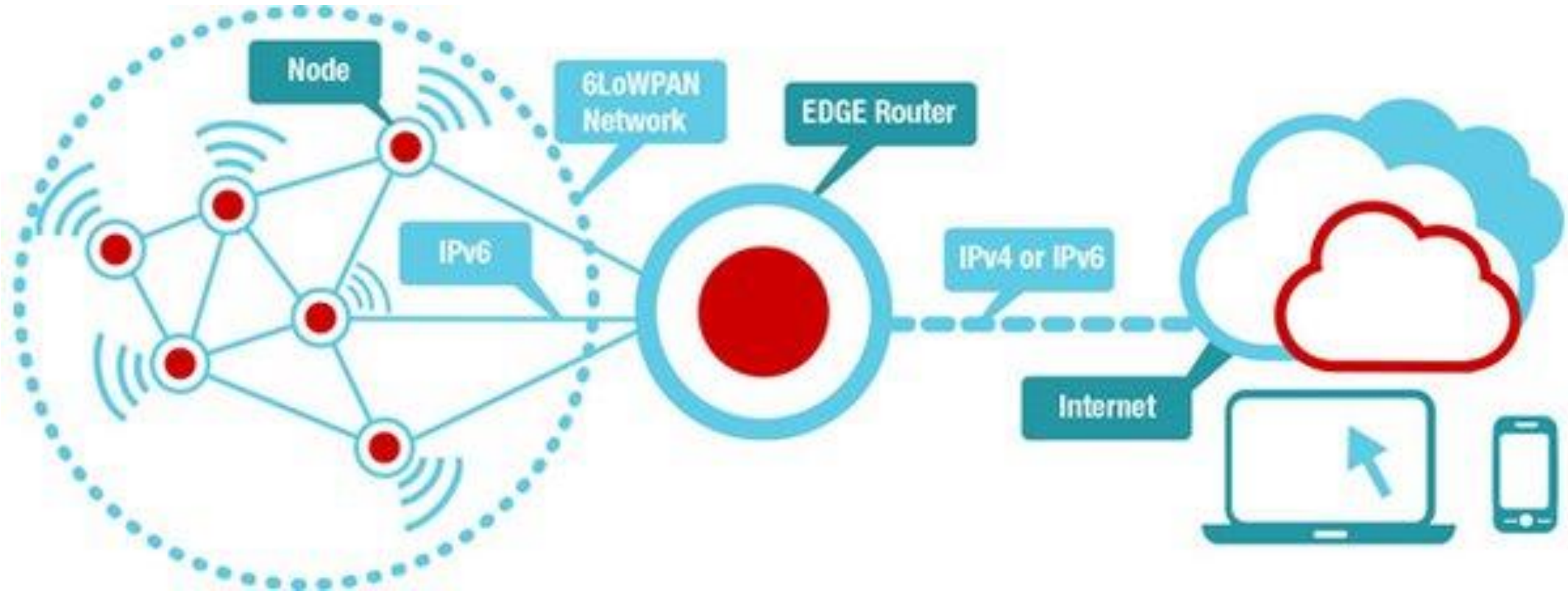
BLE



6LoWPAN (IPV6 OVER LOW-POWER WIRELESS PERSONAL AREA NETWORKS)

- A low power wireless mesh network where every node has its own IPv6 address
- 6LoWPAN came to exist from the idea that the Internet Protocol could and should be applied even to the smallest devices, and that low-power devices with limited processing capabilities should be able to participate in the Internet of Things.
- Applied in home automation, industrial monitoring, smart grid, smart home

6LoWPAN



RPL

- RPL (Routing Protocol for Low-Power and Lossy Networks) is a routing protocol for wireless networks with low power consumption and generally susceptible to packet loss.
- It is a proactive protocol based on distance vectors and operates on IEEE 802.15.4, optimized for multi-hop and many-to-one communication, but also supports one-to-one messages.

thank
you!